Critical Care Nursing

تمريض العناية المكثفة

NURS4328

Dr. Wael Subhi Mekky

Learning Objectives

- **At the end of this lecture, the student should be able to:**
  - Define the meaning of critical care nursing
  - Identify the conditions considering critical
  - Recognize the classifications of critical care unit
  - Differentiate different types of critical care units
  - Explain the principles of critical care unit working
  - Identify how critical care nursing evolution.
  - Describe the quality assurance in critical care nursing.

Definition of Terms

- **CRITICAL CARE :** is a term used to describe as the care of patients who are extremely ill and whose clinical condition is unstable or potentially unstable.

Definition of Terms

- **CRITICAL CARE UNIT :** it is the unit in which comprehensive care of a critically ill patient which is deemed to recoverable stage is carried out.

Definition of Terms

- **CRITICAL CARE UNIT :** is a specially designed and equipped facility; staffed by skilled personnel to provide effective and safe care for dependent patients with a life threatening problem.

The Critical Care Unit is not merely a room or series of rooms filled with patients attached to interventional technology; it is the home of an Organization: the Critical Care Team

Definition of Terms

- **CRITICAL CARE NURSING :** it refers to those comprehensive, specialized and individualized nursing care services which are rendered to patients with life threatening conditions and their families.

Definition of Terms

- **CRITICAL CARE NURSING:** is that specialty within nursing that deals specifically with human responses to life-threatening problems.

Definition of Terms

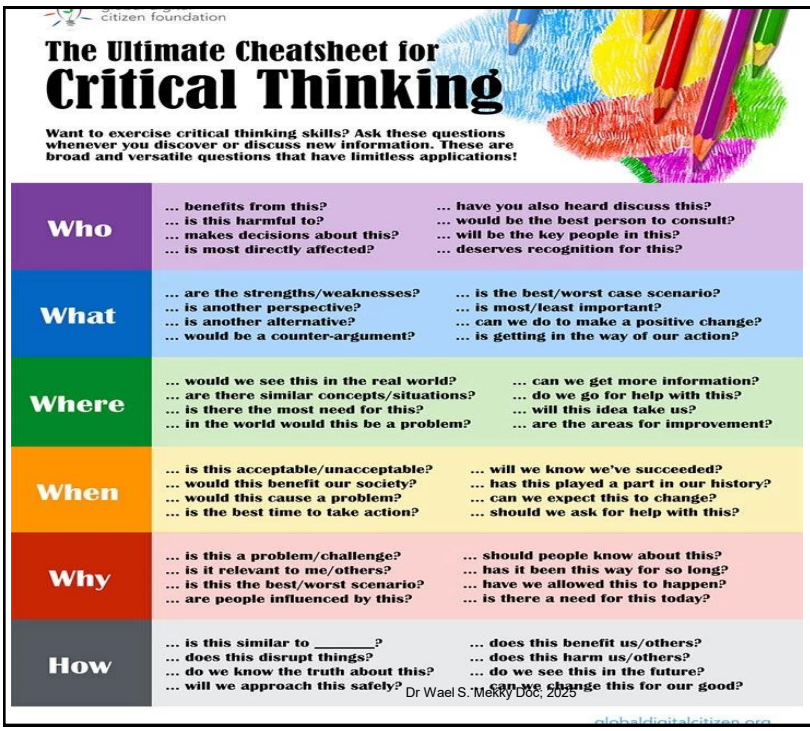
- **CRITICAL CARE NURSE:** is a licensed professional nurse who is responsible for ensuring that all critically ill patients and their families receive optimal care through: professional task statement component encompasses the five steps of the nursing process—assessment, diagnosis, planning, intervention / implementation, and evaluation.

Critical Care Nurse

A Critical Care Nurse is a licensed professional nurse who is responsible for ensuring that acutely and critically ill patients and their families receive optimal care

Principles of critical nursing care

- Priorities of patients.
- Early detection and prompt action.
- Collaborative practice.
- Communication : interpersonal, intradepartmental
- Crisis intervention and stress reduction.
- Critical thinking.



Principles of Critical Care Nursing

- Anticipatory Nursing Care
- Early Detection and Prompt Action
- Expertise
- Supportive Care
- Communication
- Collaborative Practice

Principles of Critical Care Nursing

- Preservation of Patient's Physiological Defenses
- Prevention of Infection
- Crisis Intervention
- Stress Reduction
- Ethical Principles

SEVEN Cs OF CRITICAL CARE

- **Compassion (عطف أو شفقة)**
- **Communication (with patient and family).**
- **Consideration (to patients, relatives and colleagues) and avoidance of Conflict.**
- **Comfort: prevention of suffering**
- **Carefulness (avoidance of injury)**
- **Consistency**
- **Closure (ethics and withdrawal of care).**

What are the conditions considered as Critical?

1. Any person with life threatening condition

2. Patients with :

- Acute renal failure
- AMI
- Cardiac tamponate (انسداد شرايين القلب)
- Severe shock
- Heart block
- Poly trauma
- Multiple organ failure and organ dysfunction
- Severe burns

Where Critical Care is Provided ?

- ✓ Accident & Emergency Units
- ✓ High Dependency Units
- ✓ Intensive Care Units
- ✓ Coronary Care Units
- ✓ Recovery (PACU)
- ✓ Anesthetic Rooms
- ✓ Operating Theatres

Evolution of Critical Care Nursing

Critical care units have evolved over the last many decades in response to medical advances

Historical Perspective

- Florence Nightingale recognized the need to consider the severity of illness for bed allocation of patients and she placed the seriously ill patients near the nurses' station.
- In 1923, John Hopkins University Hospital developed a special care unit for neurosurgical patients .
- After World War II: Modern medicines, concept of triage and speciality nursing came into existence

World War II: 1/9/1939 – 2/9/1945

Historical Perspective

- As surgical techniques advanced, it became necessary that post-operative patients required careful monitoring and this brought about the Recovery Room
- In 1950, the epidemic of poliomyelitis necessitated thousands of patients requiring respiratory assist devices and intensive nursing care
- At the same time came about newer horizons in cardiothoracic surgery, with refinements in intra-operative membrane oxygen techniques

Historical Perspective

- Late 1950's – beginning of Critical Care Units
- 1965 – 1st specialized ICU – The Coronary Care Unit
- Emergence of Specialized ICUs

Evolution of Critical Care

- Began with polio units (1950s), recovery rooms, and coronary care units (1960s)
- Patient outcomes improved
- Intensive Care Units (1970s)

Critical Care in the 21st Century

- **Specialization**
  - Cardiovascular, neurological, trauma, burn units
  - Pediatric
  - Neonatal
- **Variety of settings**
  - Inpatient
  - Outpatient
  - Home care
- **eICU :** Electronic ICU (eICU) is a new application of critical care from a remote monitoring site.

Professional Organizations

- Support critical care nurses and practice
- Key organizations:
  - American Association of Critical-Care Nurses
  - Society of Critical Care Medicine

American Association of Critical-Care Nurses (AACN)

- Founded in 1969
- Largest specialty organization in world
- Mission
  - Assist nurses to attain knowledge and influence
- Healthy work environment initiative
- Web site: www.aacn.org

AACN's Synergy Model

•The American Association of Critical Care Nurses (AACN) began developing their Synergy Model in the early 1990s as a way to match patient/family needs to nurse competencies

•The term synergy comes from the Greek word synergos, which means working together.

AACN's Synergy Model

AACN Certification Corporation designed synergy model describing nursing practice based on needs & characteristics of patients &demands of healthcare environment predicted for the future Today's healthcare environment is changing, so is the role of nurses

AACN's Vision

Create a healthcare system driven by patient's and family's needs in which critical care nurses make their optimum contributions

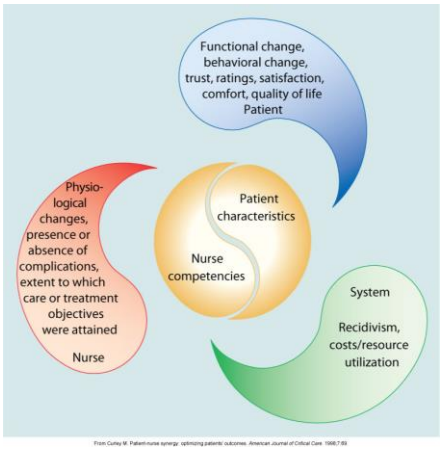
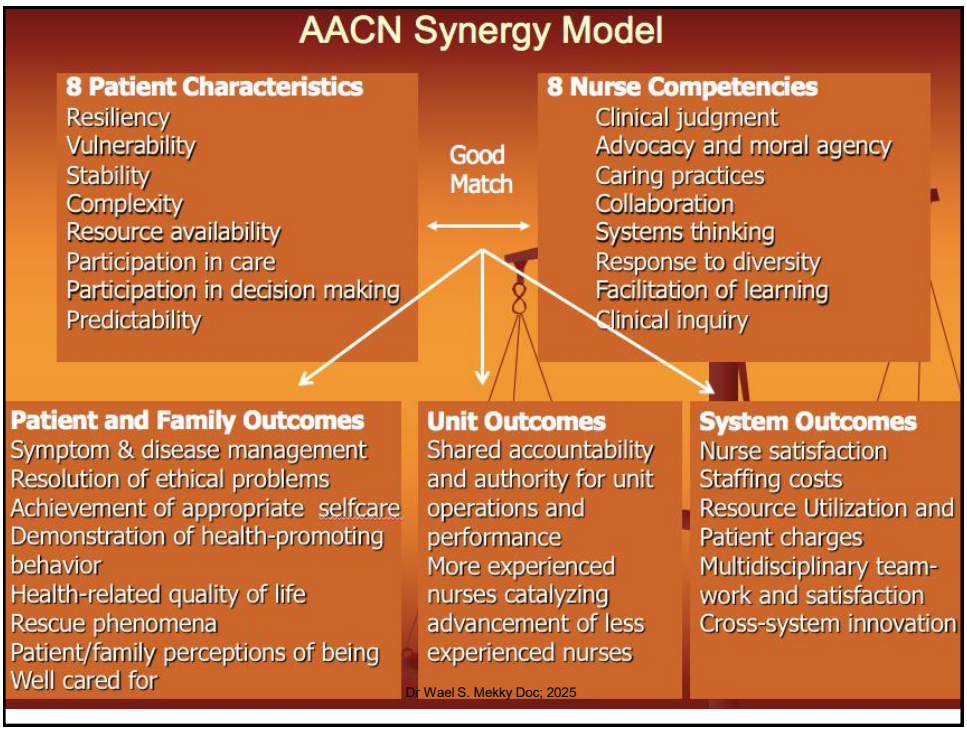


Figure 1-3. The American Association of Critical-Care Nurses Synergy Model for Patient Care. (From Curley M. Patient-nurse synergy: optimizing patients' outcomes. *American Journal of Critical Care*. 1998;7:69.)

Principles of the AACN Synergy Model for Patient Care:

- The AACN Synergy Model for Patient Care "is a conceptual framework describing a patient-nurse relationship that acknowledges the primary importance of nursing care based on the needs of patients and their families.
- The fundamental premise of the Synergy Model is that patients' characteristics drive nurse competencies. When patient characteristics and nurse competencies are in synergy, **optimal patient outcomes are more likely to occur.**



Competencies for Critical Care Nurses

BOX 1-1 Competencies of Nurses Caring for the Critically Ill

- Clinical judgment and clinical reasoning skills
- Advocacy and moral agency in identifying and resolving ethical issues
- Caring practices that are tailored to the uniqueness of the patient and family
- Collaboration with patients, family members, and health care team members
- Systems thinking that promotes holistic nursing care
- Response to diversity
- Facilitator of learning for patients and family members, team members, and the community
- Clinical inquiry and innovation to promote the best patient outcome

Data from American Association of Critical-Care Nurses. The AACN synergy model for patient care. http://www.aacn.org/wd/certifications/content/synmodel.pcms?pid=1&&menu=certification; 2015. Accessed May 10, 2015.

Standards

- Standards serve as guidelines for clinical practice.
- Establish goals for patient care
- Provide mechanisms for nurses to assess the achievement of patient goals.
- AACN Standards for Acute and Critical Care Nursing Practice delineate the **nursing process**:

Standards

- Collect data
- Determine diagnoses
- Identify expected outcomes
- Implement interventions and
- Evaluate progress toward goals.

- The standards of professional performance describe expectations of the acute and critical care nurse.

Standards of Practice (Bell, 2008)

- Evaluate quality of nursing practice
- Evaluate own practice
- Acquire knowledge and maintain competency
- Contribute to professional development of peers
- Act ethically
- Use skilled communication
- Use clinical inquiry and integrate research findings
- Consider factors related to safety, effectiveness, and cost
- Provide leadership in clinical setting

Trends and Issues

- **Critical care patients more complex**
  - Multisystem organ dysfunction
  - Complicated by aging population
- **Increasing costs for care**
  - ❖Challenged to reduce costs and length of stay
  - ❖Transfer of higher acuity patient from the critical care unit to other units or home

Importance of Evidence-Based Practice

- ❑ Implement practice based on evidence
- ❑ Evaluate research
- ❑ Hierarchy of evidence

AACN Hierarchy of Evidence

TABLE 1-1 AMERICAN ASSOCIATION OF CRITICAL-CARE NURSES' LEVELS OF RESEARCH EVIDENCE

LEVEL	DESCRIPTION
A	Meta-analysis of multiple controlled studies or metasynthesis of qualitative studies with results that consistently support a specific action, intervention, or treatment
B	Well-designed controlled studies, both randomized and nonrandomized, with results that consistently support a specific action, intervention, or treatment
C	Qualitative, descriptive, or correlational studies; integrative reviews; systematic reviews; or randomized controlled trials with inconsistent results
D	Peer-reviewed professional organizational standards, with clinical studies to support recommendations
E	Theory-based evidence from expert opinion or multiple case reports
M	Manufacturer's recommendation only

From Armola RR, Bourgault AM, Halm MA, Board RM, Bucher L, Harrington L, Heafey C, Lee RK, Shellner PK, Medina J: Upgrading the American Association of Critical-Care Nurses' evidence-leveling hierarchy. *American Journal of Critical Care*. 2009; 18:405-409.

Thank
you

Critical Care Nursing

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TYPES OF ICUs

An open: In this type, physicians admit, treat and discharge

- **A closed:** in this type, the admission, discharge and referral policies are under the control of intensivists.
- **Hybrid or Transitional:** intensives are locally present shared care between ICU staff and private physician ICU staff is a final common pathway for orders and procedures

ORGANIZATIONAL MODELS FOR ICU

Open model allows many different members of the medical staff to manage patients in the ICU

Closed model is limited to ICU-certified physicians/ intensivists managing the care of all patients

Hybrid model, which combines aspects of open and closed models by staffing the ICU with an attending physician and/or team to work in association with primary physicians

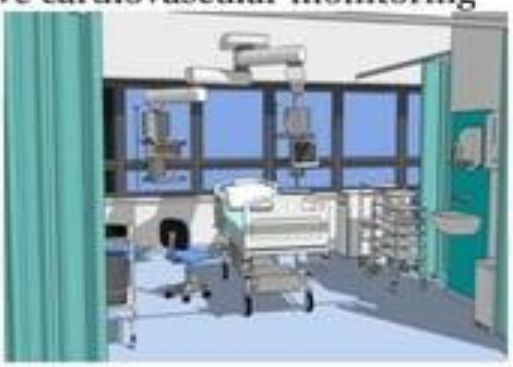
Levels of ICU

Classification Of Critical Care Units

- LEVEL - I : Provides monitoring, observation and short term ventilation. Nurse patient ratio is 1:3 and the medical staff are not present in the unit all the time.

Level I ICU

- Referred to as high dependency
- Resuscitation, short term mechanical ventilation and simple invasive cardiovascular monitoring for <24 hours



Classification Of Critical Care Units

- LEVEL - II : Provides observation, monitoring and long term ventilation with resident doctors. The nurse-patient ratio is 1:2 and junior medical staff is available in the unit all the time and consultant medical staff is available if needed.

Level II ICU

- Located in general hospital, undertake more prolonged ventilation.
- Provides a high standard of general intensive care, including complex multi-system life support

Classification Of Critical Care Units

- LEVEL - III : Provides all aspects of intensive care including **invasive** hemodynamic monitoring and dialysis. Nurse patient ratio is 1:1

- **Invasive:** means a medical procedure that enters the body, usually by cutting or puncturing the skin or by inserting instruments into the body.

Examples: Capnograph, C.V.P line (Central Venous Pressure).

Level III ICU

- Tertiary referral unit for intensive care patients
- Provides comprehensive critical care including complex multi-system life support for an indefinite period
- Demonstrated commitment to academic education and research

Classification Of Critical Care Patients

- **Level O** : normal ward care
- **Level 1**: at risk of deteriorating , support from critical care team
- **Level 2** : more observation or intervention, single failing organ or post operative care
- **Level 3:** advanced respiratory support or basic respiratory support ,multi-organ failure

Staffing

THE INTENSIVE CARE TEAM.

- Doctor
- Nurses
- Therapists
- Nutritionists
- Other support staff

Staffing

- Medical staffing, including an ICU Director/Intensivist, with sufficient experience to provide for patient care, administration, teaching, research, audits etc.
- Trained Nursing staff: 1:1 for ventilated patients and 1:2 for other patients; Nurse in charge with ICU qualification
- Allied health and ancillary staff: Respiratory services, physiotherapist, dietician, biomedical engineer, technicians, computer programmer, social worker, counsellor, housekeeping staff etc.

LABORATORY SERVICES

- A clinical laboratory should be available on a 24-hr basis to provide basic hematologic, chemistry, blood gas, and toxicology analysis.
- Laboratory tests must be obtained in a timely manner, immediately in some instances. "STAT" or "bedside" laboratories adjacent to the ICU or rapid transport systems.

Radiology and imaging services:

- The diagnostic and therapeutic radiologic procedures should be immediately available to ICU patients, 24 hrs per day.
- Portable chest radiographs affect decision making in critically ill patients.

NURSING STAFF

- The major teaching tertiary care ICU will require trained nurses in critical care.
- It may be ideal to have an in house training programme for critical Care nursing.
- The number of nurses ideally required for such units is 1:1 ratio.

NURSING STAFF

- In complex situations they may require two nurses per patient.
- The number of trained nurses should be also worked out by the type of ICU, the workload and work statistics and type of patient load.

Critical Care Nurses Role Responsibilities

- Support and respect for the patients autonomy.
- Helping the patient to obtain the necessary care.
- Continuous monitoring.
- Efficient individualized care.

Critical Care Nurses Role Responsibilities

- Counseling and information to family.
- Application of policy and procedure.
- Record all activities.
- Maintain infection control principles.
- Keep update with advance information.

Roles and Responsibilities of Critical Care Nurse

- Assessing a patient's condition and planning and implementing patient care plans
- Provides direct comprehensive bedside care to patients
- Treating wounds and providing advanced life support
- Assisting physicians in performing procedures
- Able to attach equipments on patients as ordered and interprets the data, graphs on monitors etc.

Roles and Responsibilities of Critical Care Nurse

- Observing and recording patient vital signs
- Ensuring that ventilators, monitors and other types of medical equipment function properly
- Administering intravenous fluids and medications
- Collaborating with fellow members of the critical care team
- Responding to life-saving situations, using nursing standards and protocols for treatment

Roles and Responsibilities of Critical Care Nurse

- Acting as patient advocate
- Documents appropriately
- Ensures patient safety
- Follows the policies and procedures of the unit and the institution
- Is an expert in nursing knowledge and practice
- Promotes quality assurance in nursing
- Providing education and support to patient families

ICU Equipments

DEFINITION OF INTENSIVE CARE UNIT EQUIPMENTS

- **Intensive care unit (ICU) equipment includes:**
  - ❖Patient monitoring
  - ❖Respiratory and cardiac support
  - ❖Pain management
  - ❖Emergency resuscitation devices

DEFINITION OF INTENSIVE CARE UNIT EQUIPMENTS

- Other life support equipment designed to care for patients who are seriously injured, have a critical or life-threatening illness, or have undergone a major surgical procedure, thereby requiring 24-hour care and monitoring.

Patient Monitoring Equipment's

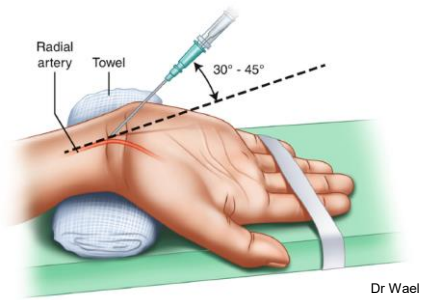
- Acute Care Physiologic Monitoring System
- Pulse Oximeter
- Intracranial Pressure Monitor
- Apnea Monitor

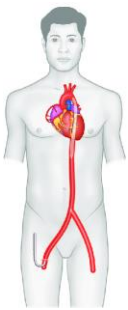
Life Support & Resuscitative Equipment's

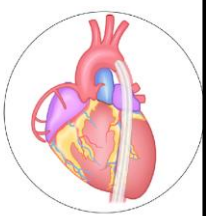
- Mechanical Ventilator
- Infusion Pump
- Syringe Pump

Life Support & Resuscitative Equipment's

- CRASH CART (Code Cart Or "MAX Cart)
- Intra-aortic Balloon Pump
- Pulmonary Artery Catheter
- Arterial line



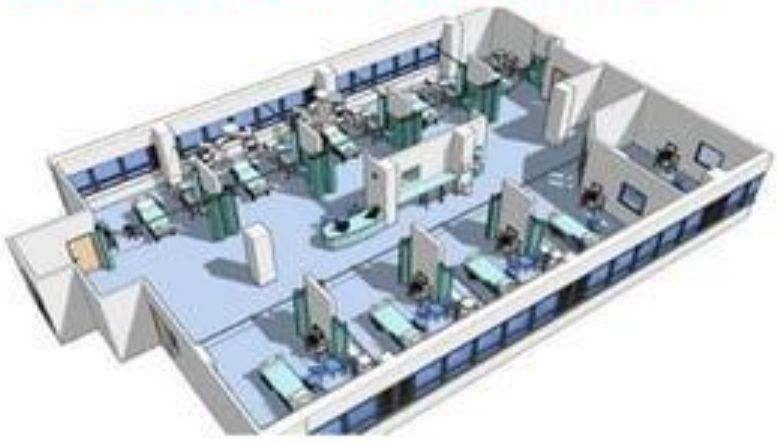




Diagnostic Equipment's

- Mobile X-rays
- Portable Clinical Lab. Devices
- Blood Analyzer

Organization/ Design of ICU



Organization of ICU

- It requires intelligent planning
- One must keep the need of the hospital and its location
- One ICU may not cater to all needs
- An institute may plan beds into multiple units under separate management by single discipline specialist such as Medical ICU, Surgical ICU, CCU, Burns ICU, Trauma ICU etc.
- The number of ICU beds in a hospital ranges from 2 to 20% of the total number of hospital beds. ICUs with fewer than the recommended number of beds are not cost- effective and excess beds are unmanageable. 10% of total ICU beds are allotted towards Isolation Room.

Organization of ICU

- Each intensive care unit should be a geographically distinct area within the hospital with controlled access
- No through traffic to other departments should occur
- Supply and professional traffic should be separated from public/visitor traffic

Organization of ICU

Location should be chosen so that the unit is adjacent to, or within direct elevator travel to and from, the Emergency Department, Operating Room, Intermediate care units, and Radiology Department. Corridors, lifts and ramps should be spacious enough to provide easy movement of bed/ trolley of a critically sick patient

Organization of ICU

Sq. ft = 0.093 Sq. Meter

Bed Space

ft = 30.5 cm

- 150- 200 sq. ft. area per open bed with 8 ft. between beds
- 225- 250 sq. ft. area per bed if in a single room. Beds should be adjustable with side rails and wheels; no head board should be there.

Isolation Room: 250 sq. ft

Sq. ft = 0.093 Sq. Meter

Patient Areas

- 3 oxygen outlets, 3 suction outlets (gastric, tracheal and underwater seal), 2 compressed air outlets and 16 power outlets per bed
- Bedside storage, hand rinse solution, equipment shelf on head end
- Storage must be provided for each patient's personal belongings, patient care supplies, linen and toiletries. Locking drawers and cabinets must be used if syringes and pharmaceuticals are stored at the bedside.
- Hooks and devices to hang infusions/ blood bags extended from the ceiling with a sliding rail to position

Patient Areas

- Multi- channel invasive monitors, ventilators, infusion pumps, portable X- Ray unit, fluid and bed warmers, portable light, defibrillators, anesthesia machines and difficult airway management equipments are necessary
- A cardiac arrest/emergency alarm button must be present at every bedside within the ICU. The alarm should automatically sound in the hospital telecommunications center, central nursing station, ICU conference room, staff lounge, and any on-call rooms.

Therapeutic Elements in ICU Environment

- Window and art that provides natural views; views of nature can reduce stress, hasten recovery, lower blood pressure and lower pain medication needs.
- Family participation, including facilities for overnight stay and comfortable waiting rooms
- Providing a measure of privacy and personal control through adjustable curtains and blinds, accessible bed controls
- Noise reduction through computerized pagers and silent alarms
- Medical team continuity that allows one team to follow the patient through his or her entire stay

Central Nursing Station

Central Nursing Station

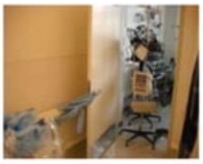
- Patients must be situated so that direct or indirect (e.g. by video monitor) visualization by healthcare providers is possible at all times. This permits the monitoring of patient status under both routine and emergency circumstances. The preferred design is to allow a direct line of vision between the patient and the central nursing station.
- In ICUs with a modular design, patients should be visible from their respective nursing substation

Central Nursing Station

Sliding glass doors and partitions facilitate this arrangement, and increase access to the room in emergency situations

X- Ray Viewing Area

Equipment Storage



Work Areas and Storage

Special Procedures Room

Reception Area

Clean Utility Room

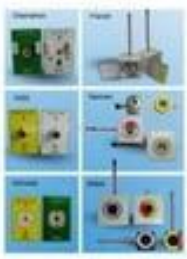
Dirty Utility Room

Pantry

Patient Module

Each Intensive Care Unit must have :-

- Electric Power
- Water Supply
- Oxygen, Compressed Air, Vacuum
- Lighting
- Environmental control systems



Thank
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Critical Care Nursing

The Patient's Experience With Critical Illness

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The Patient's Experience With Critical Illness

- The patient's experience in ICU has lasting meaning for the patient and his or her family members and significant others.
- Admission to a critical care unit (CCU) may signal a threat to the life and well-being of the patient who is admitted.
- Understanding what critical care means to patients may help nurses care for their patients.

The Patient's Experience With Critical Illness

- However, effective communication with critically ill patients is often challenging and frustrating.
- Barriers to communication may relate to the patients' physiological status; the existence of endotracheal tubes, which inhibit verbal communication; medications; or other conditions that alter cognitive function

Family Need of the Critical Care Patient

Families of critically ill patients have other needs that should be addressed frequently, including the following:

- to feel that there is hope;
- to feel that hospital personnel care about the patient;
- to know the prognosis;
- to receive information about the patient at least once a day;
- to see the patient frequently.

Patients' Experiences

- A number of authors have sought to study and describe patients' experiences related to their ICU stay.
- In a review of 26 studies, Stein-Parbury and McKinley noted that between 30% to 100% of patients studied could recall all or part of their stay in the ICU.

Patients' Experiences

- Although many of the patients recalled feelings that were negative.
- Negative experiences were related to fear, anxiety, sleep disturbance, cognitive impairment, and pain or discomfort.
- Positive experiences were related to feelings of being safe and secure.

Excessive stimulation in ICU

- Monitoring & Alarming Systems
- Cleaning
- Prolonged Lightening
- Suctioning
- Dressing changes
- Mobilization
- Physical therapy

Anxiety

- Anxiety is an emotional state of apprehension (fear & worry) in response to a real or perceived threat. Typically, the threat is associated with motor tension, increased sympathetic activity, and hypervigilance(اليقظة المفرطة). (lead to insomnia)
- Any stress that threatens one's sense of wholeness, containment, security, and control causes anxiety. Illness is one such stress.

Anxiety

- **A common cause of anxiety is a sense of isolation.**
- **Anxiety occurs when people experience the following:**
  - Threat of helplessness
  - Loss of control
  - Sense of loss of function and self-esteem
  - Failure of former defenses
  - Sense of isolation
  - Fear of dying

Anxiety, sleep deprivation

71% of patients in a medical surgical ICU get agitated at least once (46% severe agitation)

Pharmacotherapy 2000; 20: 75-82

Anxiety & Agitation

- Patients and families who face life-threatening illnesses commonly experience anxiety.
- Anxiety can be related to any number of physical, emotional, psychological, social, practical, and spiritual issues.
- Assessment of anxiety can be complex, and an interdisciplinary approach, including nursing, social services, psychology, and religious people may be needed to evaluate the patient accurately and treat the anxiety properly.

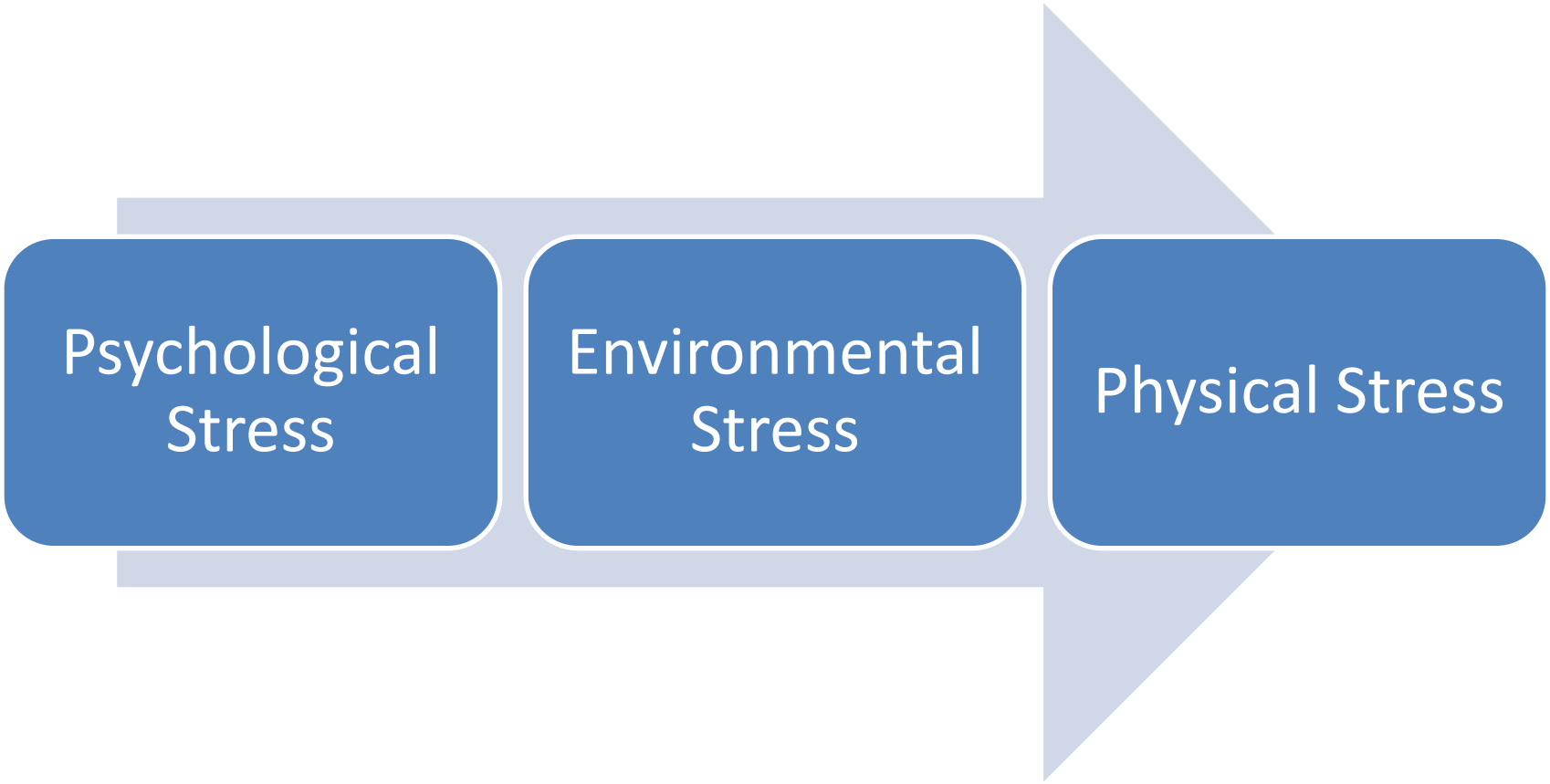
Anxiety & Agitation

- Nonpharmacological interventions may include counseling, taking care of practical matters and arranging for spiritual concerns to be addressed (eg, arranging for a visit from a religion member).
- If medication is needed, short- or long-acting benzodiazepines (Xanax & Valium) and atypical antidepressants (Fluoxetine or Prozac) may be helpful.

Stress

- Stress exists when an organism is faced with any stimulus that causes disequilibrium between psychological and physiologic functioning.
- All hormone levels can be altered by stress. Extreme levels of stress damage human tissue and may interfere with adaptive responses.
- Patients admitted to the ICU are subject to multiple **physical**, **psychological**, and **environmental** stressors.

STRESS IN ICU?



(1) Psychological Stress in ICU

- Loss of control
- Fear of death or serious illness
- Fear of pain
- Overwhelming isolation
- Feelings of helplessness
- Loss of normal circadian rhythms (الساعة البيولوجية)
- The disruption of normal sleep patterns
- Sleep deprivation
- Disorientation and panic (Phobia)

(2) Environmental Stress in ICU

- Foreign environments
- **Room temperature**
- **Continuous ambient lighting**
- Family not continuously available for comfort
- Significant noise from personnel and medical equipment

(3) Physical Stress in ICU

- Attached to equipment's with tubes or wires
- Intubated and ventilated
- Treatment or diagnostic procedures
- Confined (restricted) to bed
- Uncomfortable bed and pillow
- Unable to control stool habit
- Inability to communicate

Acute Stress Response

- Increased heart rate and contractility, vasoconstriction, and increase in blood pressure.
- Blood flow is redirected to vital organs.
- Pain sensations are temporarily attenuated.
- Body temperature and nutrient consumption fall.

Acute Stress Response

- A sensation of thirst may be prominent.
- Other physiologic effects include increase in ventilation and respiratory rate, hyperglycemia, insulin resistance, and coagulopathies.
- If this phase is prolonged, it can result in impaired oxygen and nutrients delivery to tissues secondary to alterations in microcirculatory blood flow.

Delirium

- Delirium may be an unintended consequence of a patient's ICU experience that can lead to a prolonged stay and higher mortality.
- Often undetected, delirium is increasingly recognized as a factor in morbidity and mortality that can affect up to 80% of patients.

Delirium

- Delirium is an acute change in awareness or cognitive status that may manifest as agitation, withdrawal, or confusion. “Confusion” is an all-inclusive term that refers to inappropriate behavior, disorientation, or hallucinations.
- Terminal delirium is common in patients near death and may manifest as day–night reversal.

Delirium

- Management of delirium is focused more on symptom control and relief of the distress of the patient and family.
- Benzodiazepines or neuroleptics (eg, haloperidol) are helpful in managing this symptom.

Delirium in 87%

with fluctuating mental status, inattention, disorganized thinking with or without agitation

JAMA 2001; 286: 2703-2710

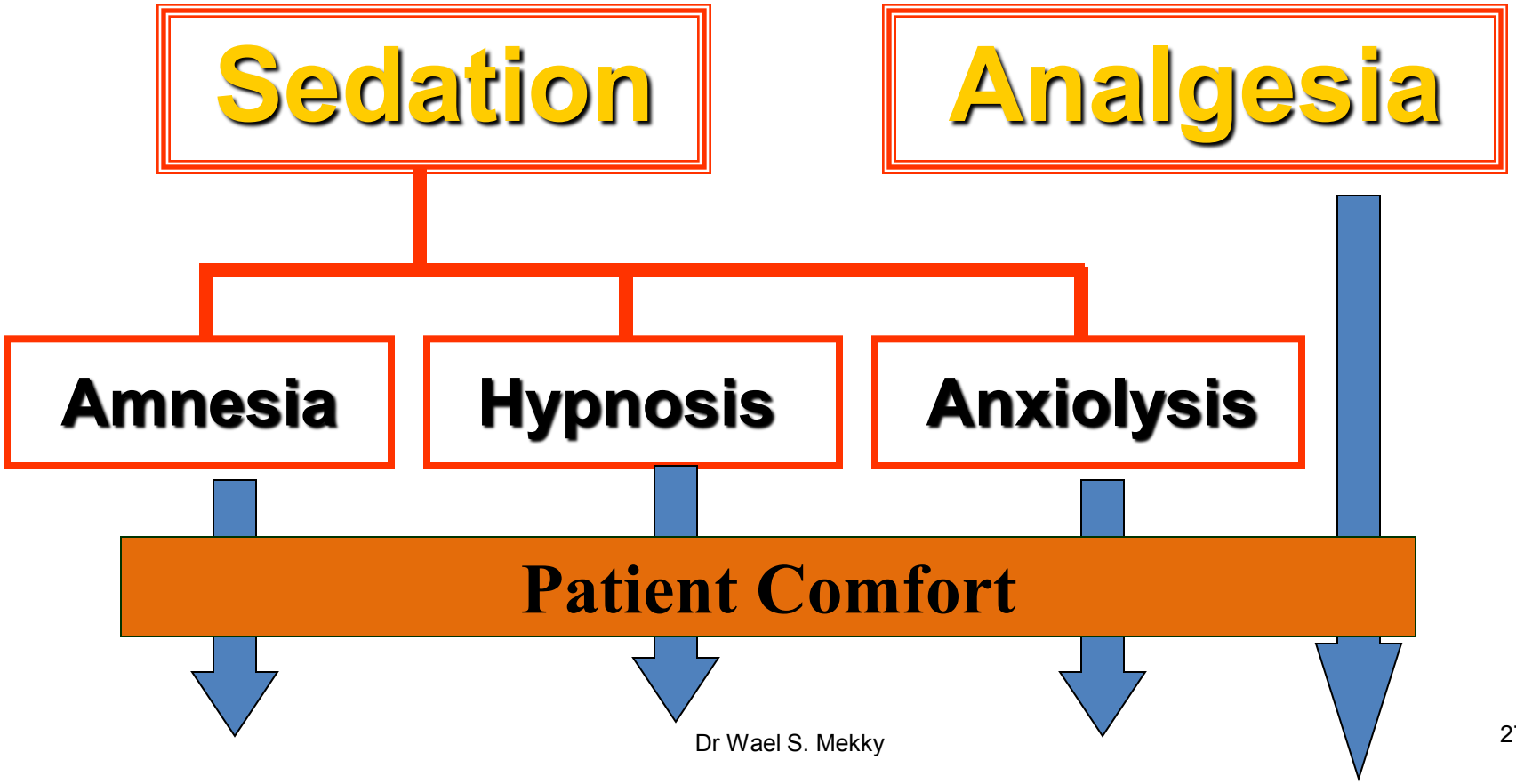
Correctable Causes of Agitation

Full bladder	Uncomfortable bed position	Inadequate ventilator flow rates	Mental illness
Uremia	Drug side effects	Disorientation	Sleep deprivation
Noise	Inability to communicate	Cold room	Uncomfortable mattress or pillow
	Traction on endotracheal tube		

Sedation of agitated patients should start only after providing adequate analgesia and treating reversible physiological causes

Pain, hypoxemia, hypoglycemia, hypotension, withdrawal from alcohol and other drugs

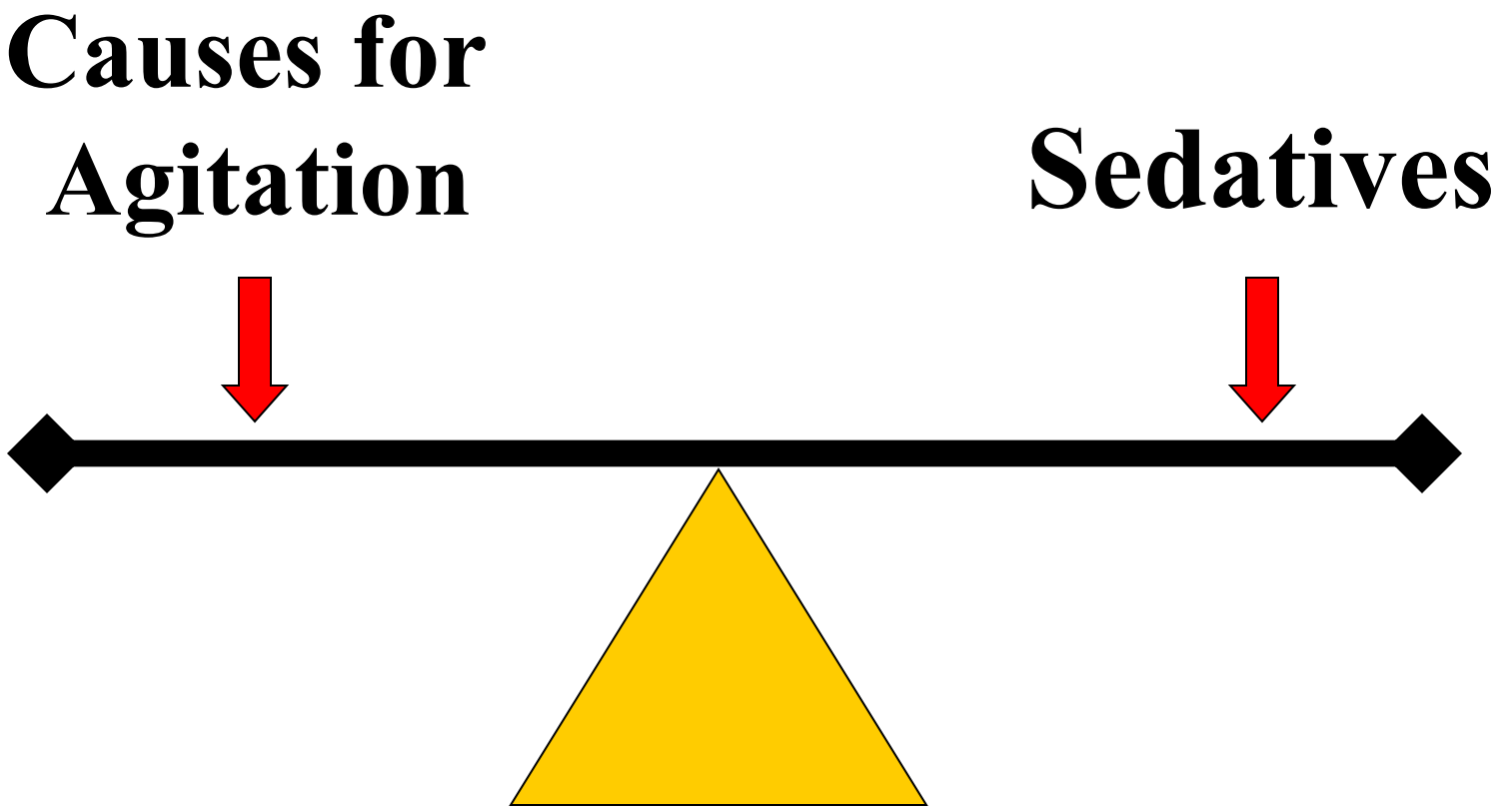
"ICU Sedation"



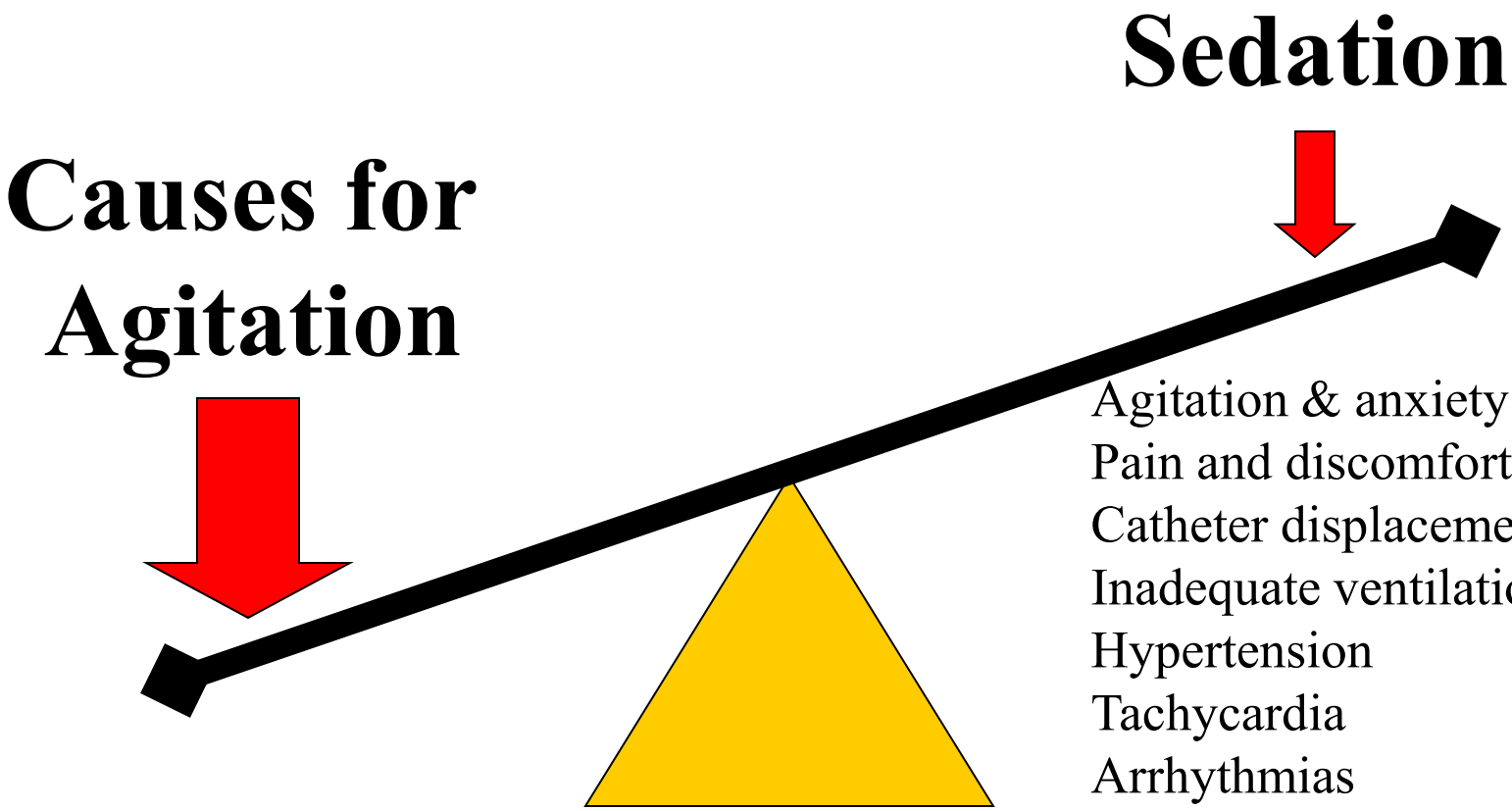
ICU Sedation Goal

- **Stabilize hemodynamics & modulate stress response**
- **Reduce motor activity – tolerance of procedures, facilitate nursing management**
- **Facilitate mechanical ventilation**
- **Facilitate sleep patterns**

Sedation



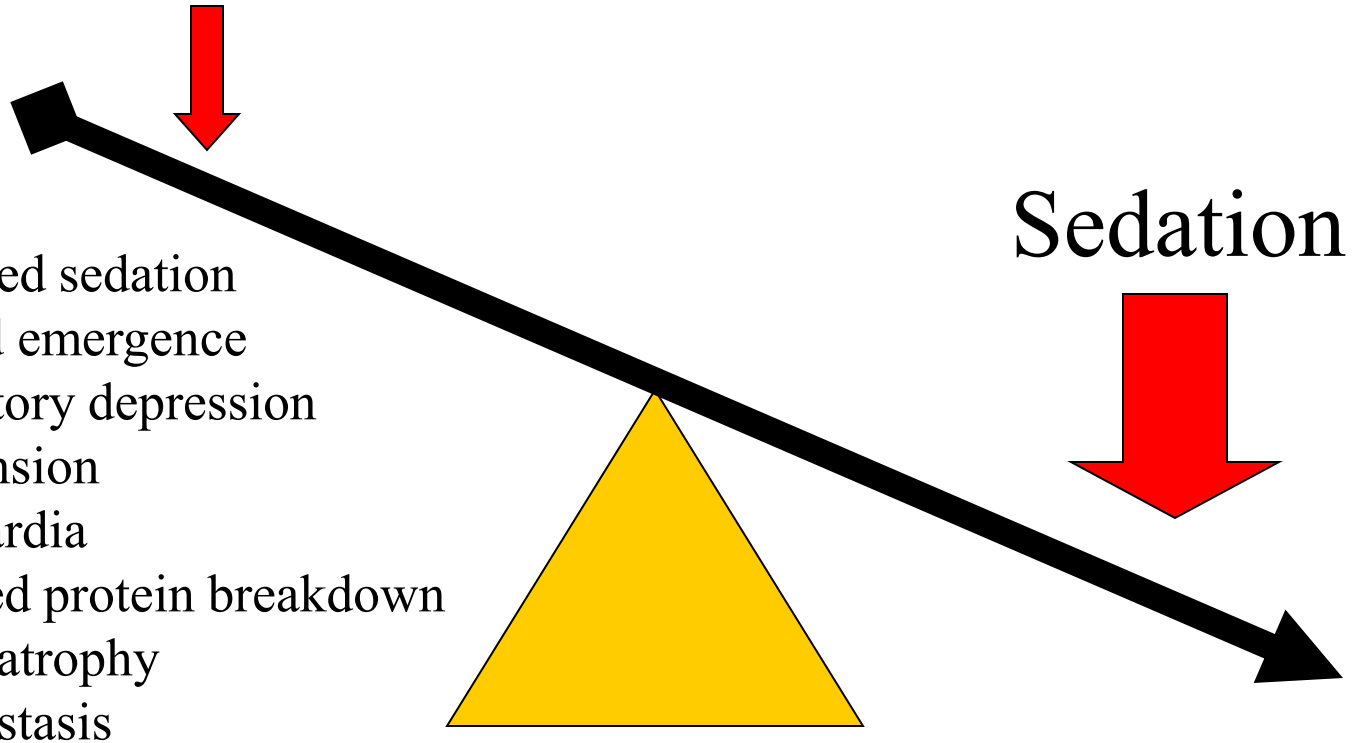
Undersedation



Agitation & anxiety
Pain and discomfort
Catheter displacement
Inadequate ventilation
Hypertension
Tachycardia
Arrhythmias
Myocardial ischemia
Wound disruption
Patient injury

Over-sedation

Causes for Agitation



Prolonged sedation
Delayed emergence
Respiratory depression
Hypotension
Bradycardia
Increased protein breakdown
Muscle atrophy
Venous stasis
Pressure injury
Loss of patient-staff interaction
Increased cost

Undersedation

Underdosing
Tolerance
Withdrawal

Oversedation

Overdosing
Drug accumulation
Impaired elimination

Drug interactions
Adverse side effects

Assessment of the Critically Ill Patient

Assessment of Critically Ill Patient

- Assessment is the first step in caring for a patient, and assessing the critically ill patient is an essential part of their care.
- Careful assessment is fundamental in order to recognize when a patient is becoming compromised.
- The nurse acts as patient advocate, monitoring, anticipating potential problems, planning, implementing and constantly evaluating care, and communicating with other medical staff involved in the patient care.

Reasons for doing Assessment

- To establish a starting line information on the client
- To determine the client's normal function
- To determine the client's risk for dysfunction
- To determine the client's strengths

Types of Assessment

1- Initial Assessment:

Aim: Initial identification of normal function, functional status, and collection of data concerning actual or potential dysfunction. Baseline for reference and future comparison.

Time Frame: Within the specified time frame after admission to a hospital.

Types of Assessment

2- Focus Assessment:

Aim: Status determination of a specific problem identified during previous assessment.

Time Frame: Ongoing process, integrated with nursing care, a few minutes to a few hours between assessments.

Types of Assessment

3- Time - Lapsed Reassessment: إعادة التقييم بعد انقضاء الوقت

Aim: Comparison of client's current status to baseline obtained previously, detection of changes in all functional health patterns after an extended period of time has passed.

Time Frame: Several months (3,6,9 months or more) between assessment.

Types of Assessment

4- Emergency Assessment:

Aim: Identification of life – threatening situation

Time Frame: At anytime

FANCAP Assessment

- **F: Fluid:** movement of fluid and electrolytes among body tracts “Electrolyte imbalance, Dehydration, or electrolytes stability”
- **A: Aeration (تهوية):** assess the patient clinically and laboratory “Type of respiration, normal and extra breathing sounds, color of skin & mucous membrane, rate of respiration, **ABGs** determination is the best indicator for the patient condition.

FANCAP Assessment

- **N: Nutrition:** assess the patient through physical examination and laboratory investigation and ask the following question: “Can use the normal GI tract, what about the GI system condition, does the patient need feeding, what about vitamins and minerals alternatives”.
- **C: Communication:** Assess neurological function, look at the patient eyes, (is he cooperative, is he oriented).

FANCAP Assessment

- **A: Activity:** bed rest with mild activity, complete bed rest, up with help, up in chair or cannot walk.
- **P: Pain:** from the disease process, psychological pains from loneliness, hopelessness, fear.

Thank you

Critical Care Nursing

What is Pain?

Dr. Wael Subhi Mekky

WHAT IS PAIN ?

Pain

- Pain is a common and distressing symptom in (ICU) patients and represents a major clinical, social, and economic problem.
- It has been reported that almost 80% of patients experience different intensities of pain during their ICU stay and identify it as one of the greatest sources of stress.

Consequences of Untreated Pain

- **What happens if pain isn't properly treated?**

- Poor appetite and weight loss
- Disturbed sleep
- Withdrawal from talking or social activities
- Sadness, anxiety, or depression
- Physical and verbal aggression, resists care
- Difficulty walking or transferring; may become bed bound

Consequences of Untreated Pain, cont.

- Skin ulcers
- Incontinence
- Increased risk for use of chemical and physical restraints
- Decreased ability to perform ADL's
- Impaired immune function

Effects of pain

- ✓Increased heart rate
- ✓ Diaphoresis
- ✓increased blood glucose levels
- ✓dilatation of pupils
- ✓decreased GI motility
- ✓increased musle tension
- ✓ increased respiratory rate

Descriptions of Pain

Categories of Pain by Duration

Acute Pain

Brief duration, goes away with healing, usually 6 months or less.

- Not necessarily more severe than chronic
- May be sudden onset or slow in onset
- Acute pain serves as a warning of disease or a threat to the body.
- Examples are Trauma, Headache, and pain after surgery or injury.

Descriptions of Pain

Categories of Pain by Duration

Chronic Cancer Pain

Pain is expected to have an end, with cure or with death.

- Aggressive treatment
- Addiction not a concern

Descriptions of Pain

Categories of Pain by Duration

Chronic Non-Malignant Pain

Pain has no predictable ending

- Difficult to find specific cause
- Often can't be cured
- Frequently undertreated

Categories of Pain by Type

Somatic

Source: Skin, muscle, and connective tissue

Examples: Sprains, headaches, arthritis

Description: Localized, sharp/dull, worse with movement or touch.

Categories of Pain by Type

Visceral

Source:	Internal organs
Examples:	Tumor growth, gastritis, chest pain
Description:	Not localized, refers, constant and dull, less affected with movement

Categories of Pain by Type

Bone Pain

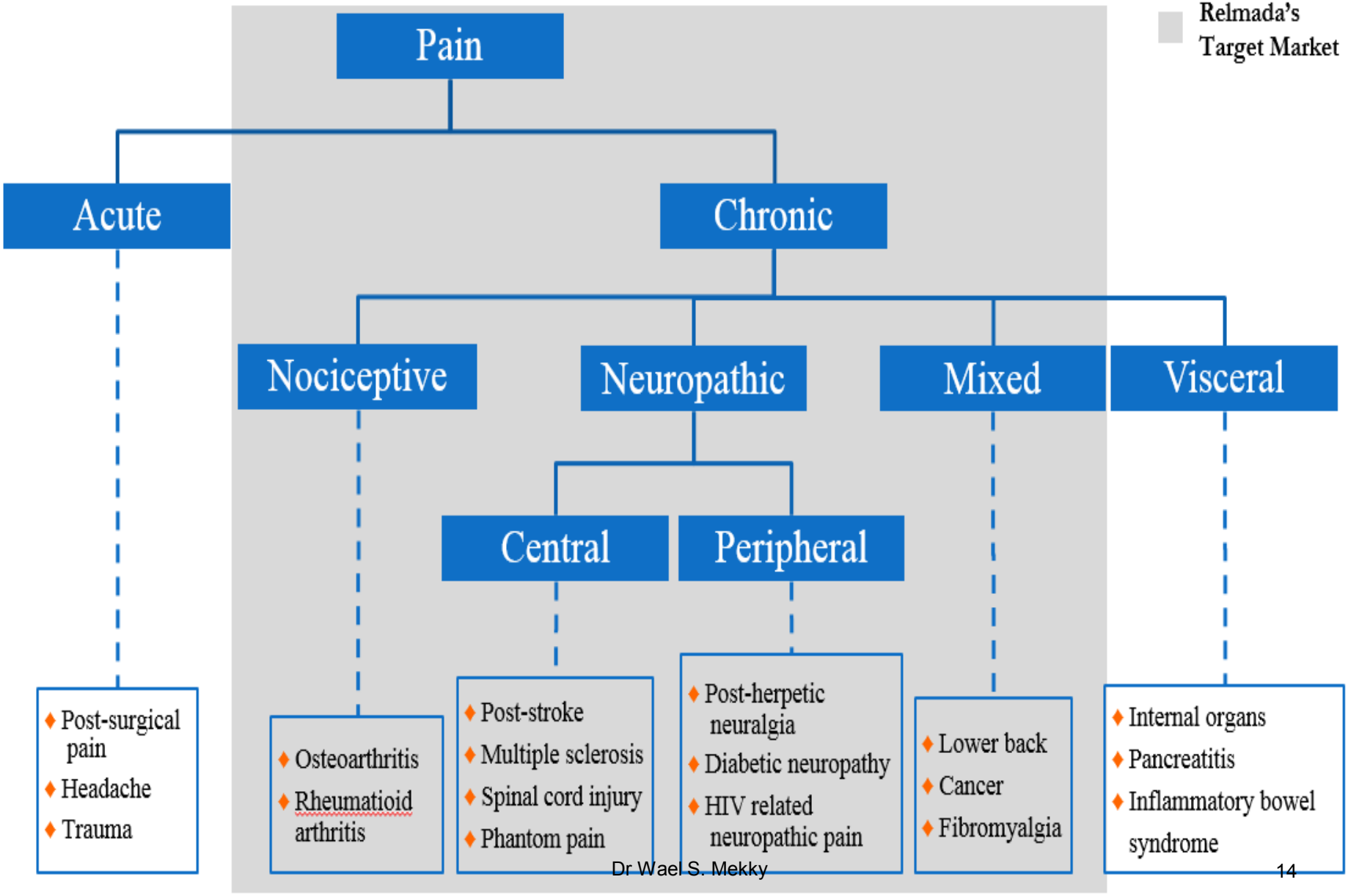
Source:	Sensitive nerve fibers on the outer surface of bone
Examples:	Cancer spread to bone, fx, and severe osteoporosis
Description:	Tends to be constant, worse with movement

Categories of Pain by Type

Neuropathic

Source:	Nerves
Examples:	Diabetic neuropathy, phantom limb pain, cancer spread to nerve plexis
Description:	Burning, stabbing, pins and needles, shock-like, shooting

Types of Pain



Acute Pain Assessment

- The assessment of acute pain in the ICU can be challenging. Unfortunately, many ICU patients cannot provide full or even partial information regarding their pain. However, the inability of ventilated, sedated ICU patients to report pain should not prevent its assessment and management.

Acute Pain Assessment

- In heavily sedated or paralyzed patients, nurse's must use signs of heightened sympathetic activity like hypertension, tachycardia, lacrimation, diaphoresis, and restlessness as alternative indicators for the presence of pain.

Patients who cannot communicate should be assessed through subjective observation of pain-related behaviors (movement, facial expression, and posturing) and physiological indicators (HR, BP, RR) and the change in these parameters following analgesic therapy

- Hypertension
- Tachycardia
- Lacrimation
- Sweating
- Pupillary dilation

Pain Assessment Criteria

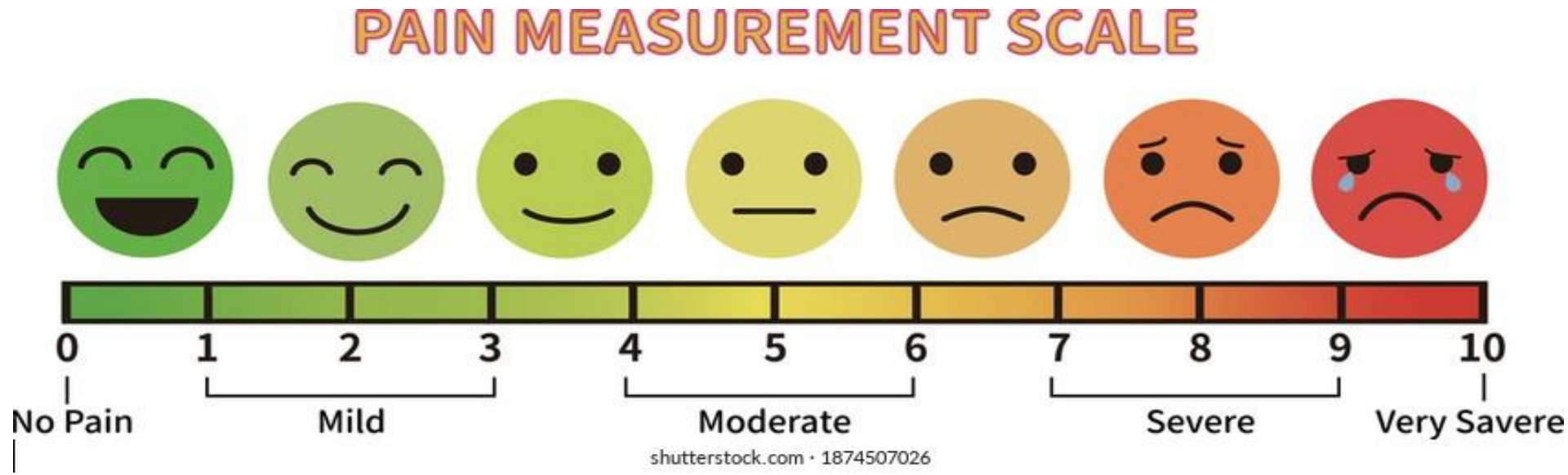
- P.Q.R.S.T. assessment of the pain done according to the following criteria:

- **P:** (Precipitating as causes & palliating factors that makes it better).

- **Q:** (Quality of pain) how would describe the pain "burning, stabbing, dull, crushing or squeezing" description may indicate the cause.

Pain Assessment Criteria

- **R:** (Region & radiation) where is the pain, point the area of pain, does it travel anywhere.
- **S:** (Severity) does it make stop what patient doing and can be measured by scale of 1 – 10 (see below figure).
- **T:** (Time factor) how often does the pain occur, how long does it last, and when did you first experience the pain.


PAIN MEASUREMENT SCALE
0
1
2
3
4
5
6
7
8
9
10
No Pain
Mild
Moderate
Severe
Very Savere
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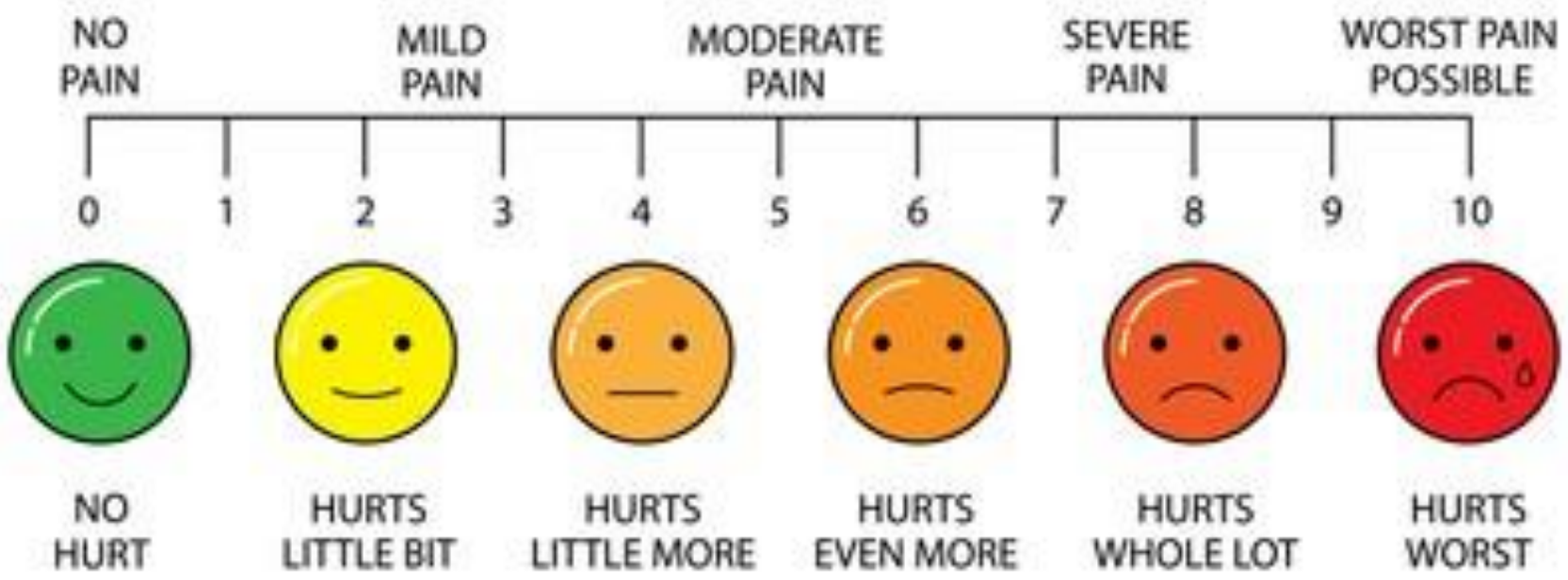
PAIN MEASUREMENT SCALE



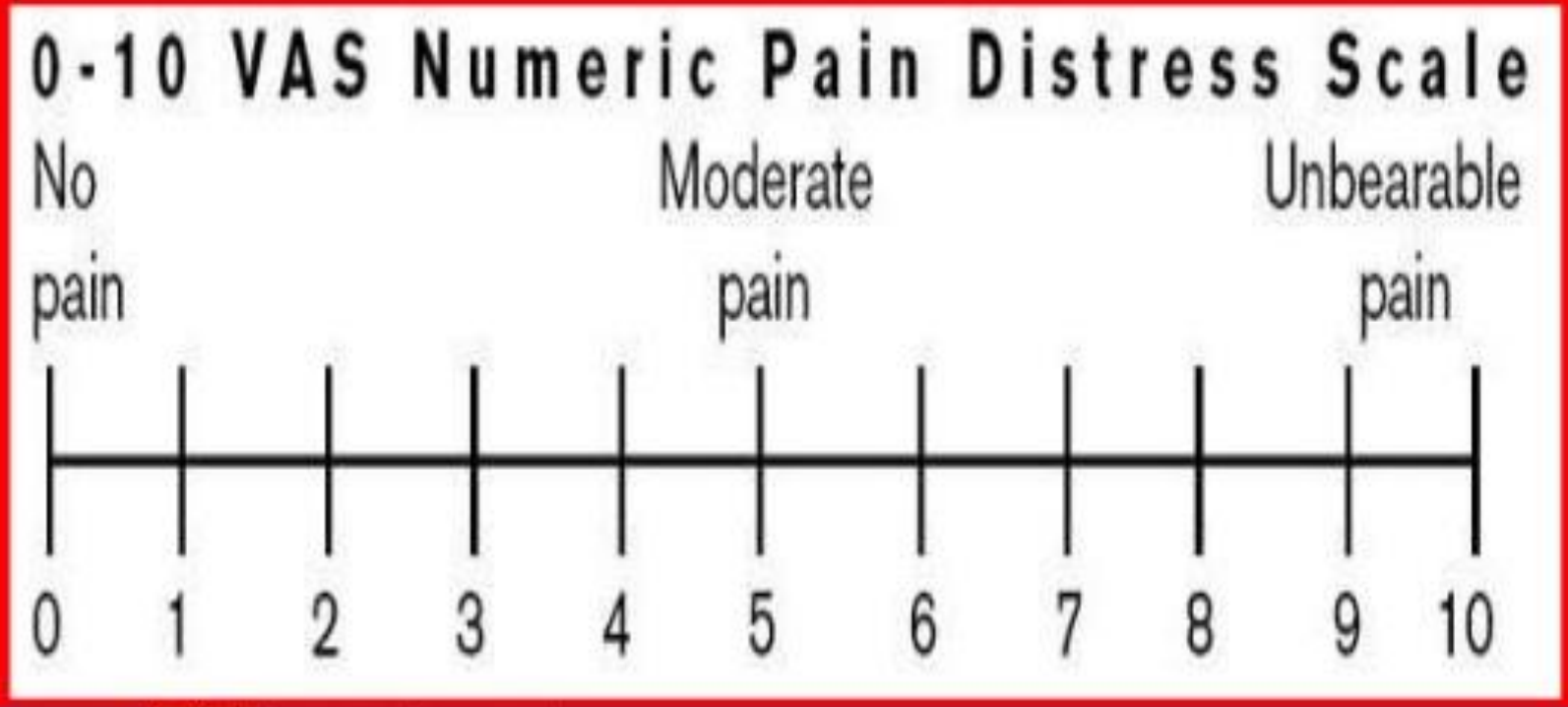
Acute Pain Assessment

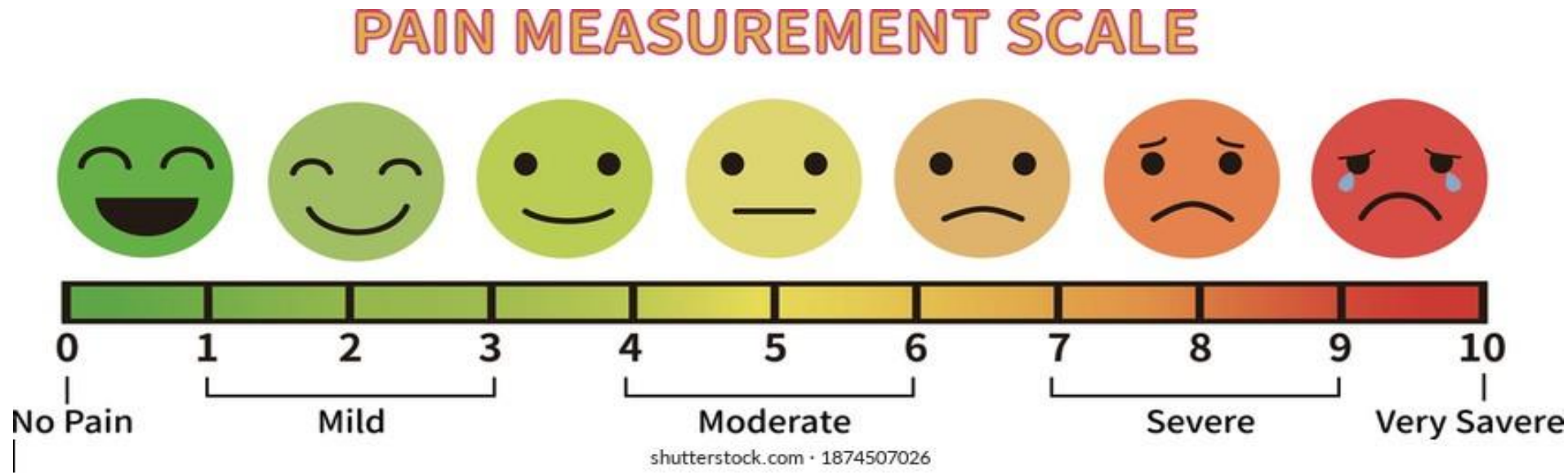
- A number of scales and assessment tools for the evaluation of pain in ICU patients have been developed, such as **the visual analogue scale**, **the numeric rating scale**, **behavioral pain scale**, and **critical care pain observation scale .**

1- The Visual Analogue Scale

INSTRUMENTS FOR ASSESSING PAIN PERCEPTION

- **VISUAL ANALOGUE SCALES(VAS)**




PAIN MEASUREMENT SCALE
0
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No Pain
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Severe
Very Savere
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2- The Numeric Rating Scale

2-The Numeric Rating Scale

- Numerical rating scale involves asking the patient to rate his or her pain from 0 to 10 (11 point scale) or from 0 to 100 (101 point scale) with the understanding that 0 is equal to no pain and 10 or 100 is equal to worst possible pain. This does not require the patient to write or use a ruler and he or she provides a verbal response which the healthcare provider can then document.

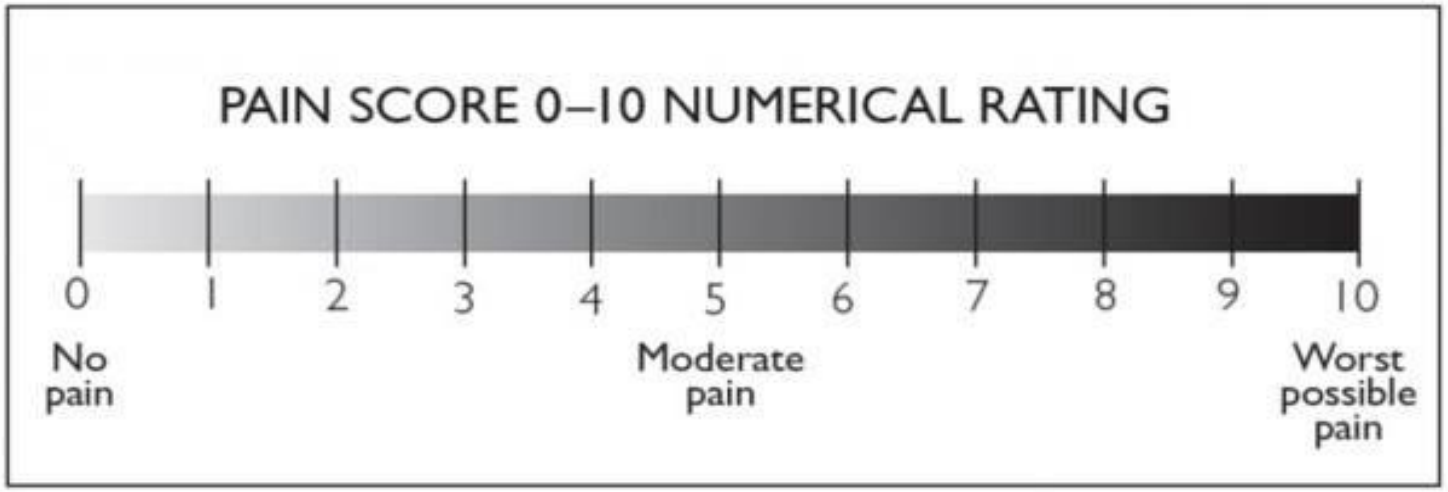
2-The Numeric Rating Scale

- Numerical rating scale are valid and demonstrate positive and significant correlations with other measures of pain intensity.

  It is extremely easy to administer and score and therefore can be used with a greater variety of patients (e.g. older adults and patients with motor problems) than with a Visual Analogue Scale. It is also useful for telephone assessments.

2-The Numeric Rating Scale

- The simplicity of the measure means that individuals comply better than with other tools.



3- The Behavioral Pain Scale

3-The Behavioral Pain Scale

- The Behavioral Pain Scale (BPS) was developed by ***Payen et al*** to assess pain among unconscious, mechanically ventilated patients. The BPS is based on the total score of three behavioral expressions: 1) facial expression, 2) upper limb movements, and 3) compliance with mechanical ventilation. The BPS allows the assessor to derive a score between 3 (no pain) and 12 (highest pain score)

3-The Behavioral Pain Scale

Behavioral Pain Scale (BPS) 3-12

Item	Description	Score
Facial expression	Relaxed	1
	Partially tightened (eg, brow lowering)	2
	Fully tightened (eg, eyelid closing)	3
	Grimacing	4
Upper limbs	No movement	1
	Partially bent	2
	Fully bent with finger flexion	3
	Permanently retracted	4
Compliance with ventilation	Tolerating movement	1
	Coughing but tolerating ventilation for most of the time	2
	Fighting ventilator	3
	Unable to control ventilation	4

Payen JF, et al. *Crit Care Med.* 2001;29(12):2258-2263.

4- Critical Care Pain Observational Scale

4-Critical Care Pain Observational Tool

- The Critical Care Pain Observational Tool (CPOT) are the most valid and reliable behavioral pain scales for assessing pain in adult, ICU patients unable to communicate pain.
- The CPOT includes evaluation of four different behaviors (facial expressions, body movements, muscle tension, and compliance with the ventilator for mechanically ventilated patients) rated on a scale of zero to two with a total score ranging from 0 to 8. The CPOT is feasible, easy to complete, and simple to understand.

4-Critical Care Pain Observational Tool

Medscape® www.medscape.com

Indicator	Description	Score	
Facial expression	No muscular tension observed	Relaxed, neutral	0
	Presence of frowning, brow lowering, orbit tightening, and levator contraction	Tense	1
	All of the above facial movements plus eyelid tightly closed	Grimacing	2
Body movements	Does not move at all (does not necessarily mean absence of pain)	Absence of movements	0
	Slow, cautious movements, touching or rubbing the pain site, seeking attention through movements	Protection	1
	Pulling tube, attempting to sit up, moving limbs/ thrashing, not following commands, striking at staff, trying to climb out of bed	Restlessness	2
Muscle tension Evaluation by passive flexion and extension of upper extremities	No resistance to passive movements	Relaxed	0
	Resistance to passive movements	Tense, rigid	1
	Strong resistance to passive movements, inability to complete them	Very tense or rigid	2
Compliance with the ventilator (intubated patients)	Alarms not activated, easy ventilation	Tolerating ventilator or movement	0
	Alarms stop spontaneously	Coughing but tolerating	1
	Asynchrony: blocking ventilation, alarms frequently activated	Fighting ventilator	2
OR			
Vocalization (extubated patients)	Talking in normal tone or no sound	Talking in normal tone or no sound	0
	Sighing, moaning	Sighing, moaning	1
	Crying out, sobbing	Crying out, sobbing	2
Total, range			0-8

Source: Am J Crit Care © 2006 American Association of Critical-Care Nurses

The Nurse's Role In Pain Management

1- **Identifying goals for pain management**

2- **Establishing the nurse-patient relationship and teaching**

3- **Providing physical care**

4- **Managing anxiety related to pain.**

Pain Management Approach

- Should be interdisciplinary and multimodal.
- Care is individualized and may depend on:
  - Pain source and intensity
  - Patient's age
  - Developmental, physical, emotional and cognitive status
  - Cultural beliefs
  - Treatment preferences
  - Concurrent medical conditions

Multimodal Analgesia

- This term describes the use of multiple modalities that are used to provide pain relief with various parts of the pain pathway targeted.
  - Decreased dependence on single modality agents decreases the risk of side effects.
  - **May include:**
    - **Pharmacological** (Opioids, NSAIDS, Gabapentanoids)
    - **Relaxation techniques** (Coughing & deep breathing)
    - **Regional analgesia** (Nerve blocks, Epidural catheters)

Treatments May Include

Non-pharmacologic Methods

- Heat/cold
- Massage & Relaxation
- Weight reduction,
- Exercises,
- Stop smoking
- Acupressure/acupuncture
- Repositioning

Pharmacologic Methods

- NSAIDS
- Anti-seizure Drugs
- Anti-depressants
- Opioid analgesics
- Local anesthetics
- Neurolytics

Acute vs. Chronic Pain Management

Acute Pain

- Most often treated with:
  - NSAIDS
  - Opioids
  - Local anesthetics
  - Splinting
  - Positioning changes
  - Ice

Chronic Pain

- Most often treated with:
  - Anti-seizure medications
  - Anti-depressant Drugs
  - NSAIDS
  - Implantable devices
  - Psychological therapy
  - Acupuncture
- When all else fails and benefits outweigh risks
  - Opioids

PHARMACOLOGIC
MANAGEMENT

Pharmacologic Pain Management

1- **Nonsteroidal Anti-inflammatory Drugs:**

Drugs in this class inhibit cyclooxygenase (COX) enzymes, which are involved in synthesis of prostaglandins and related inflammatory mediators in response to injury.

- **Adverse effects:** 1-gastrointestinal (GI) ulcer 2- GI bleeding 3 -inhibition of platelet function, 4- renal injury 5-bronchospasm in aspirin-sensitive patients (triad of asthma, nasal polyposis, and aspirin allergy).
- **Example:** Ibuprofen, ketodolac (Toradol), Diclofenac, Aspirin.

Pharmacologic Pain Management

Opioid Analgesics:

Morphine, and fentanyl are most commonly used .

- **Fentanyl** is a synthetic Opioid pain reliever.
- **Morphine:** used to treat moderate to severe pain

Side effect:

- ✓ Abdominal or stomach pain
- ✓ Blurred vision
- ✓ Decreased urination
- ✓ Change in the ability to see colors, especially blue or yellow

Pharmacologic Pain Management

- **Ketamine**:

  General anesthetic and analgesic that has some hallucinogenic effects.

*** Methadone:**

Reduces withdrawal symptoms in people addicted to heroin or other narcotic drugs without causing the "high" associated with the drug addiction

Tramadol:

A narcotic-like pain reliever

Used to treat moderate to severe pain

Pharmacologic Pain Management

- **Symptoms of overdose:**

- Change in consciousness
- Decreased awareness or responsiveness
- Difficulty with breathing
- Lack of muscle tone
- Lightheadedness دوار
- Loss of consciousness
- Pinpointed pupils of the eyes
- Severe sleepiness
- Slow or irregular heartbeat
- Unusual tiredness

Pharmacologic Pain Management

- **Gabapentin** :

  Gabapentin is an anti-epileptic drug, also called an anticonvulsant. It affects chemicals and nerves in the body that are involved in the cause of seizures and some types of pain.

- Gabapentin is used in adults to treat neuropathic pain (nerve pain) caused by herpes virus or shingles (herpes zoster).

- The **Neurontin brand** of **Gabapentin** is also used to treat seizures in adults and children who are at least 3 years old.

Pharmacologic Pain Management

- **Alpha-2 Adrenergic Agonists** :

  α2) adrenergic activation represents another inherent pain-control network in the CNS

  class of drugs can inhibit somatic pain.

- **Dexmedetomidine:** is a useful sedative agent with analgesic properties, and ability to recover respiratory function in mechanically ventilated patients facilitating early weaning.

- **Clonidine**

Motor Activity Assessment Scale (MAAS)

Seven categories to describe the patient's reaction to stimulation

***Devlin et al. Crit Care Med 1999;27:1271-1275**

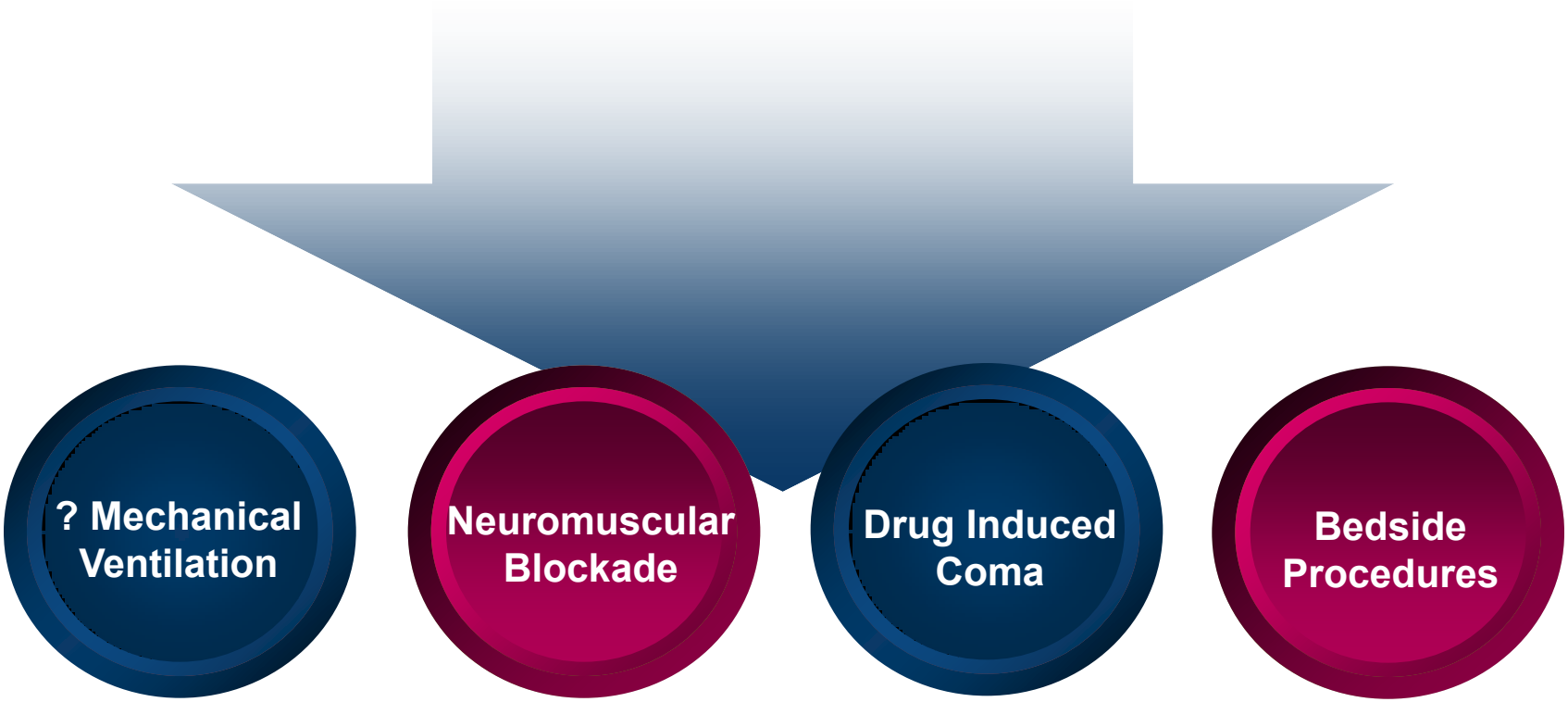
Score	Description	Definition
0	Unresponsive	Does not move with noxious stimulus*
1	Responsive only to noxious stimuli	Open eyes OR raises eyebrows OR turns head toward stimulus OR moves limbs with noxious stimuli
2	Response to touch or name	Opens eyes OR raises eyebrows OR turns head towards stimulus OR moves limbs when touched or name is loudly spoken
3	Calm and cooperative	No external stimulus is required to elicit movement AND patient is adjusting sheets or clothes purposefully and follows commands

*Noxious stimuli = Suctioning OR 5 sec of vigorous orbital, sternal, or nail bed pressure

Score	Description	Definition
4	Restless and cooperative	No external stimulus is required to elicit movement AND patient is picking at sheets or tubes or uncovering self and follows commands
5	Agitated	No external stimulus is required to elicit movement AND attempting to sit up OR moves limbs out of bed AND does not consistently follow commands (e.g. will lie down when asked but soon reverts back to attempts to sit up or move limbs out of bed
6	Dangerously agitated Uncooperative	No external stimulus is required to elicit movement AND patient is pulling at tubes or catheters OR thrashing side to side or striking at staff OR trying to climb out of bed AND does not calm down when asked

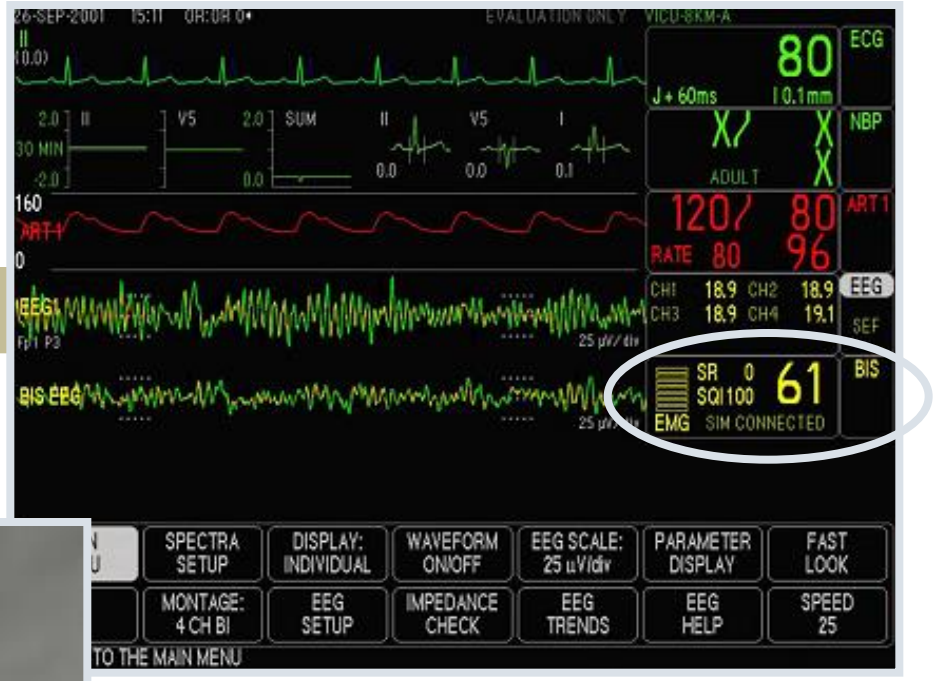
BIS in the ICU: Key Applications

Objective assessment of sedation during:



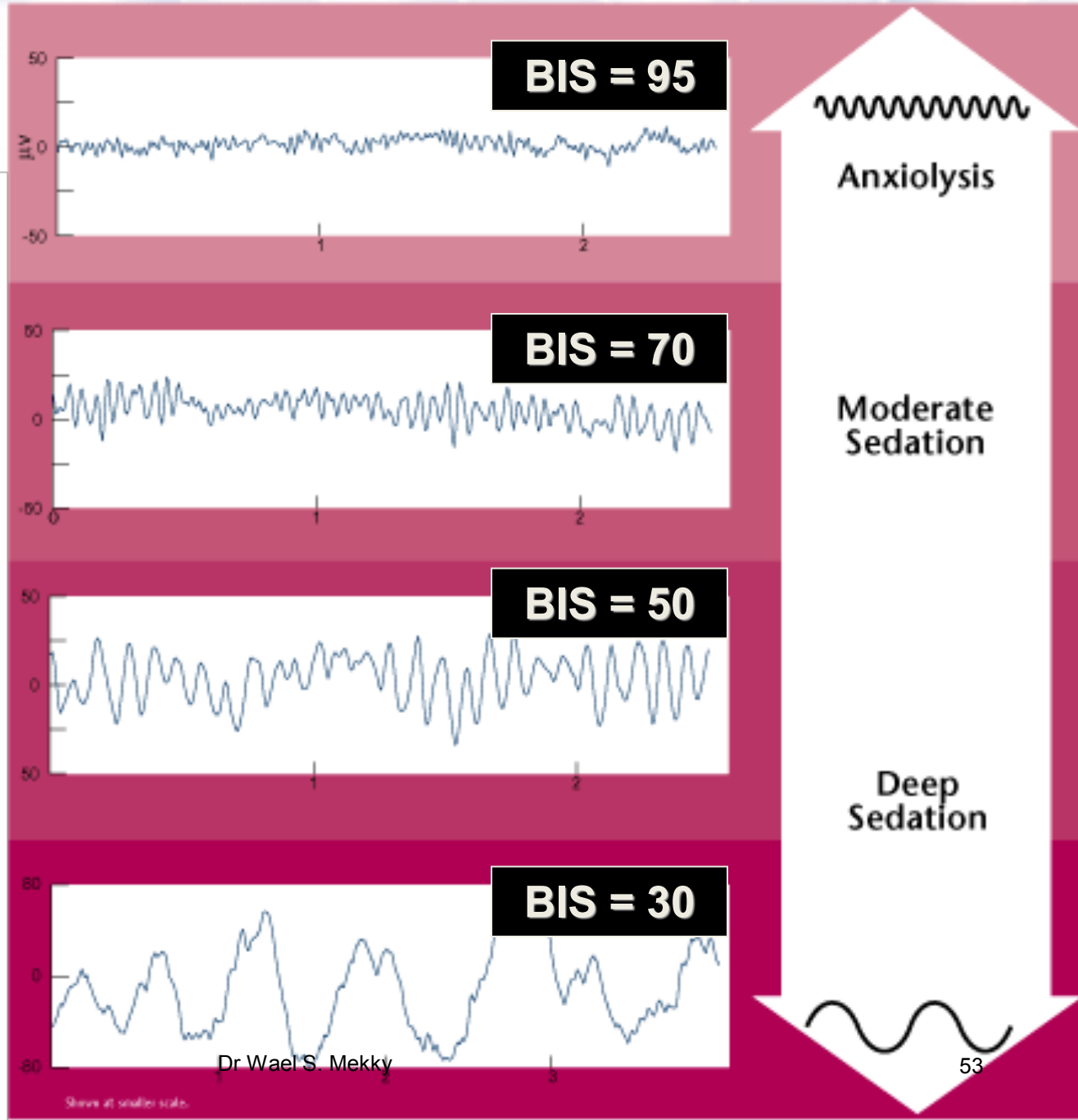
GE BIS Display / BIS Sensor

GE BIS Display

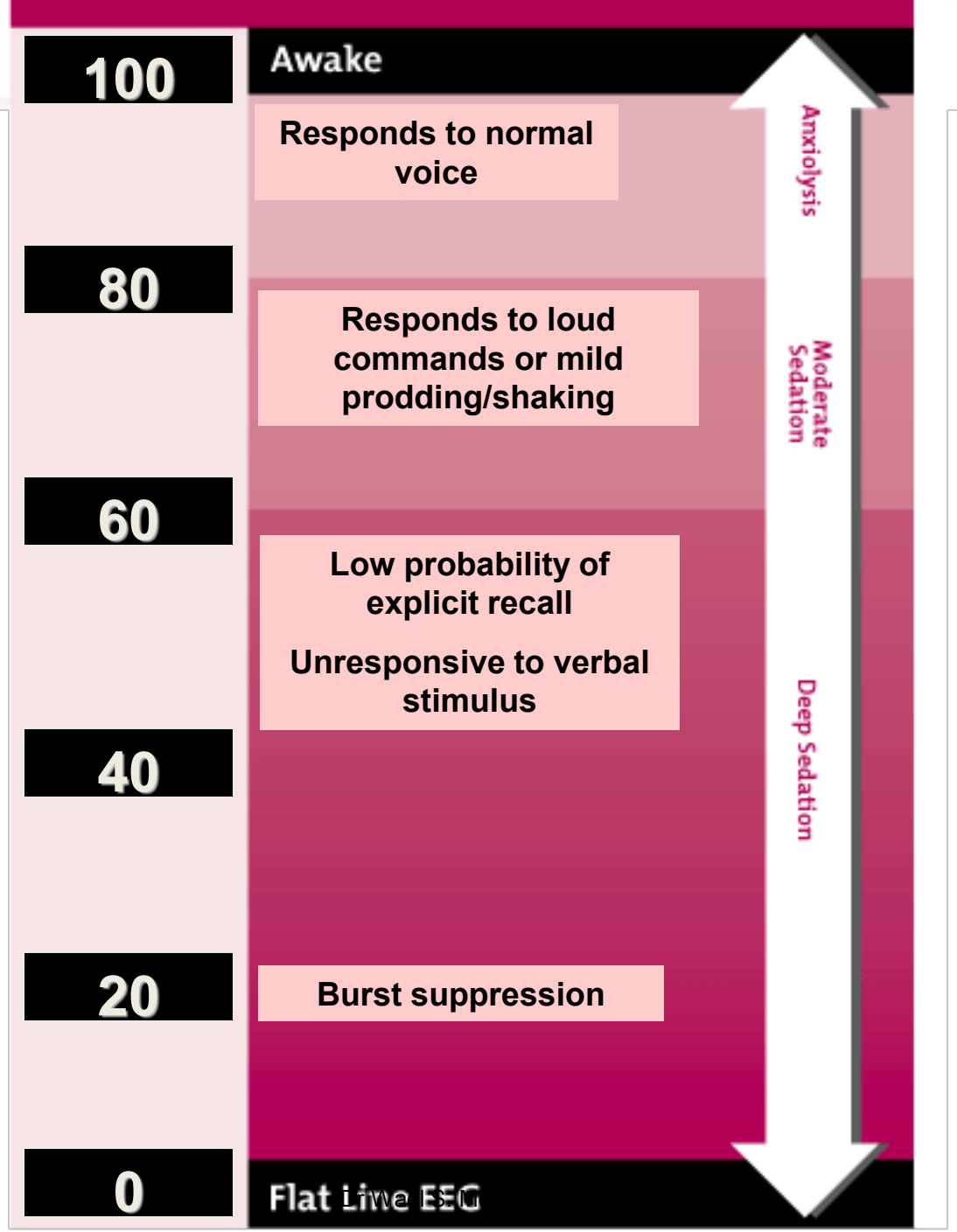


BIS Sensor

BIS converts the "raw" EEG signal to a number 0-100



BIS


100
Awake
Responds to normal voice
80
Responds to loud commands or mild prodding/shaking
60
Low probability of explicit recall
Unresponsive to verbal stimulus
40
20
Burst suppression
0
Flat Line EEG
Anxiolysis
Moderate Sedation
Deep Sedation

Thank you

Critical Care Nursing

Dr. Wael S. Mekky

Glasgow Coma Scale (GCS)

Glasgow Coma Scale (GCS)

- The **Glasgow Coma Scale** (GCS) allows healthcare professionals to consistently evaluate the level of consciousness of a patient.
- It is commonly used in the context of head trauma, but it is also useful in a wide variety of other non-trauma related settings. Regular assessment of a patient's GCS can identify early signs of deterioration.

Glasgow Coma Scale (GCS)

- There are three aspects of behavior that are independently measured as part of an assessment of a patient's (GCS) which are:

1- Eye-opening,

2- Verbal Performance and,

3- Motor Responsiveness.

- The highest possible score is **15** (fully conscious) and the lowest possible score is 3 (coma or dead).

Glasgow Coma Scale

- **The assessment is carried out in three stages:**

1) **Eye Opening Response (4 points)**
2) **Verbal Response (5 points)**
3) **Motor Response (6 points)**

Glasgow Coma Scale

EYE OPENING			VERBAL RESPONSE			MOTOR RESPONSE		
Spontaneous	>	4	Orientated	>	5	Obey commands	>	6
To sound	>	3	Confused	>	4	Localising	>	5
To pressure	>	2	Words	>	3	Normal flexion	>	4
None	>	1	Sounds	>	2	Abnormal flexion	>	3
			None	>	1	Extension	>	2
						None	>	1

GLASGOW COMA SCALE SCORE

Mild	Moderate	Severe
13-15	9-12	3-8

1) Eye Opening Response

A. Spontaneous-open with blinking at baseline = 4 points
B. To verbal stimuli, command, speech = 3 points
C. To pain only (not applied to face) = 2 points
D. No response = 1 point

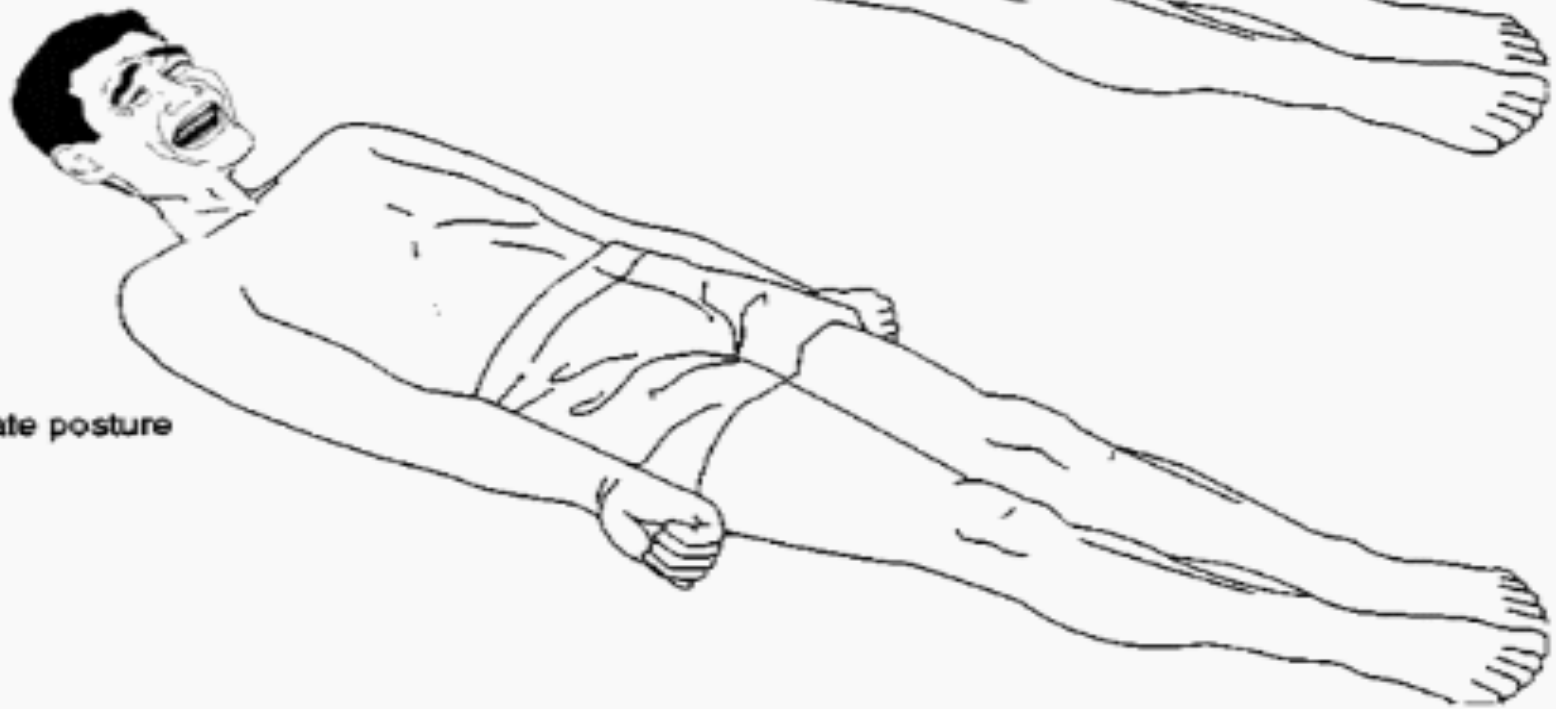
2) Verbal Response

- 1- Oriented = 5 points
- 2-Confused conversation, but able to answer questions = 4 points
- 3-Inappropriate words = 3 points
- 4-Incomprehensible speech = 2 points
- 5-No response = 1 point

3) Motor Response

1. Obeys commands for movement = 6 points
2. Purposeful movement to painful stimulus= 5 points
3. Withdraws in response to pain = 4 points
4. Flexion in response to pain (decorticate posturing)= 3 points
5. Extension response in response to pain (decerebrate posturing) =2 points
6. No response = 1 point

Decorticate posture



Decerebrate posture

Eye opening

Criterion	Observed	Rating	Score
Open before stimulus	✔	Spontaneous	4
After spoken or shouted request	✔	To sound	3
After finger tip stimulus	✔	To pressure	2
No opening at any time, no interfering factor	✔	None	1
Closed by local factor	✔	Non testable	NT

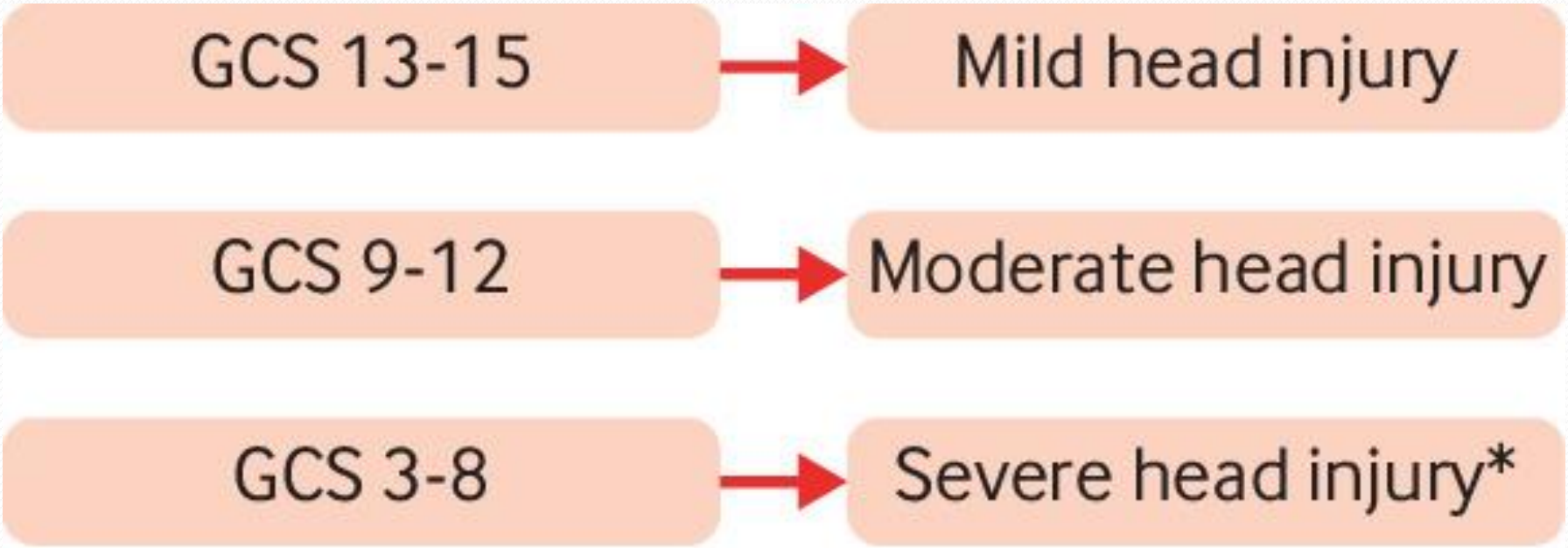
Verbal response

Criterion	Observed	Rating	Score
Correctly gives name, place and date	✔	Orientated	5
Not orientated but communication coherently	✔	Confused	4
Intelligible single words	✔	Words	3
Only moans / groans	✔	Sounds	2
No audible response, no interfering factor	✔	None	1
Factor interferring with communication	✔	Non testable	NT

Best motor response

Criterion	Observed	Rating	Score
Obey 2-part request	✔	Obeys commands	6
Brings hand above clavicle to stimulus on head neck	✔	Localising	5
Bends arm at elbow rapidly but features not predominantly abnormal	✔	Normal flexion	4
Bends arm at elbow, features clearly predominantly abnormal	✔	Abnormal flexion	3
Extends arm at elbow	✔	Extension	2
No movement in arms / legs, no interfering factor	✔	None	1
Paralysed or other limiting factor	✔	Non testable	NT

Final assessment of GCS:



Example (1):

- **What is the patient's score based on these findings:** when you arrive to the patient's bedside the patient's eyes are closed but they open when you speak to the patient. and Confused conversation, but able to answer questions, Purposeful movement to painful stimulus?

- Answer:
- GCS.....[E.....,V.....,M.....]

Answer (1):

- The patient's eyes are closed but they open when you speak to the patient which equal: **E = 3**
- Confused conversation, but able to answer questions which equal: **V = 4**
- Purposeful movement to painful stimulus which equal: **M = 5**
- **Answer/** GCS 12 [E 3 ,V 4, M 5]
- **GCS 12 means Moderate Head Injury**

Example (2):

What is the patient's score based on these findings: when you arrive to the patient's bedside the patient's eyes open To pain only, and Inappropriate words and Extension response in response to pain ?

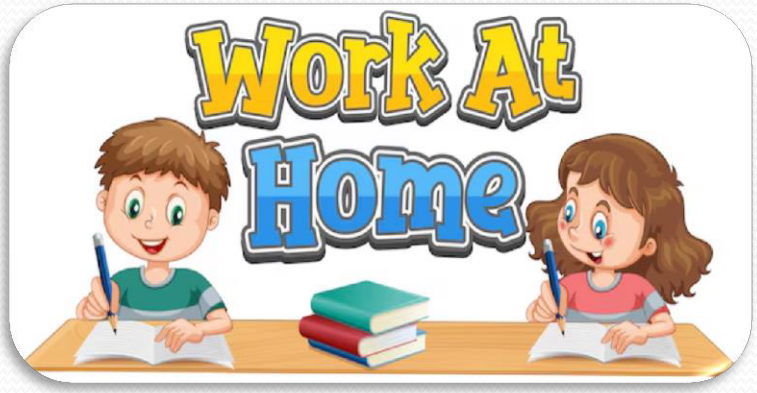
- Answer:
- GCS.....[E.....,V.....,M.....]

Answer (2):

- The patient's eyes open to pain only which equal: **E = 2**
- Inappropriate words which equal: **V = 3**
- Extension response in response to pain which equal: **M = 2**
- **Answer**/ GCS 7 [E 2 ,V 3, M 2]
- **GCS 7 means Severe Head Injury**

Assignment??

- when you arrive to the patient's bedside the patient's eyes are closed, but they open when you speak to the patient, the patient doesn't respond appropriately to questions asked and says words that don't make sense. the patient can't obey a motor command. when you apply a central stimulus the patient moves to locate and remove the stimulus.?
- A. GCS 12 (E3 V4 M5)
- B. GCS 8 (E2 V4 M2)
- C. GCS 11 (E3 V3 M5)
- D. GCS 10 (E3 V3 M4)



15
106
96/-
22
94
SpO2
96
TEMP
10 kmh
morph.

#THANKYOU

Critical Care Nursing

Acid – base balance

Dr. Wael Subhi Mekky

Acid - Base Balance

Or

- Homeostasis of Blood pH

Or

- Regulation of Blood pH

Definition Of Terms

- **Acid** – H^+ ion (Proton) donor
- **Base** – H^+ ion (Proton) acceptor
- **pH** – Negative log of H^+ activity
- **Acidemia** – Arterial blood pH < 7.35
- **Alkalemia** – Arterial blood pH > 7.45
- **Acidosis and Alkalosis** – Pathological states leading to acidemia and alkalemia

Acid - Base Balance

- Acid Base balance is a **homeostatic mechanism**
- Carried out to **regulate the altered pH of blood** and other body compartments **to its normal constant range.**

(Background)

- The major physiological function of the body is maintenance of homeostasis; it's defined as the body's system for maintaining that consistency.
- An important and integral goal of homeostasis is acid-base regulation.
- Acid–base disturbances are commonly encountered in clinical practice.

(Background)

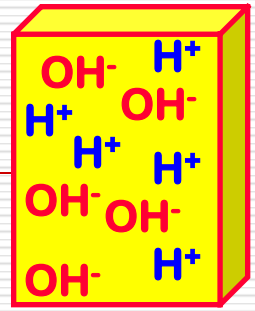
- Identification of the specific acid–base imbalance is important in identifying the underlying cause of the disorder and in determining appropriate treatment.
- Maintenance of acid-base homeostasis is a vital function of the living organism, that the plasma pH is an indicator of hydrogen ion (H^+) concentration.
- Homeostatic mechanisms keep pH within a normal range (7.35–7.45) by buffer systems -the kidneys and the lungs.

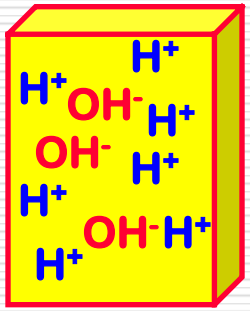
(Background)

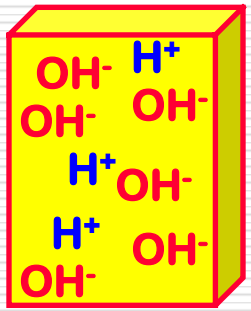
- **The PH and H+ concentration are inversely related:** the greater the concentration, the more acidic the solution and the lower the pH.
- The lower the H+ concentration, the more alkaline the solution and the higher the pH.

Acids vs. Bases

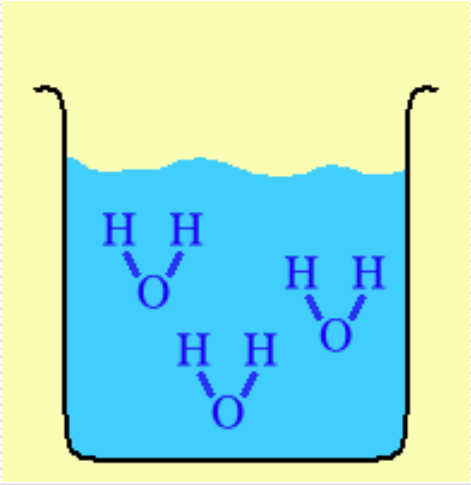
- Pure water is Neutral
  - ($H^+ = OH^-$)
  - pH = 7
- Acid
  - (H+ > OH-)
  - pH < 7
- Base
  - (H+ < OH-)
  - pH > 7







- H^+ ion is an acid
- OH^- ion is a base



Acids vs. Bases

- **An Acid:**

- It's a substance that can donate or contribute H+ ions.

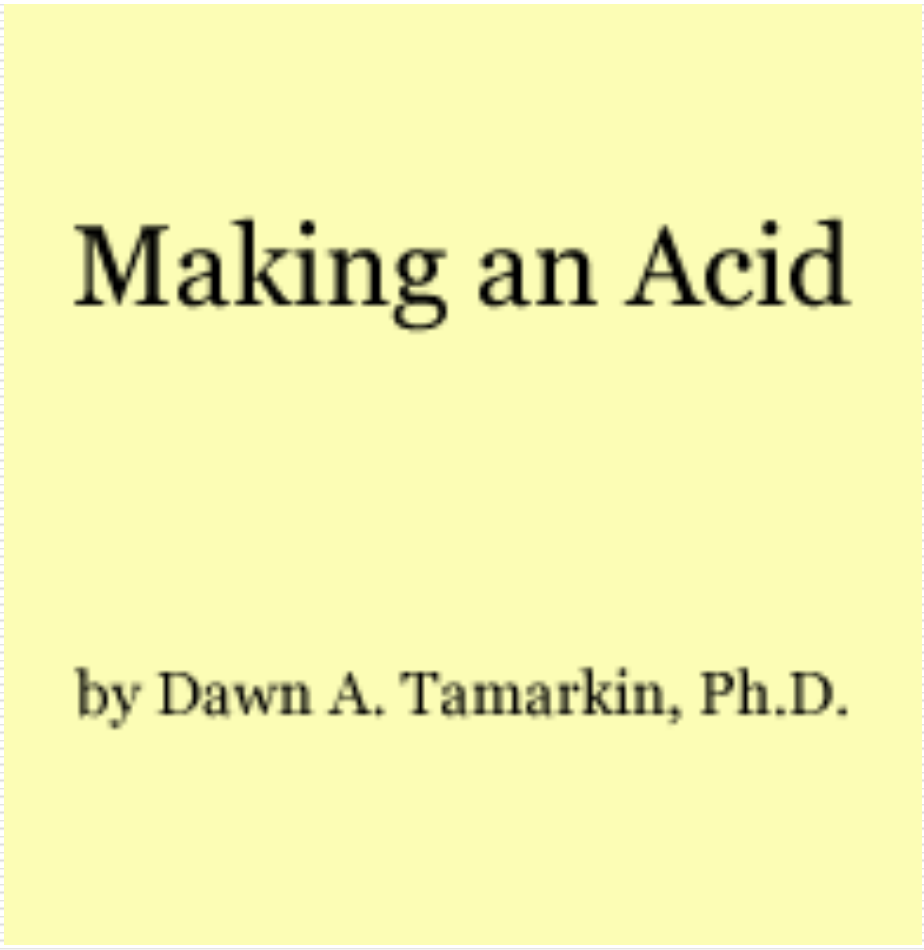
Acid: H^+ donor

- An acid is molecules containing hydrogen atoms that can release hydrogen ions in solutions are referred to as an acid.

- If **H+** concentration **increases**; the **pH** will **decrease**.

Acid

- H^+ ion is an acid



Acids vs. Bases

- **A Base:**

- It's a substance that can accept or combine with H+ ions to remove these ions from a solution.

Base: H^+ acceptor

- A base is an ion that can accept a hydrogen ion.

- If **H+** concentration **decreases,** the pH will **increases.**

Base

- OH^- ion is a base

pH

- The term "pH" originates from Latin and is an acronym for "potentia hydrogenii" - the power of hydrogen.
- The pH scale is commonly used to represent hydrogen ion activity. Because pH is the negative logarithm of [H+] ($pH = -\log_{10}[H^+]$), an increase in pH indicates a decrease in [H+] and vice versa.

pH

- Different compartment of human body has specific pH.
- pH has role in Enzyme activity.
- pH of blood and other body compartments are precisely regulated.
- pH is always tried to be maintained to its normal constant range.

pH

- pH of the medium directly affects the enzyme activities
- Optimum pH is an essential requisite for enzyme activities and normal metabolism.

pH

- It is prerequisite for regulating blood/body pH:

- **To maintain normal/optimal Enzyme activities**
- **Normal metabolism**
- **Normal Coordination**
- **Normal Health**

pH and the pH Scale

- Acid Base Balance maintains the blood pH at normal constant narrow range of (7.35-7.45).
- pH range compatible with life is (6.8 - 8.0).

Acidosis vs. Alkalosis

- **Acidosis:** the process that cause academia (a condition in which the hydrogen ion concentration of the blood is elevated, the blood has an acid excess or base deficit (pH < 7.35).
- **Alkalosis:** the processes that cause alkalemia (a condition in which the hydrogen ion concentration of the blood is reduced, the blood has an acid deficit or base excess (pH > 7.45)

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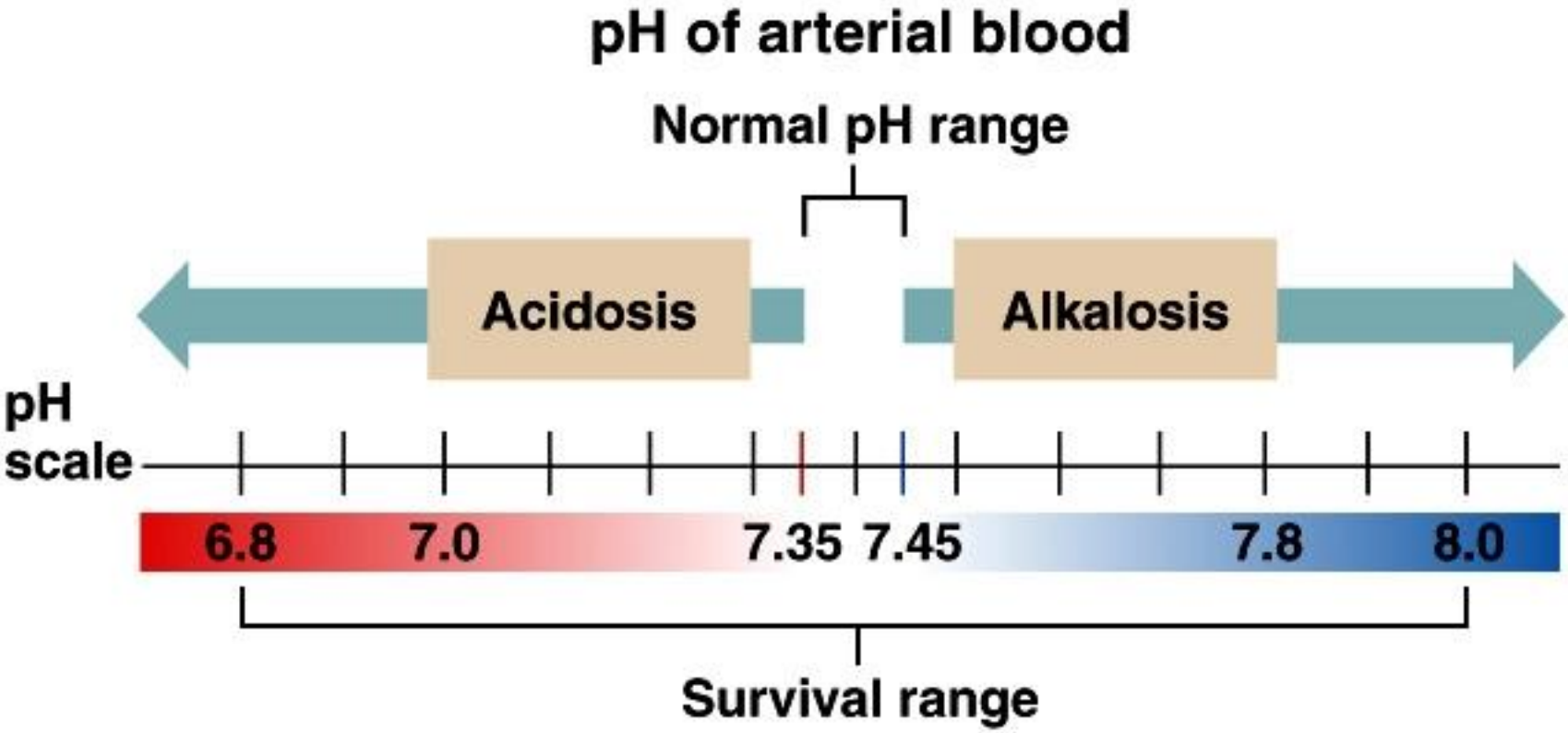
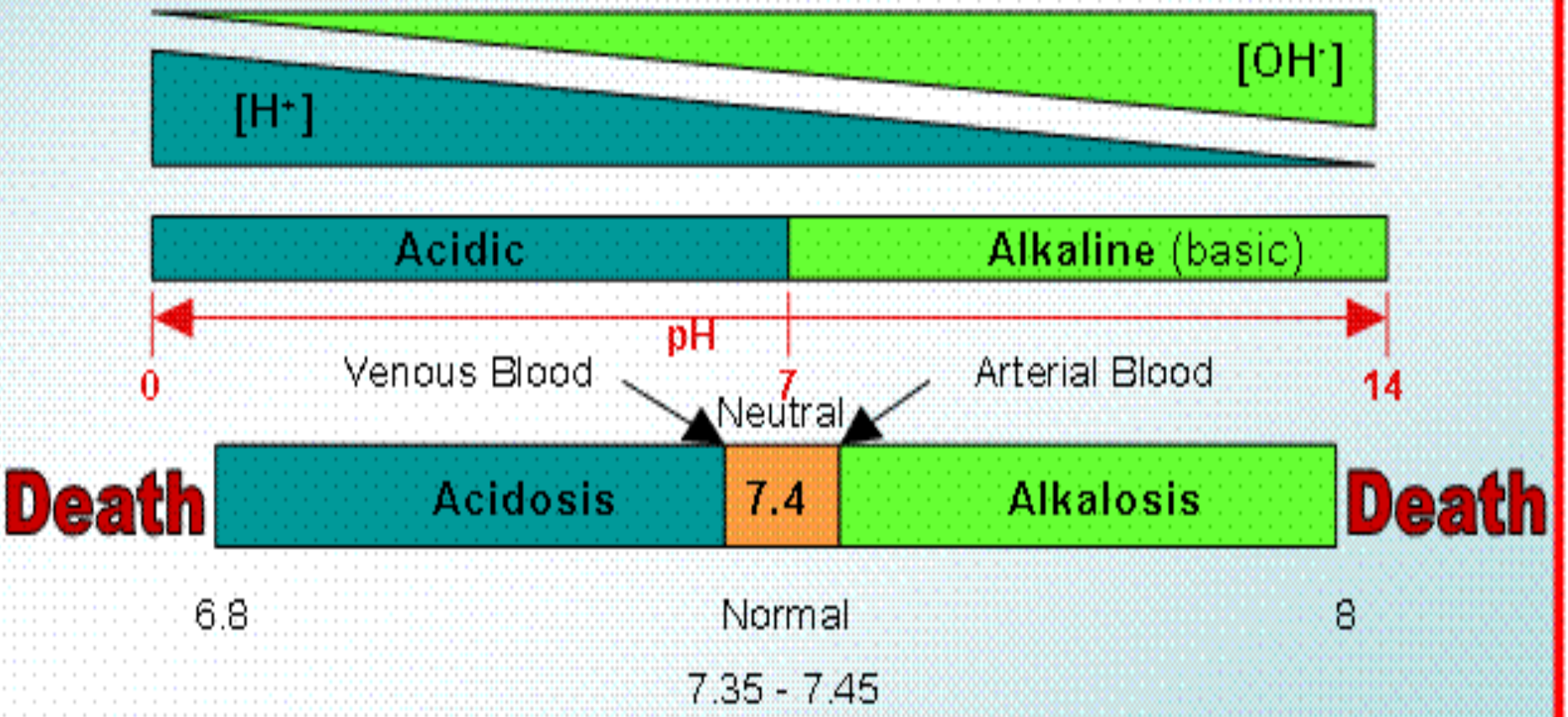


Acid Base Balance



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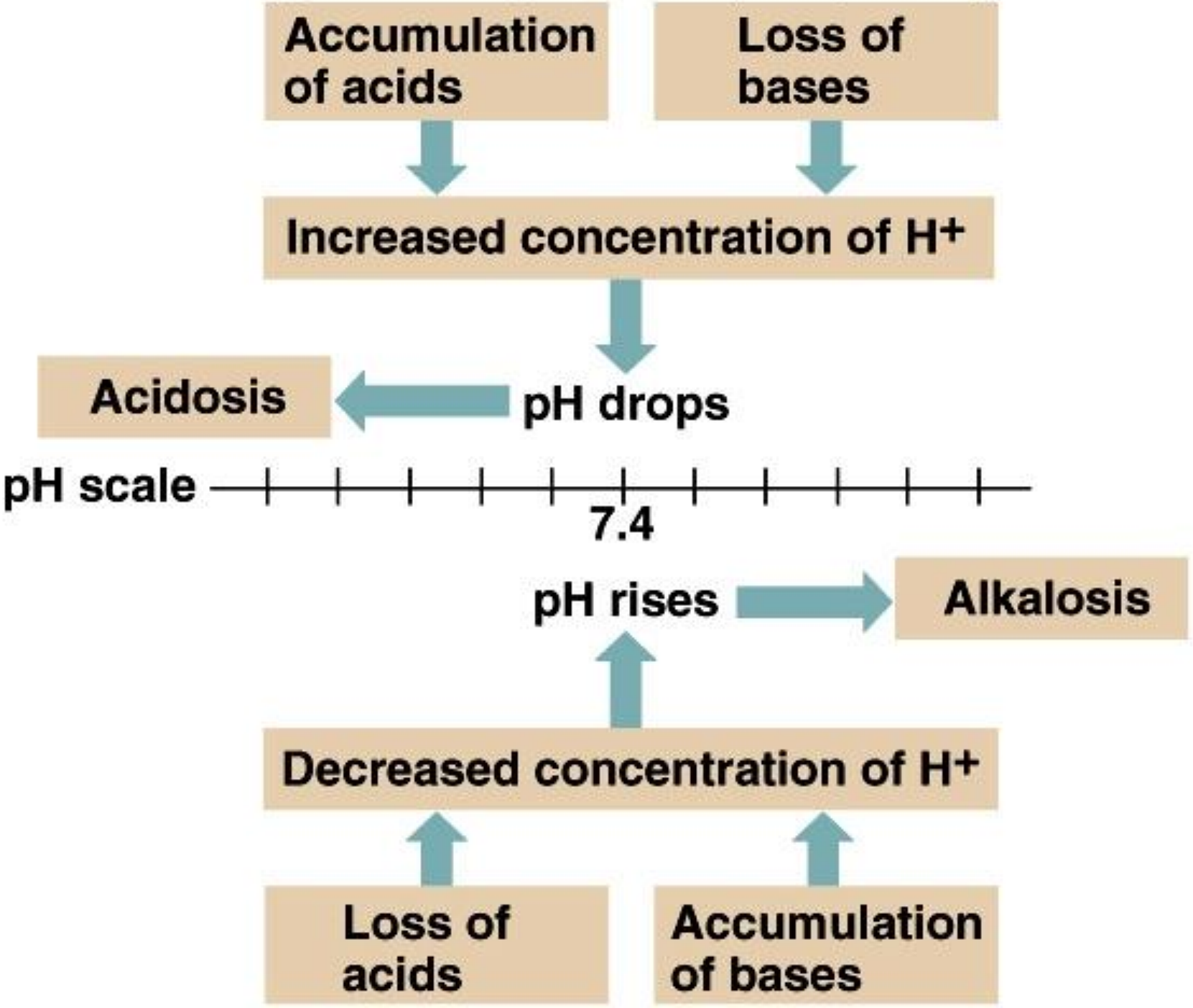
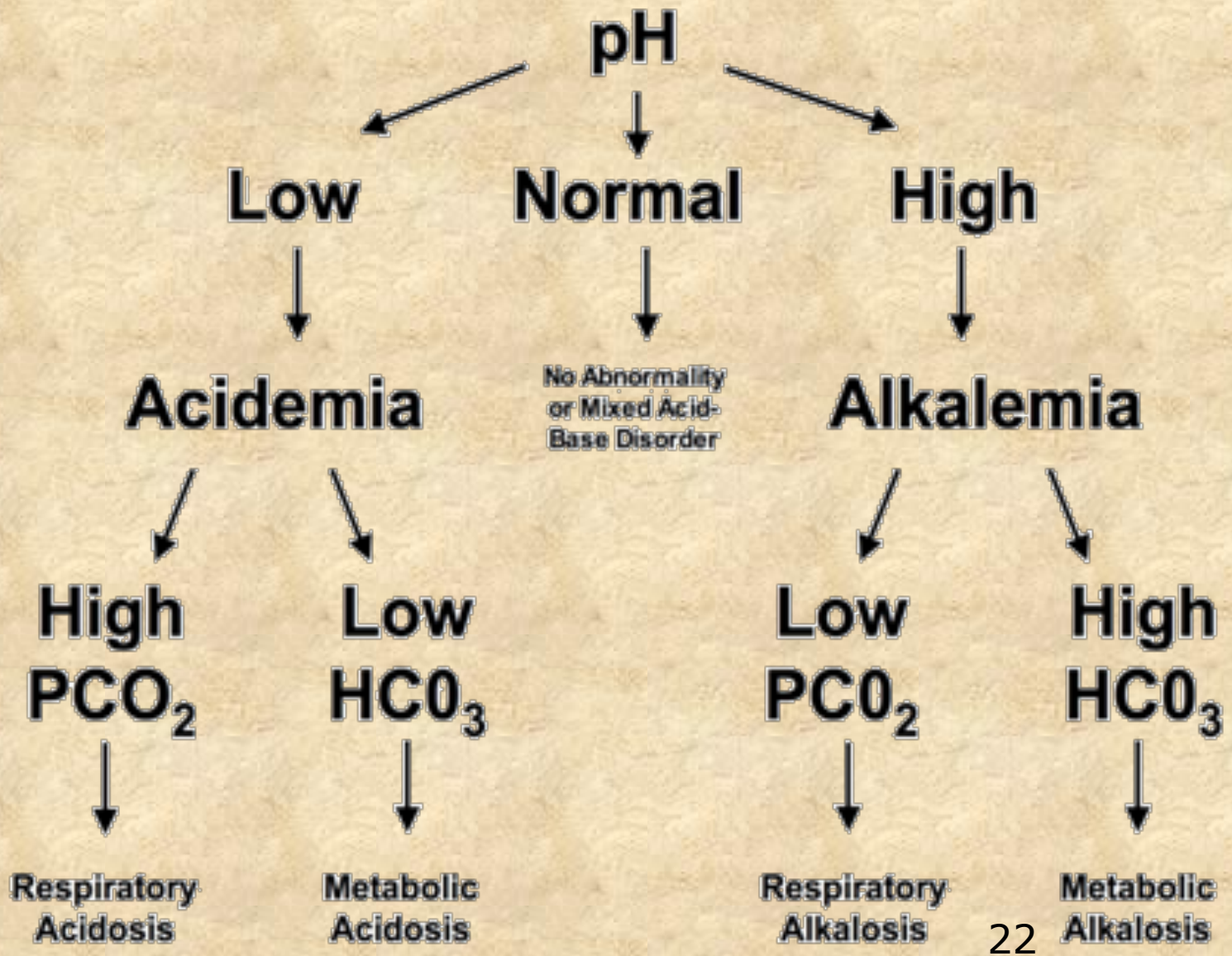


Figure 1: Identifying the Primary Process



Normal Blood Gases Values

	Arterial blood		Mixed venous blood	
		range		range
pH	7.40	7.35-7.45	pH	7.33-7.43
pCO_2	40 mmHg	35 – 45	pCO_2	41 – 51
pO_2	95 mmHg	80 – 95	pO_2	35 – 49
Saturation	95 %	80 – 95	Saturation	70 – 75
BE	±2		BE	
HCO_3^-	24 mEq/l	22 - 26	HCO_3^-	24 - 28

Normal Values

pH	7.35-7.45
CO_2	35-45
pO_2	80-100
HCO_3	22-26
O_2 Sat.	95-100%

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Effect of Altered pH

- Small changes in pH can produce major disturbances
- Acid-base balance can also affect Electrolytes (Na^+, K^+, Cl^-)
- Can also affect Hormones.

Effect of Altered pH

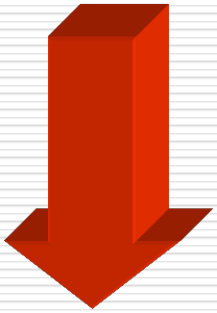
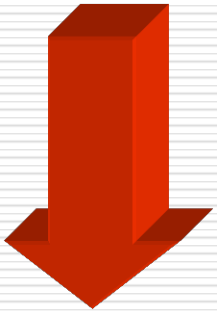
- Altered pH may seriously disturbs the vital processes.
- Might lead to death.

Effect of Altered pH

- Most enzymes function only with narrow pH ranges.
- Makes **Enzymes inactive**.
- **Inactivated Enzymes** affect metabolic reactions and metabolic pathways.
- Leads to **metabolic syndromes.**

Effect of Altered pH

pH also affect excitability of Nerve and Muscle cells


pH 
Excitability

pH
Excitability

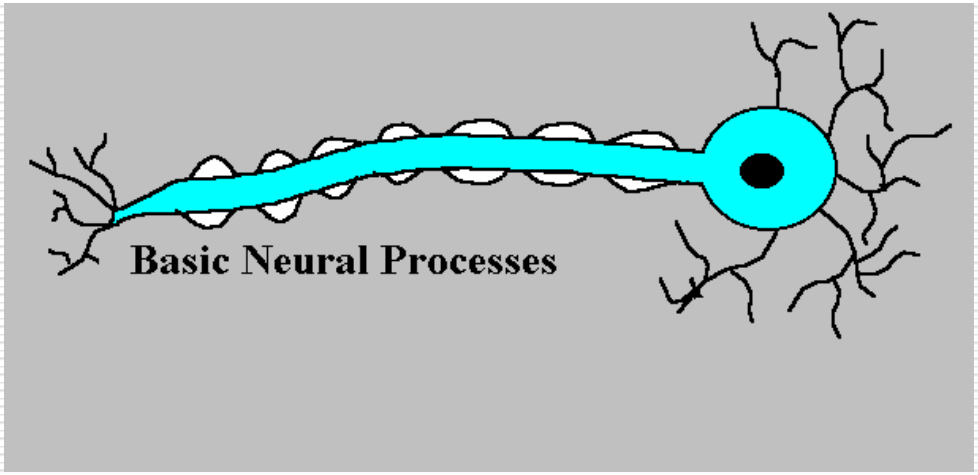
ACIDOSIS / ALKALOSIS

- **pH changes have dramatic effects on normal cell function**

1) Changes in excitability of nerve and muscle cells

2) Influences Enzyme activity

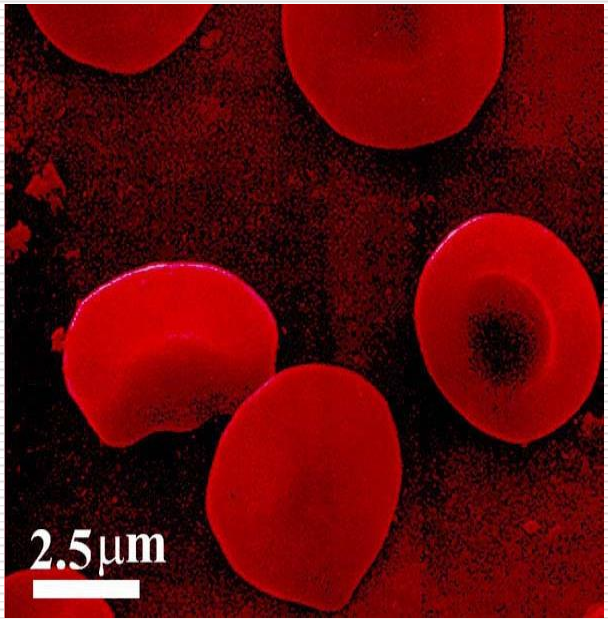
3) Influences **K^+** levels/Retention of K^+

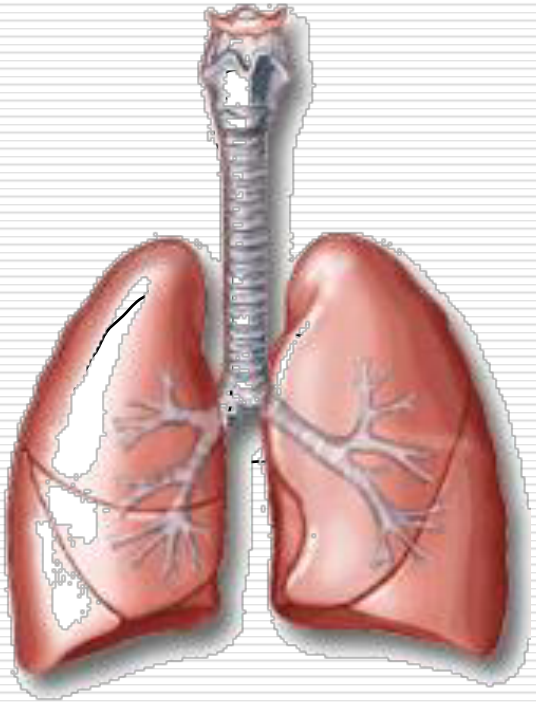


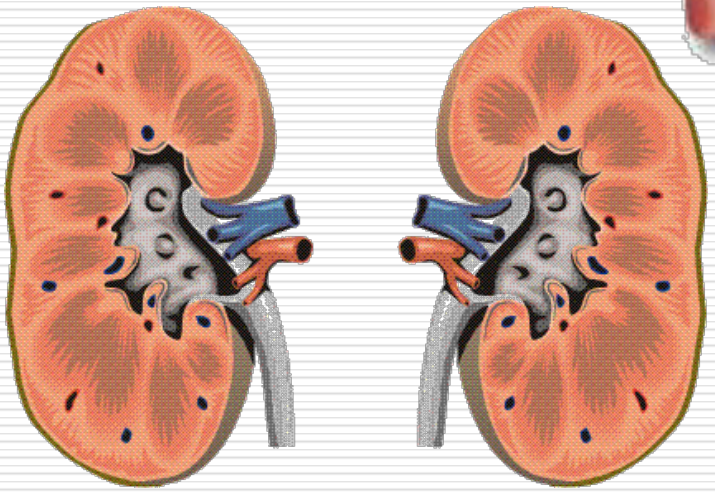
CHANGES IN CELL EXCITABILITY

- **pH decrease** (more acidic) **depresses** the central **nervous system**
  - Can lead to **loss of consciousness**
- **pH increase** (more basic)causes **over excitability** of **nervous system.**
  - Tingling sensations, nervousness, muscle twitches

Factors Regulating Acid-Base Balance







Acid - Base Balance

- ✓ Acid Base balance is **regulated By:**
- ☐ **First Line of Defense**
  - ❖ **Blood Buffer System**
- ☐ **Second Line of Defense**
  - ■ **Respiratory Mechanism**
- ☐ **Third Line of Defense**
  - ➢ **Renal Mechanism**

Acid - Base Balance

1) **Buffers Regulation**

- **React very rapidly (less than a second)**

2) **Respiratory Regulation**

- **Reacts rapidly (seconds to minutes)**

3) **Renal Regulation**

- **Reacts slowly (minutes to hours)**

Blood Buffer System

- First line of defense in mechanism of Acid Base Balance.
- **Buffers** is any mechanism of resistant significant changes in pH Accomplished by converting:
  - Strong acid ⟶ Weak acid
  - Strong base ⟶ Weak base

Blood Buffer System

- **Buffer** systems prevent major changes in the pH of body fluids by removing or releasing H+, they can act quickly to prevent excessive changes in H+ concentration.
- **Buffer** system is assessed when **arterial blood gases(ABG)** are measured.

Blood Buffer System

- Normally, there are **20 parts** of **bicarbonate (HCO_3^-)** to **one part** of **carbonic acid (H_2CO_3).**
- If this ratio is altered, the pH will change.
- **Carbon dioxide (CO_2)** is a potential acid; when dissolved in water, it becomes **carbonic acid ($CO_2 + H_2O = H_2CO_3$).**

Blood Buffer System

- Carbonic acid a weak acid described as volatile acid or **respiratory acid**, because it is the only acid excreted as a **gas** by the lungs.
- Thus, when CO_2 is increased, the carbonic acid content is also increased, and vice versa.

H+ ions are buffered by both Extracellular & Intracellular

- **Extracellular Buffers:**
- Bicarbonate Buffer
  - HCO_3/H_2CO_3
- Inorganic Phosphates Buffer
- Plasma Protein Buffer

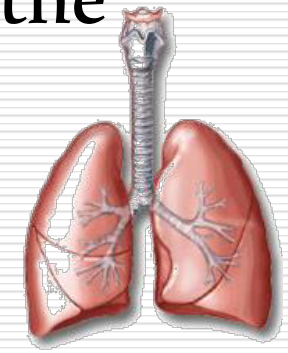
H+ ions are buffered by both Extracellular & Intracellular

- **Intracellular Buffers:**
- Proteins
- Organic & Inorganic Phosphates
- In the RBC, hemoglobin (HB-).

Respiratory Regulation

- **Respiratory:** referring to breathing (respiration); give and take of oxygen and carbon dioxide in the body's tissues, from the lungs to the level of the cells.

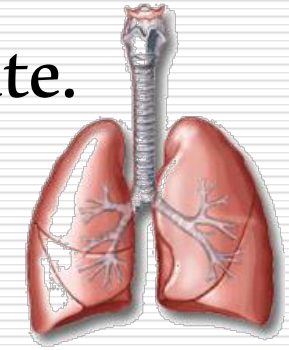
- Respiratory system controls the pH by the **rate** and **depth** of respiration to increase or decrease the release of CO_2.



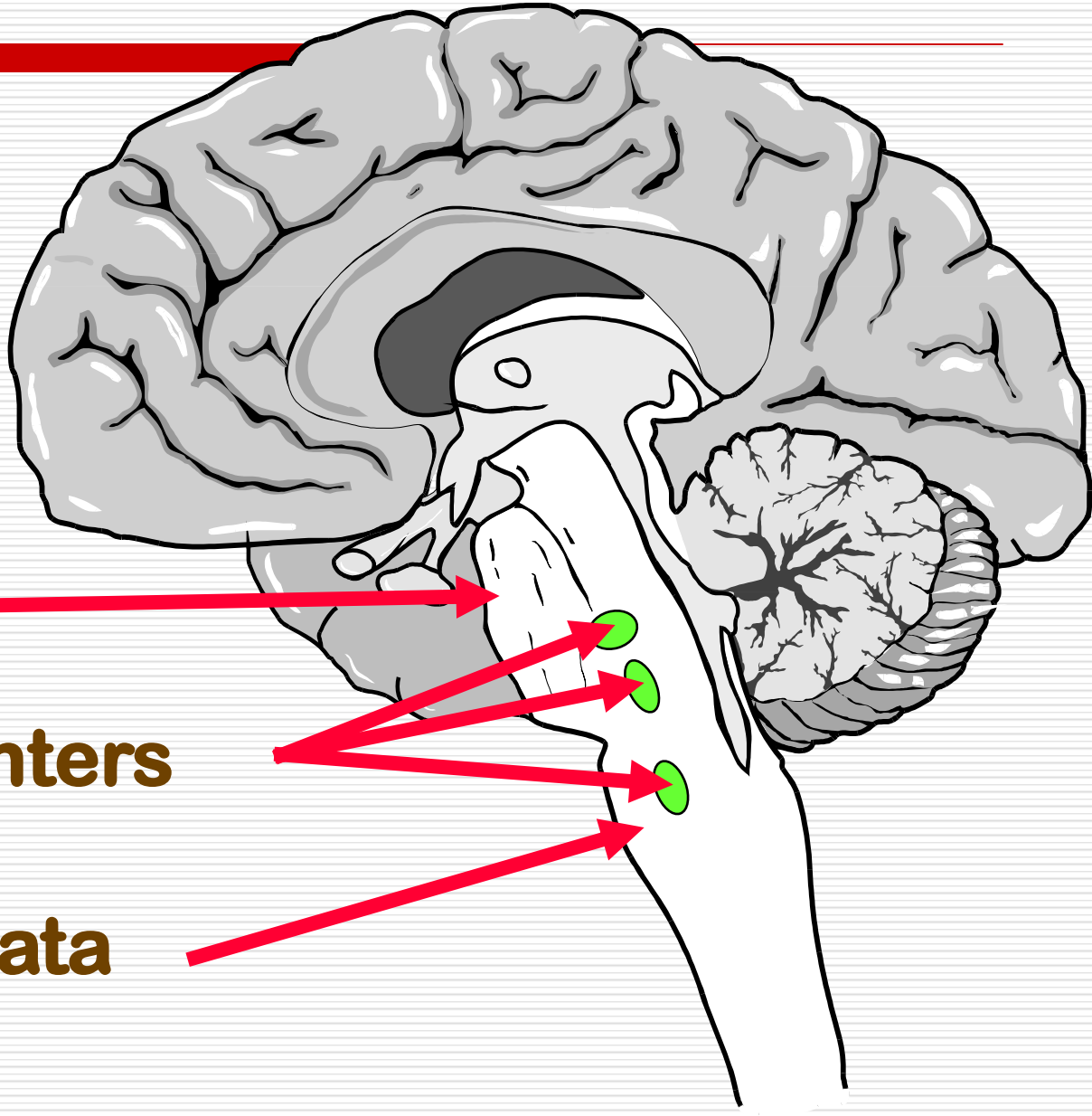
Respiratory Regulation

- When the H+ ion increase this cause chemoreceptor to stimulate the respiratory center of the brain (**medulla**) to increase the rate and depth of respiration.

- **S**o if the blood buffer system isn't able to maintain pH in its normal range, the respiratory system will adjust the frequency and depth of respiration to compensate.



RESPIRATORY CENTER



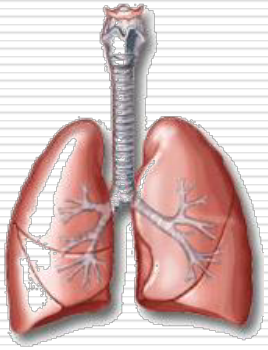
Pons

Respiratory centers

Medulla oblongata

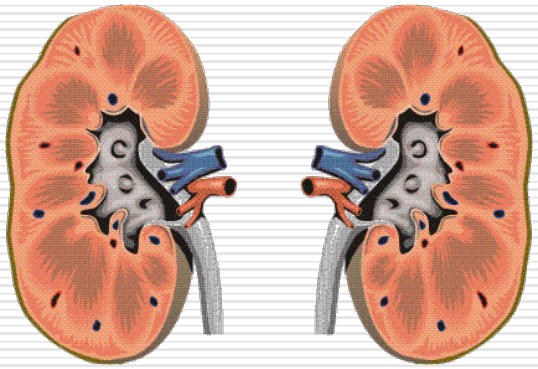
Respiratory Regulation

- In an acidic environment (a low pH and an excess of H+), The respiratory rate increases & cause hyperventilation which increase the amount of CO_2 exhaled that return pH to normal.
- In an alkaline environment (a high pH and a H+ deficit), decreasing the depth and frequency of ventilation will cause the body to retain CO_2, thus increasing H+ and restoring homeostasis.



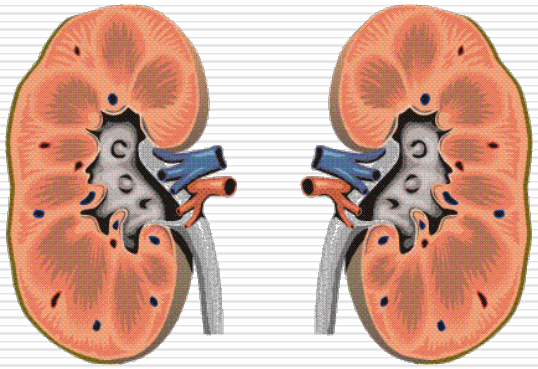
Renal Regulation

- The kidneys play the primary role in maintaining long term control of Acid-Base balance.
- The kidneys do this by selecting which ions to retain and which to excrete



Renal Regulation

- The kidneys can neutralize more acid or base than both respiratory system and chemical buffers.
- If the pH of the solution being filtered in the nephron is **acidic**, the kidneys increase the amount of H+ secreted into the urine, thereby decreasing the acidity and increasing pH in the blood/body.



Renal Regulation

- If the pH in the kidneys is alkalotic, the tubules in the nephrons will decrease the amount of H+ secreted, thereby increasing the acidity and lowering pH in the blood/body.
- While the kidneys are best able to compensate for large variations in pH, their effect is not seen for **hours or even day.**
- **SO If kidneys fail, pH balance fails.**

Acid-Base Imbalance

Acid-Base Imbalances

- pH< 7.35 Acidosis
- pH > 7.45 Alkalosis

Acid-Base Imbalance Causes

- Respiratory
  - abnormal processes which tend to alter pH because of a primary change in **pCO_2** levels
    - acidosis
    - alkalosis
- Metabolic
  - abnormal processes which tend to alter pH because of a primary change in **$[HCO_3^-]$**
    - acidosis
    - alkalosis

Respiratory system - CO_2

- Differences in the stimulation of respiration by pCO_2, H^+ and pO_2
- Alveolar ventilation
- Disturbances
  - Acidemia
    - → respiratory centre of the brain
    - → ↑alveolar ventilation
    - → ↓CO_2
  - Alkalemia
    - → respiratory centre of the brain
    - → ↓ alveolar ventilation
    - → ↑ CO_2

4 Types of Primary Acid-Base Disorders

Acid Base Imbalances	Biochemical Change
Respiratory Acidosis	Increased H2CO3 $\uparrow PCO_2$
Respiratory Alkalosis	Decreased H2CO3 $\downarrow PCO_2$
Metabolic Acidosis	$\downarrow [HCO_3^-]$
Metabolic Alkalosis	$\uparrow [HCO_3^-]$

Acid-Base Balance Abnormalities

I. Respiratory

Acidosis

- High pCO_2, low pH
- Pneumonia, cystic fibrosis, etc
- Kidneys

Retain bicarbonate

Alkalosis

- Low pCO_2, high pH
- Hyperventilation
- Kidneys

Secrete bicarbonate

Kidneys compensate for the problem

Acid-Base Balance Abnormalities

II. Metabolic

Acidosis

- Low bicarbonate
- Low pH
- Too much alcohol
- Excessive loss of bicarbonate (diarrhea)
- Hyperventilation

Alkalosis

- High bicarbonate
- High pH
- Vomiting
- Excessive base intake
- Hypoventilation

Lungs compensate for metabolism

respiratory acidosis

excrete acid

conserve bicarbonate

Oh, Oh, Oh
I'm Very Sick

Dr. Wael S. Mekky Doc; 2025

Respiratory Acidosis (RA)

- **Primary Carbonic acid excess**
- **Increased H_2CO_3/Increased pCO_2**
- **Defect in respiratory centre of brain**
- **Defect in respiratory organ system**
- **Decreased elimination of H_2CO_3 by the lungs.**

Respiratory Acidosis (RA)

- Hypoventilation
- Increased blood levels of CO_2 above 45 mm Hg.
- Hypercapnia – high levels of pCO_2 in blood

Respiratory Acidosis (RA)

- primary disorder is a **↓pH** due to **↑$PaCO_2$** (>40 mmHg), i.e. hypercapnia
- time course:
  - acute (↓pH)
  - chronic (↓pH or normalisation of pH)
    - renal compensation – retention of HCO_3^-, 3-4 days
- causes:
  - decreased alveolar ventilation
  - (presence of excess CO_2 in the inspired gas)
  - (increased production of CO_2 by the body)

RESPIRATORY ACIDOSIS (RA)

- Respiratory acidosis develops when the lungs don't expel **CO_2** adequately.
- This can happen in diseases that severely affect the lungs.

Chronic conditions:

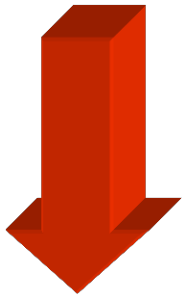
- Depression of respiratory center in brain that controls breathing rate – drugs or head trauma
- Paralysis of respiratory or chest muscles
- Emphysema
- Asthma

Chronic conditions:

- Pneumonia
- Pulmonary edema
- Obstruction of respiratory tract
- Congestive Cardiac Failure

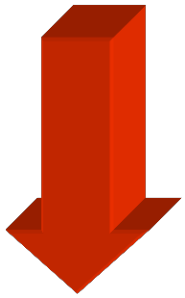
HYPOVENTILATION
Causes Respiratory Acidosis

- Hypo = “Under”



Elimination of CO_2

H^+

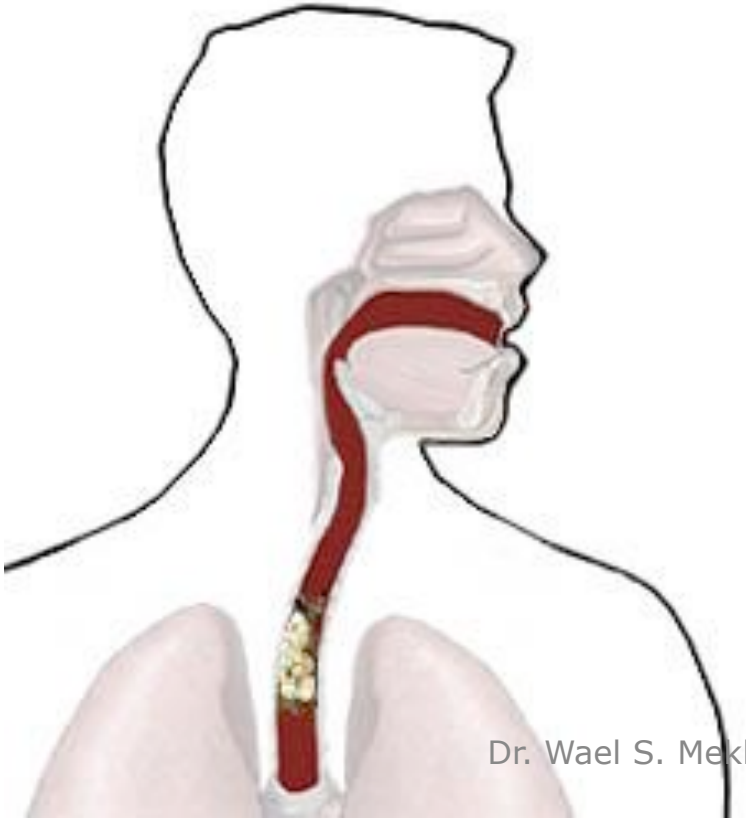


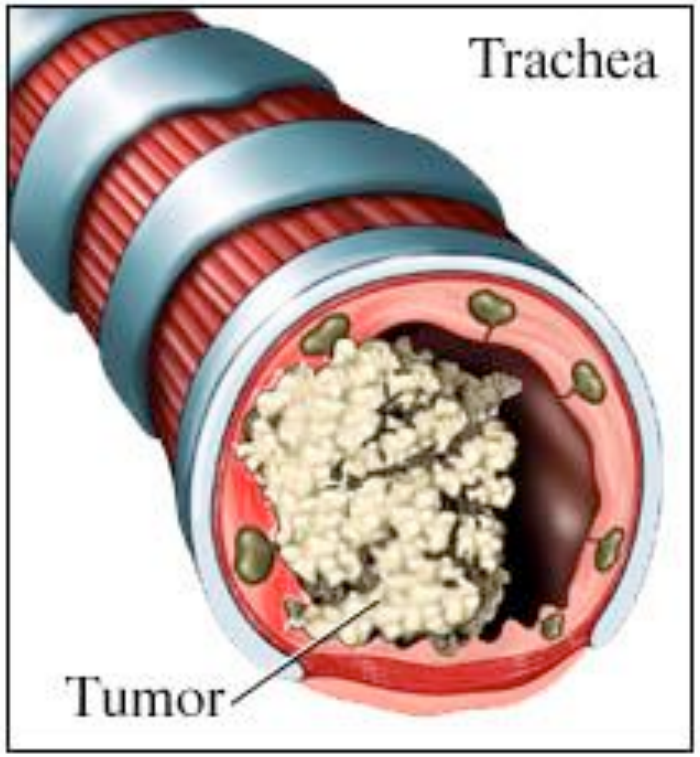
pH

RESPIRATORY ACIDOSIS

1) Obstruction of air passages

—Vomit, Anaphylaxis, Tracheal Cancer





Respiratory Acidosis

- **Acute conditions:**
  - Adult Respiratory Distress Syndrome
  - Pulmonary edema
  - Pneumothorax

Compensation for Respiratory Acidosis

- Kidneys eliminate hydrogen ion and retain bicarbonate ions.

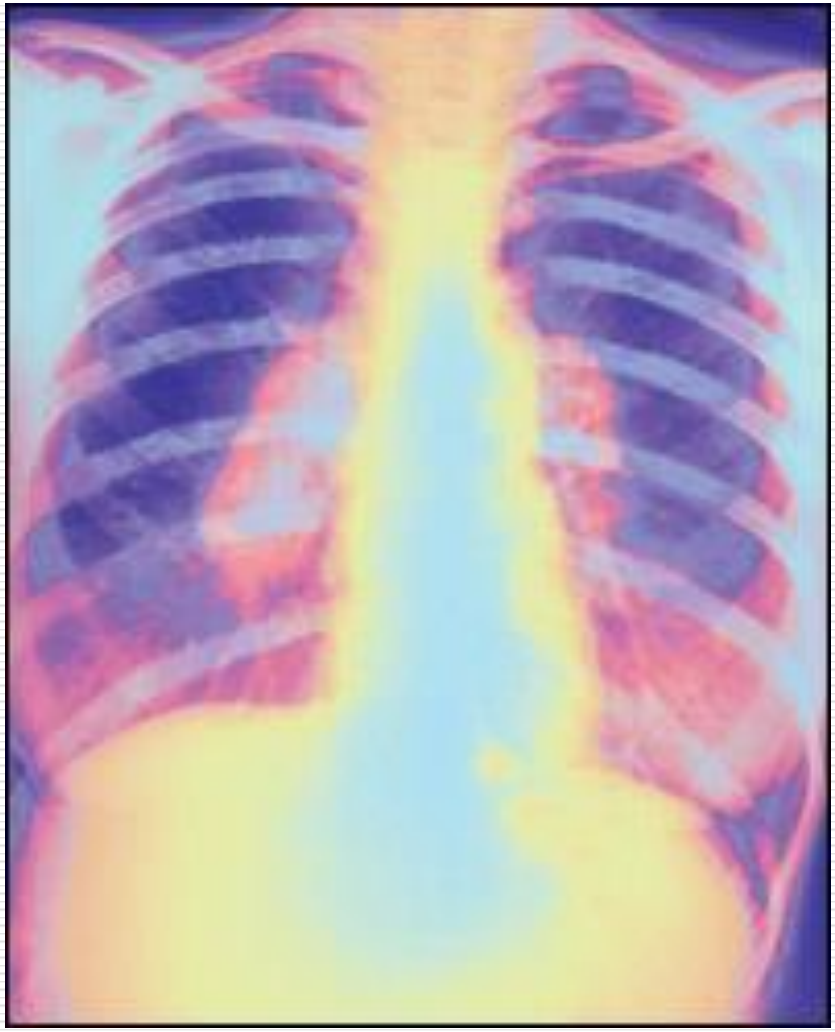
Signs and Symptoms of Respiratory Acidosis

- Breathlessness
- Restlessness
- Lethargy and disorientation
- Tremors, convulsions, coma
- Respiratory rate rapid, then gradually depressed
- Skin warm and flushed due to vasodilation caused by excess CO_2

Treatment of Respiratory Acidosis

- Restore ventilation
- IV lactate solution
- Treat underlying dysfunction or disease.

RESPIRATORY ALKALOSIS



Respiratory Alkalosis

- **Primary Carbonic acid deficit**
- **Decreased H_2CO_3**
- pCO_2 less than 35 mm Hg (hypocapnea)
- Most common acid-base imbalance
- Primary cause is **hyperventilation**
- Washes out excessive quantity of H_2CO_3 through expiration process of lungs.

Respiratory Alkalosis

- Stimulation of respiratory centre in brain
- Hyperventilation

Respiratory Alkalosis

- **Conditions that stimulate respiratory center:**
  - Oxygen deficiency at high altitudes
  - Pulmonary disease and Congestive heart failure – caused by hypoxia
  - Respiratory center lesions
  - Acute anxiety
  - Fever, anemia
  - Early salicylate intoxication
  - Cirrhosis
  - Gram-negative sepsis/Meningitis

RESPIRATORY ALKALOSIS

- **Anxiety** is an emotional disturbance
- The most common cause of **hyperventilation,** and thus **respiratory alkalosis**, is noted in **anxiety.**

RESPIRATORY ALKALOSIS

- **Respiratory center lesions**
  - Damage to brain centers responsible for monitoring breathing rates
    - Tumors
    - Strokes

RESPIRATORY ALKALOSIS

- **High Altitude المرتفعات الشاهقة**
  - Low concentrations of O_2 in the arterial blood stimulates ventilation in an attempt to obtain more O_2
  - Too much CO_2 is "blown off" in the process

RESPIRATORY ALKALOSIS

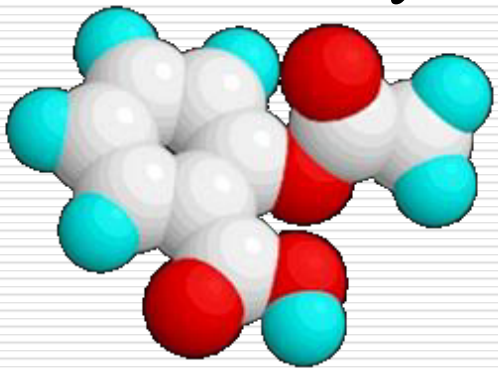
- **Fever**
  - Rapid shallow breathing blows off too much CO_2

RESPIRATORY ALKALOSIS

Salicylate Poisoning

(Aspirin overdose)

- Ventilation is stimulated without regard to the status of O_2, CO_2 or H^+ in the body fluids.



HYPERVENTILATION
Causes Respiratory Alkalosis

- **Hyper = "Over"**

Elimination of CO_2

H^+

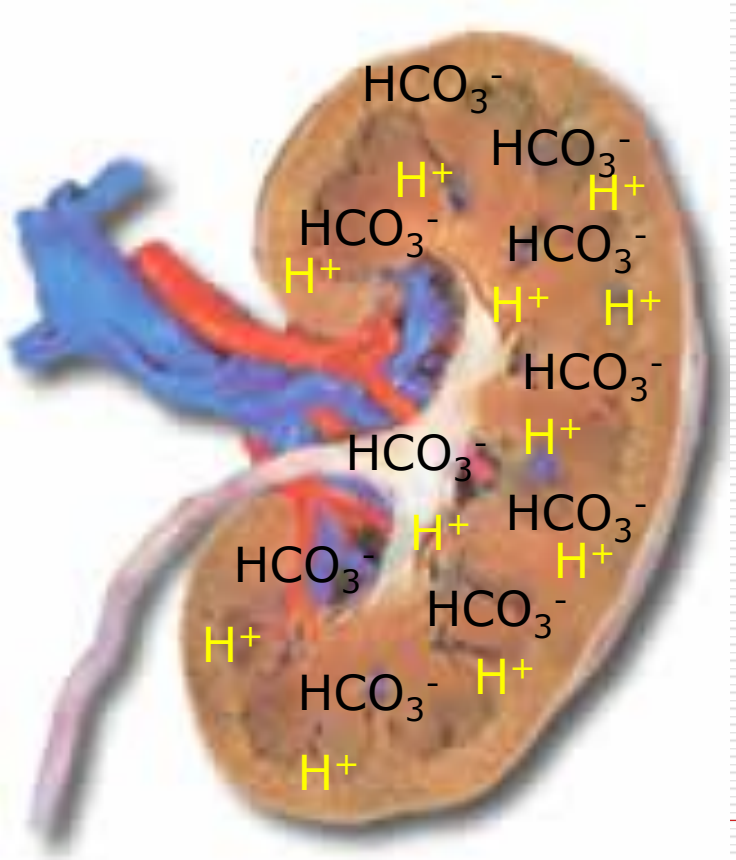
pH

Compensation of Respiratory Alkalosis

- If **kidneys** are functioning normal:
- The conditions of respiratory acidosis or alkalosis are compensated.
- **Kidneys conserve hydrogen ion &**
- **Excrete bicarbonate ion**

Compensation of Respiratory Alkalosis

- Kidneys compensate by:
  - **Retaining hydrogen ions**
  - **Increasing bicarbonate excretion**



Treatment of Respiratory Alkalosis

- Treat underlying cause
- Breathe into **a paper bag**
- IV Chloride containing solution Cl^- ions replace lost bicarbonate ions

Breathe into a paper bag

- Is a technique that can help you regulate hyperventilation.
- It works by putting some of the lost carbon dioxide back into your lungs and body.
- This helps to balance oxygen flow in your body.
- However, breathing into a paper bag must be done properly and may not work for everyone.

METABOLIC ACIDOSIS

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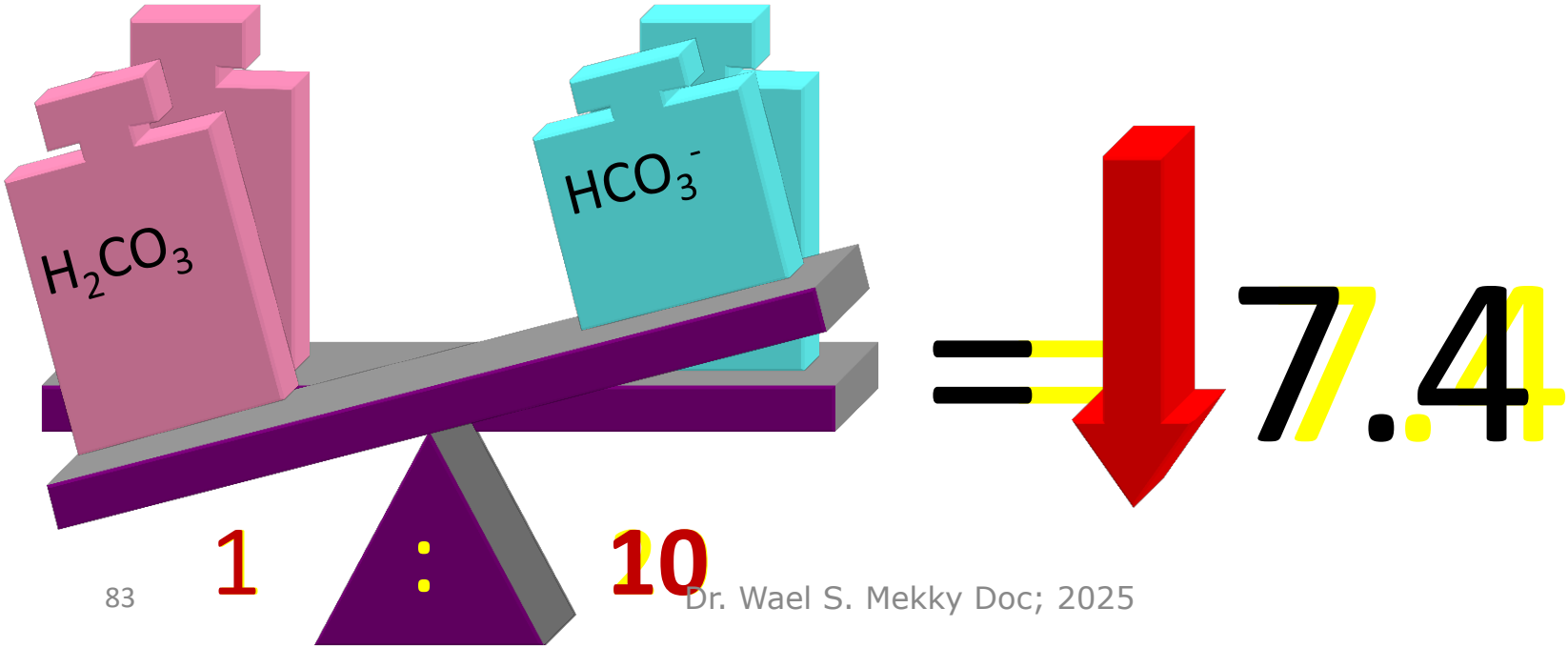
Metabolic Acidosis

- **Primary Alkali deficit**
- **Bicarbonate deficit** - blood concentrations of bicarbonate drop below 22mEq/L
- **Causes:**
  - **Loss of bicarbonate through diarrhea or renal dysfunction.**
  - **Overproduction of acids (lactic acid or ketones)**
  - **Failure of kidneys to excrete H^+**

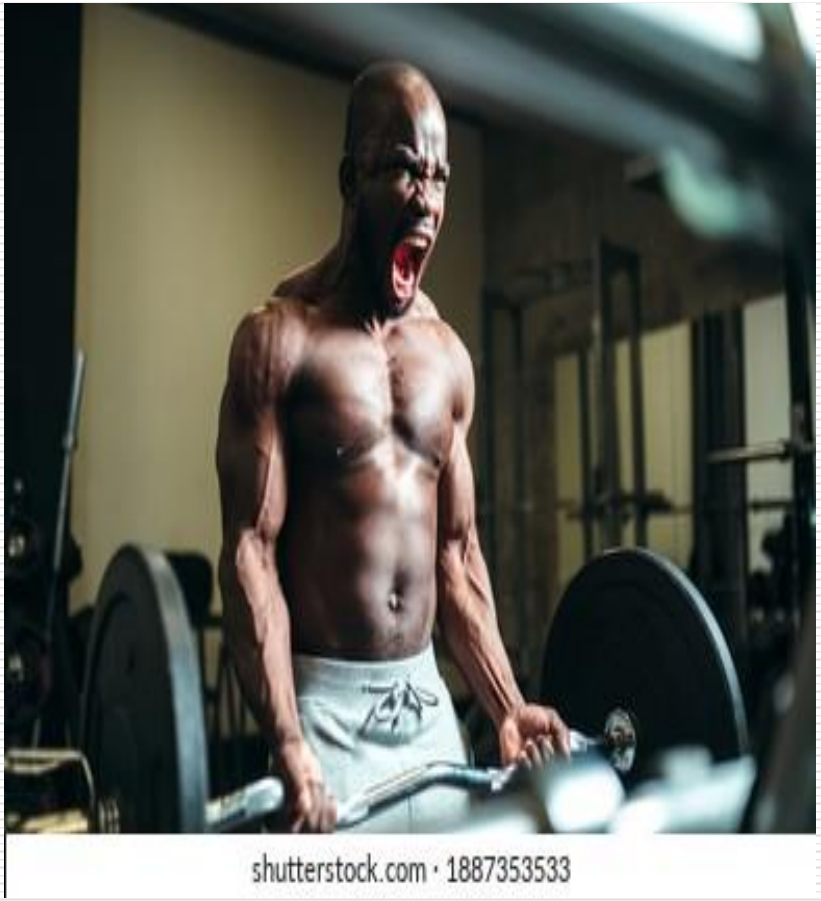
METABOLIC ACIDOSIS

- Occurs when there is a decrease in blood **pH** and bicarbonate level
- Excessive H^+ or decreased HCO_3^-



METABOLIC ACIDOSIS

- Any acid-base imbalance not attributable to **CO_2** is classified as **metabolic**
  - Metabolic production of **Acids**
  - Or loss of **Bases**



METABOLIC ACIDOSIS

- The causes of metabolic acidosis can be grouped into **five** major categories:
  - **1) Ingesting** an acid or a substance that is metabolized to acid
  - **2) Abnormal Metabolism**
  - **3) Kidney Insufficiencies**
  - **4) Strenuous Exercise** تمارين شاقة
  - **5) Severe Diarrhea**

METABOLIC ACIDOSIS

1) Ingesting An Acid

- Most substances that cause acidosis when ingested are considered poisonous
- Examples include **wood alcohol** (methanol) and antifreeze (ethylene glycol)
- However, even an **overdose** of **aspirin** (acetylsalicylic acid) can cause **metabolic acidosis**

METABOLIC ACIDOSIS

- **2) Abnormal Metabolism**
  - The body can produce excess acid as a result of several diseases
    - **Ketoacidosis**
    - **Type I Diabetes Mellitus**
    - **Uncontrolled DM**
    - **Prolonged Starvation**
    - **Lactic acidosis**
    - **Shock**
    - **Hemorrhage**
    - **Violent Exercise** التمارين العنيفة

METABOLIC ACIDOSIS

- Unregulated diabetes mellitus causes **ketoacidosis**
  - Body metabolizes fat rather than glucose
  - Accumulations of metabolic acids (**Keto Acids**) cause an increase in plasma H^+

METABOLIC ACIDOSIS

3) Kidney Insufficiencies

- This type of kidney malfunction is called **renal tubular acidosis** or **uremic acidosis** and may occur in people with kidney failure or with abnormalities that affect the kidneys' **ability to excrete acid.**

METABOLIC ACIDOSIS

4) Strenuous Exercise التمارين الشاقة

- Muscles resort to anaerobic glycolysis during strenuous exercise
- Anaerobic respiration leads to the production of **large amounts** of **lactic acid**

$$C_6H_{12}O_6 \xrightarrow{\text{Enzymes}} 2C_3H_6O_3 + ATP\ (energy)$$

Lactic Acid

METABOLIC ACIDOSIS

5) Severe Diarrhea

- Fluids rich in **HCO_3^-** are released and reabsorbed during the digestive process
- During **diarrhea** the **HCO_3^-** is **lost from the body** rather than **reabsorbed.**

METABOLIC ACIDOSIS

- **5) Severe Diarrhea**
  - The loss of HCO_3^- without a corresponding loss of H^+ lowers the pH
  - Less HCO_3^- is available for buffering H^+
  - Prolonged deep (from duodenum) vomiting can result in the same situation.

Symptoms of Metabolic Acidosis

- Headache, lethargy
- Nausea, vomiting, diarrhea
- Coma
- Death

Compensation for Metabolic Acidosis

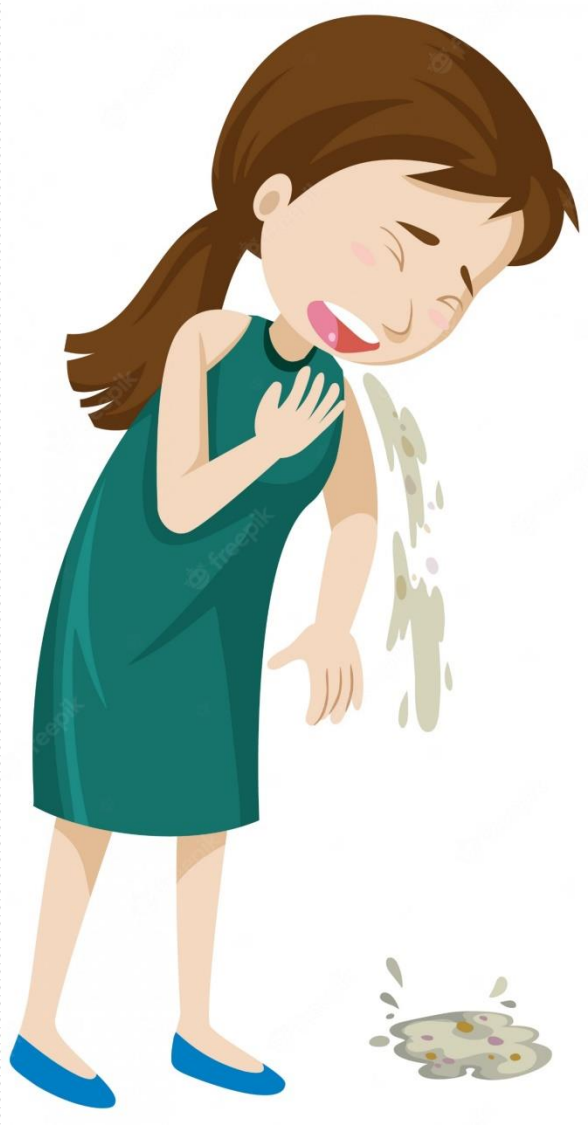
- Increased ventilation.
- Renal excretion of hydrogen ions if possible.
- K^+ exchanges with excess H^+ in ECF.
- H^+ into cells, K^+ out of cells.

Treatment of Metabolic Acidosis

- **IV lactate solution:**

Lactated Ringer's solution (LR), or **Ringer's lactate**, is an isotonic crystalloid intravenous fluid used for fluid resuscitation, dehydration, and replacing electrolyte imbalances. It contains sodium, chloride, potassium, calcium, and lactate, which the liver converts to bicarbonate to **treat metabolic acidosis.** It is commonly used during surgery, burns, and sepsis.

METABOLIC ALKALOSIS



Metabolic Alkalosis

- **Bicarbonate Excess** - concentration in blood is greater than 26mEq/L

- **Causes:**
  - **Excess vomiting = loss of stomach acid**
  - Excessive use of alkaline drugs
  - Certain diuretics

Metabolic Alkalosis

- Endocrine disorders
- Heavy ingestion of antacids
- Severe dehydration
- Cushing's Syndrome
- Prolonged exposure to **X-rays** and **UV rays**

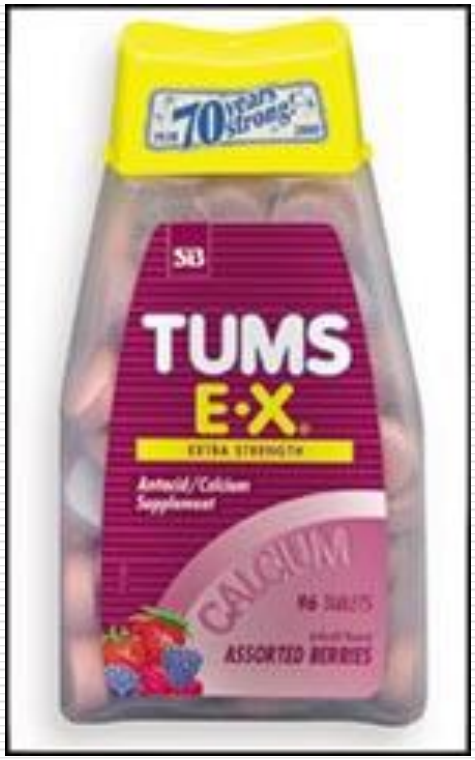
METABOLIC ALKALOSIS

- **May be caused by:**
  - An **increase** of bicarbonate
  - A **decrease** in hydrogen ions
- Imbalance again cannot be due to **CO_2**
- Increase in **pH** which has a non-respiratory origin

7.4

METABOLIC ALKALOSIS

- **Can be the result of:**
  - **1) Ingestion of Alkaline Substances**
  - **2) Vomiting (loss of HCl)**



METABOLIC ALKALOSIS

- Baking soda (**$NaHCO_3$**) often used as a remedy for gastric hyperacidity.
  - **$NaHCO_3$** dissociates to **Na^+** and **HCO_3^-**

Compensation for Metabolic Alkalosis

- Alkalosis most commonly occurs with renal dysfunction, so can't count on kidneys.
- Respiratory compensation difficult – hypoventilation limited by hypoxia.

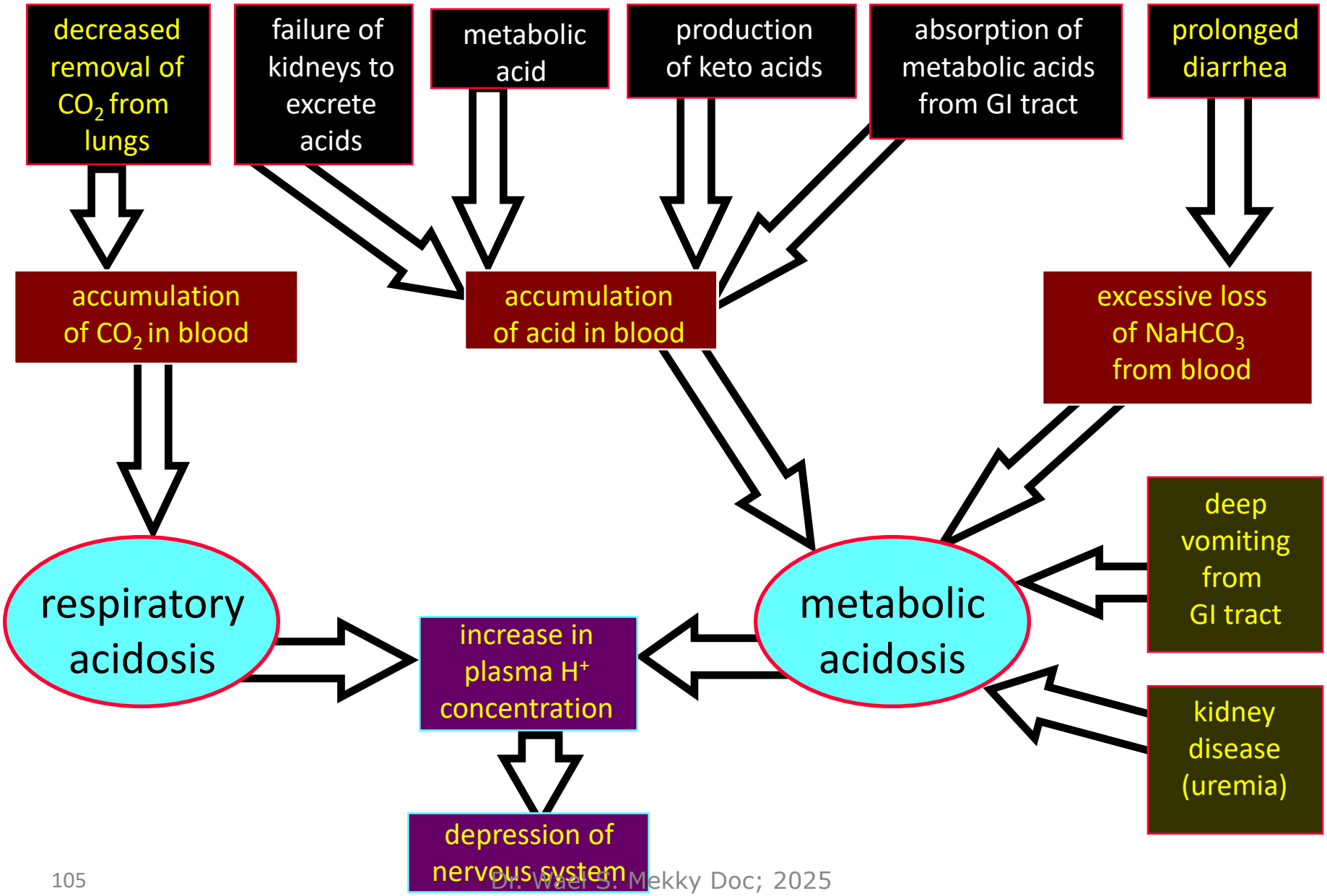
Symptoms of Metabolic Alkalosis

- Respiration slow and shallow
- Hyperactive reflexes ; Tetany
- Often related to depletion of electrolytes
- Atrial tachycardia
- Dysrhythmias

Treatment of Metabolic Alkalosis

- Electrolytes to replace those lost
- IV chloride containing solution
- Treat underlying disorder

ACIDOSIS



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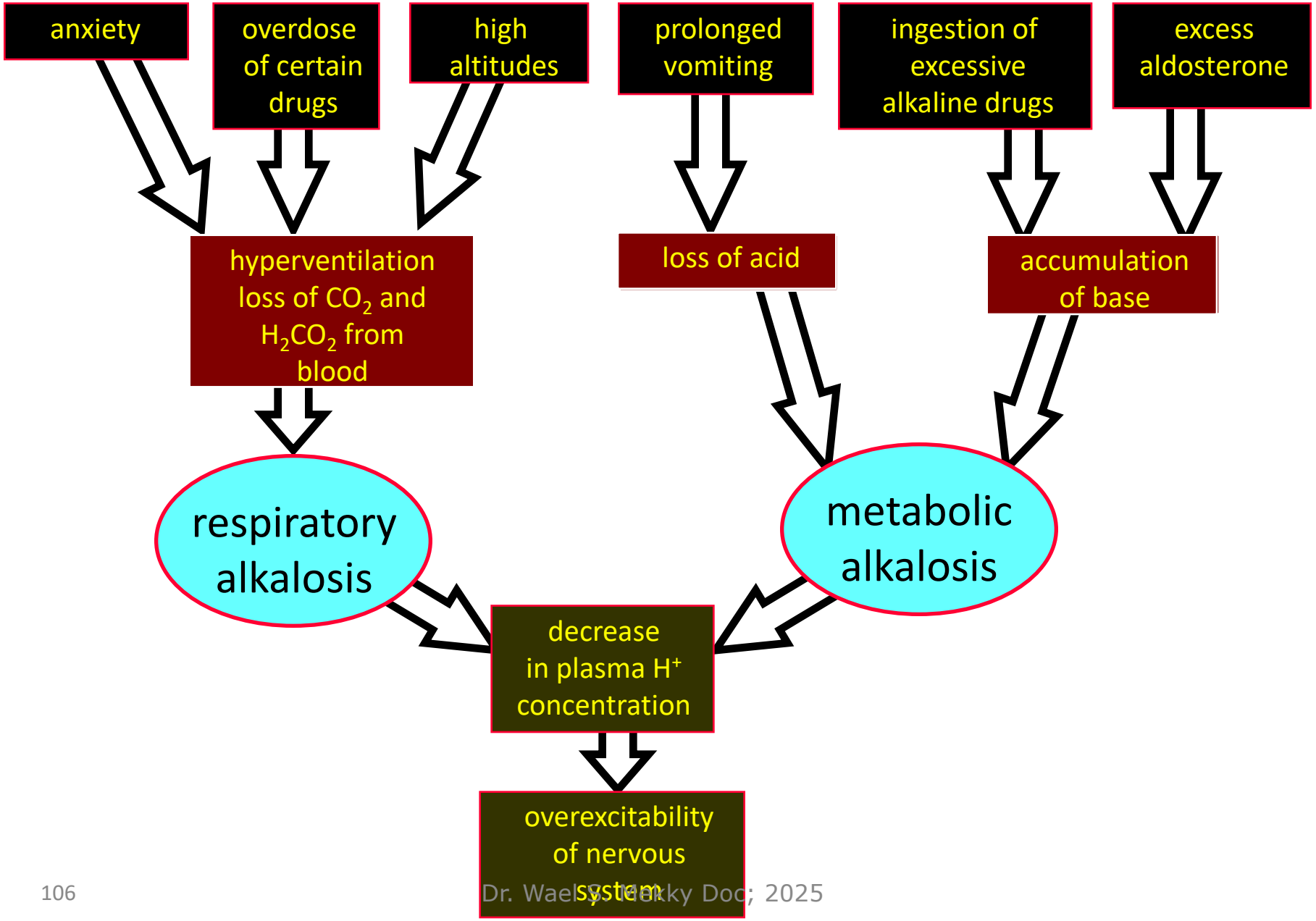
ALKALOSIS



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Organ dysfunction And Acid Base Imbalance

- CNS -

- Respiratory acidosis (suppression) and alkalosis (stimulation)

- Pulmonary –

- Respiratory acidosis (COPD) and alkalosis (hypoxia, pulmonary embolism)

- Cardiac –

- Respiratory alkalosis, respiratory acidosis, metabolic acidosis (pulmonary edema)

- GIT –

- Metabolic alkalosis (vomiting) and acidosis (diarrhea)

Organ dysfunction And Acid Base Imbalance

- Liver –

- Respiratory alkalosis, metabolic acidosis (liver failure)

- Kidney –

- Metabolic acidosis (RTA) and alkalosis (1st Aldosterone)

- Endocrine -
  - **Diabetes mellitus** – metabolic acidosis
  - **Addison's Disease**/Adrenal insufficiency – **metabolic acidosis.**(Decreased H^+ ions excretion)
  - **Cushing's Syndrome** – **metabolic alkalosis**

    (Increased Cortisol- Increased H^+ ions excretion)
  - **Primary aldosteronism** – metabolic alkalosis

Organ dysfunction And Acid Base Imbalance

- Drugs/toxins
  - Toxic alcohols – metabolic acidosis
  - ASA/Aspirin – metabolic acidosis and respiratory alkalosis(Causes Hyperventilation)
  - Theophylline overdose – respiratory alkalosis

ACID – BASE DISORDERS

Clinical State	Acid-Base Disorder
Pulmonary Embolus	Respiratory Alkalosis
Cirrhosis	Respiratory Alkalosis
Pregnancy	Respiratory Alkalosis
Diuretic Use	Metabolic Alkalosis
Vomiting	Metabolic Alkalosis
Chronic Obstructive Pulmonary Disease	Respiratory Acidosis
Shock	Metabolic Acidosis
Severe Diarrhea	Metabolic Acidosis
Renal Failure	Metabolic Acidosis
Sepsis (Bloodstream Infection)	Respiratory Alkalosis, Metabolic Acidosis

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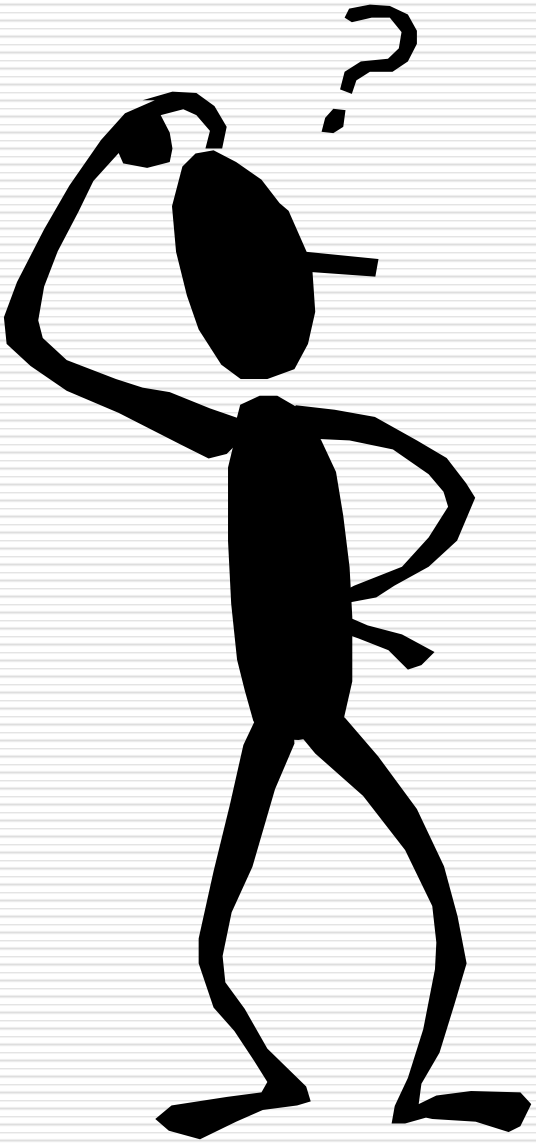
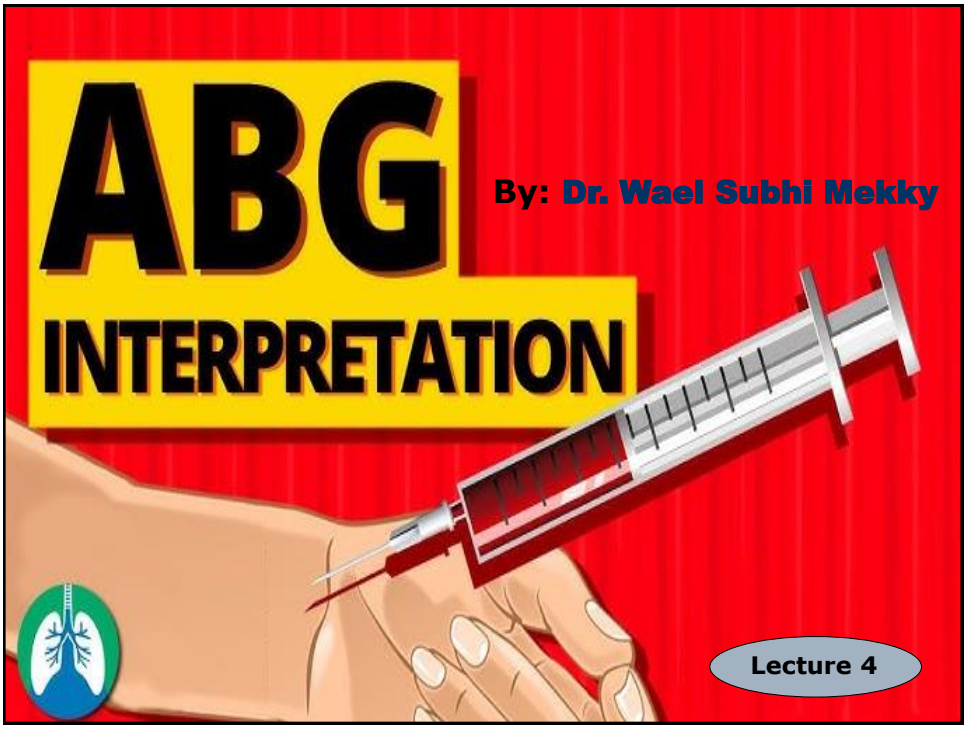
Any Questions?





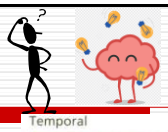
- **Arterial Blood Gas (ABG)**

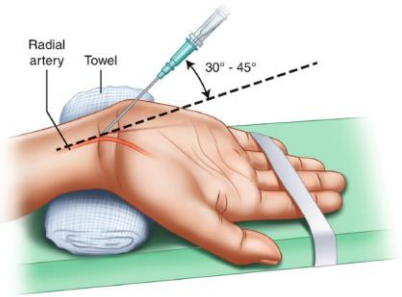
Analyzer determines Acid Base Balance and Imbalance.

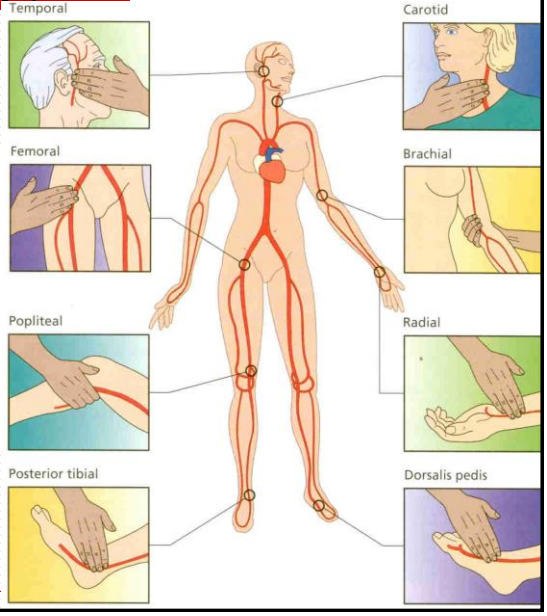
Arterial Blood Gas (ABG)

- The purpose of arterial blood gases is to assess the degree to which lungs are able to provide adequate oxygen & remove CO_2 & degree to which the kidneys are able to reabsorb or excrete HCO_3;
- It's used to establish the diagnosis and severity of respiratory failure; and guide therapy in ICU patients.

Remember...





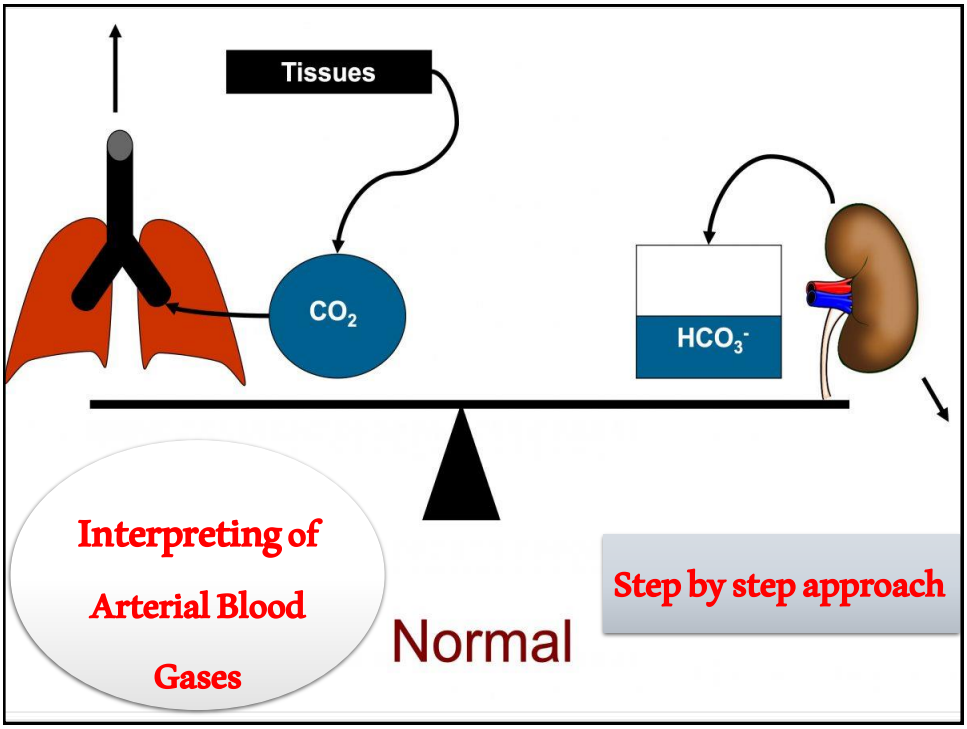


It's not magic understanding ABG's, it just takes a little practice!

Diagnosis of Acid-Base Imbalances

1. **Note** whether **the pH** is low (acidosis) or high (alkalosis)
2. Decide which value, pCO_2 or HCO_3^-, is outside the normal range
3. If the cause is a **change in pCO_2** the problem **is respiratory.**
4. If **the change is in HCO_3^-** the problem is **metabolic.**

6



Step 1: Analyze the pH

- The first step in analyzing ABGs is to look at the pH (Normal blood pH is 7.4, plus or minus 0.05, forming the range 7.35 to 7.45).
- If pH < 7.35, then acidemia
- If pH > 7.45, then alkalemia
- If pH within normal range, then acid base disorder not likely present.

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Step 2:Analyze the CO2

- The second step is to examine the pCO_2 (Normal pCO_2 level is 35-45mmHg;
- below 35 is alkalotic,
- above 45 is acidic.

9

Step 3:Analyze the HCO3

- The third step is to look at the HCO_3 level (A normal HCO_3 level is 22-26 mEq/L)
- If the HCO_3 is below 22, the patient is acidotic.
- If the HCO_3 is above 26, the patient is alkalotic).

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Step 4:Match the CO2 or HCO3 with pH

- Next match either the pCO_2 or the HCO_3 with the pH to determine the acid- base disorder.
- For example, if the pH is acidotic, and the CO_2 is acidotic, then the acid-base disturbance is being caused by **the respiratory system.**
- Therefore, we call it a respiratory acidosis.

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Step 4:Match the CO2 or HCO3 with pH

- However, if the pH is alkalotic and the HCO_3 is alkalotic, the acid-base disturbance is being caused by **the metabolic (or renal) system**.
- Therefore, it will be a metabolic alkalosis.

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Step 5: Does the CO2 or HCO3 go the opposite direction of the pH?

- If so, there is compensation by that system.
- For example, the pH is acidotic, the CO_2 is acidotic, and the HCO_3 is alkalotic. The CO_2 matches the pH making the primary acid-base disorder respiratory acidosis.

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Step 5: Does the CO2 or HCO3 go the opposite direction of the pH?

- The HCO_3 is opposite of the pH and would be evidence of compensation from the metabolic system.
- One abnormal value will be the initial change and the other will be the **compensatory response**.

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Step 6: Analyze the pO2 and the O2 saturation

- Finally, evaluate the PaO_2 and O_2 sat. If they are below normal there is evidence of **hypoxemia.**

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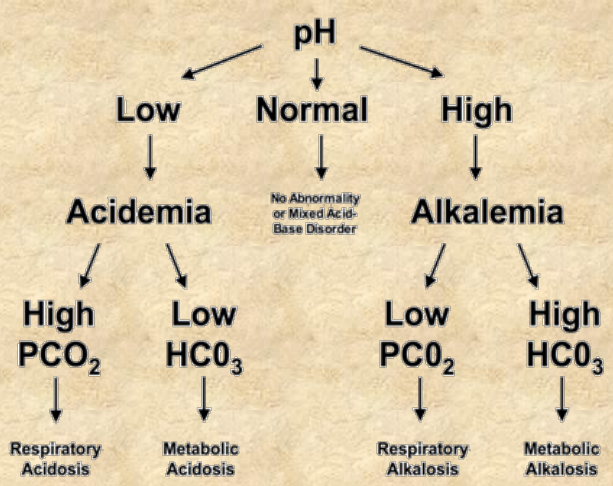
Figure 1: Identifying the Primary Process



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	pH	$PaCO_2$	HCO_3
Respiratory Acidosis	↓	↑	↑
Respiratory Alkalosis	↑	↓	↓
Metabolic Acidosis	↓	↓	↓
Metabolic Alkalosis	↑	↑	↑

Normal Arterial Blood Gas (ABG) Lab Values:

- Arterial pH: 7.35 – 7.45
- HCO_3^-: 22 – 26 mEq/L
- PCO_2: 35 – 45 mmHg
- TCO_2: 23 – 27 mmol/L
- PO_2: 80 – 100 mmHg
- Base Excess: -2 to +2
- Anion Gap: 12 – 14 mEq/L

Normal Values

pH	7.35-7.45
CO_2	35-45
pO_2	80-100
HCO_3	22-26
O_2 Sat.	95-100%

Compensation Of Acid Base Imbalance

- The body response to acid-base imbalance is called **compensation**
- May be **complete compensation** if altered pH brought back within normal limits
- **Partial compensation** if pH range is still outside norms.
- **Uncompensated** if pH range is very out from norms.

Compensation Of Acid Base Imbalance

- If **underlying problem is respiratory**, renal mechanisms can bring about **metabolic compensation.**
- If **underlying problem is metabolic**, hyperventilation or hypoventilation can help : **respiratory compensation**.

ACID – BASE DISORDERS

Clinical State	Acid-Base Disorder
Pulmonary Embolus	Respiratory Alkalosis
Cirrhosis	Respiratory Alkalosis
Pregnancy	Respiratory Alkalosis
Diuretic Use	Metabolic Alkalosis
Vomiting	Metabolic Alkalosis
Chronic Obstructive Pulmonary Disease	Respiratory Acidosis
Shock	Metabolic Acidosis
Severe Diarrhea	Metabolic Acidosis
Renal Failure	Metabolic Acidosis
Sepsis (Bloodstream Infection)	Respiratory Alkalosis, Metabolic Acidosis

Example

- A patient is in intensive care because he suffered a severe myocardial infarction 3 days ago. The lab reports the following values from an arterial blood sample:
  - pH = 7.3 (7.35-7.45)
  - HCO_3^- = 20 mEq / L (22 - 26)
  - pCO_2 = 32 mm Hg (35 - 45)

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Diagnosis

- Metabolic acidosis

With compensation

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CASE STUDY 1

* Mr. Frank is a 60 year old with pneumonia. He is admitted with dyspnea, fever, and chills. His blood gas is below:

PH 7.28
CO2 56
P02 70
HCO3 25
SaO2 89%

What is your interpretation?

Answer of Case Study 1

- Mr. frank has **uncompensated respiratory acidosis with hypoxemia** as a result of his pneumonia.

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CASE STUDY 2

* Ms. Kamel is a 24 year old college student. She has a history of Crohn's disease and is complaining of a four day history of bloody-watery diarrhea. A blood gas is obtained to assess her acid-base balance:

PH 7.28

CO_2 43

PO_2 88

HCO_3 20

SaO_2 96%

What is your interpretation?

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Answer of Case Study 2

- Mr. Kamel has an uncompensated metabolic acidosis.
- This is due to excessive bicarbonate loss from her diarrhea.

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CASE STUDY 3

* Mr. Shadi is a 80 year old nursing home resident admitted with urosepsis. Over the last 2 hours he has developed shortness of breath and is becoming confused. His ABGs shows the following results:

PH 7.02

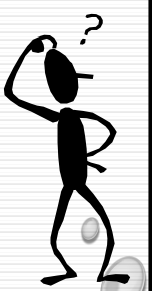
CO_2 55

PO_2 77

HCO_3 14

SaO_2 89%

What is your interpretation?



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Answer of Case Study 3

- Mr. Shadi has metabolic and respiratory acidosis with hypoxemia.
- The metabolic acidosis is caused by his sepsis.
- The respiratory acidosis is secondary to respiratory failure.

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CASE STUDY 4

* Mrs. Sasha is a thin, elderly-looking 61 year old COPD patient. She has an ABG done as part of her routine care in the pulmonary clinical. The results are as follows:

PH 7.37

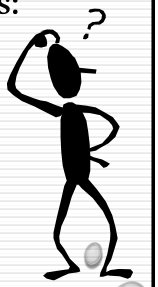
CO_2 63

PO_2 58

HCO_3 35

SaO_2 89%

What is your interpretation?



Answer of Case Study 4

- Mrs. Sasha has a **fully compensated respiratory acidosis with hypoxemia**.
- Full compensation is evidenced by the normal pH in spite of her acid/base disorder.

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CASE STUDY 5

* Mr. Sameer is a 55 yea old with Gerd. He takes about 16 Tums antacid tablets a day. An ABG is obtained to assess his acid/base balance:

PH 7.46

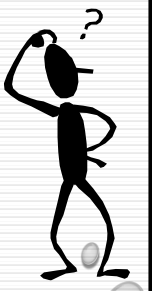
CO_2 42

PO_2 86

HCO_3 29

SaO_2 97%

What is your interpretation?



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Answer of Case Study 5

- Mr. Sameer has overmedicated himself with tums, effectively absorbing too much stomach acid.
- His ABG shows a **partially compensated metabolic alkalosis**.

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CASE STUDY 6

* Mrs. Dalal is found pulseless and not breathing this morning. After a couple of minutes of CPR she responds with a pulse and starts breathing on her own. A blood gas is obtained:

PH 6.89

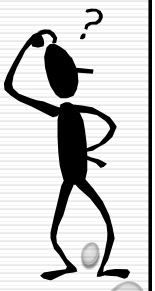
CO_2 70

PO_2 42

HCO_3 13

SaO_2 50%

What is your interpretation?



Answer of Case Study 6

- Mrs. Dalal has **severe metabolic and respiratory acidosis with hypoxemia.**

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CASE STUDY 7

* Mr. Osman was admitted for a drug overdose. She is being mechanically ventilated and a blood gas is obtained to assess her for weaning,. The ABG results are as follows:

PH 7.54

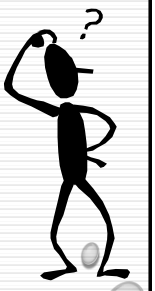
CO_2 19

PO_2 100

HCO_3 16

SaO_2 98%

What is your interpretation?



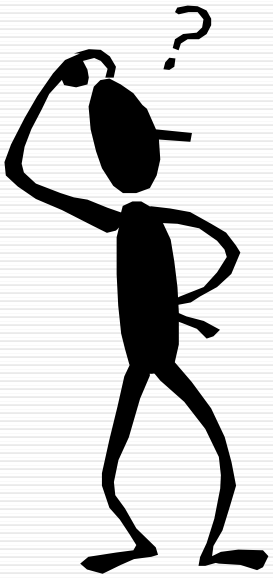
B6.WaelSimekkyDoc;2025

Answer of Case Study 7

- Mr. Osman is being over-ventilated which caused **a partially-compensated respiratory alkalosis.**

37

Any Questions?



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END
ACID - BASE BALANCE
& ABGs

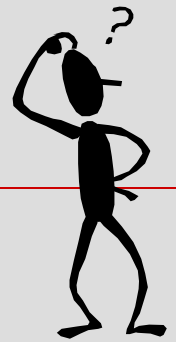
Dr. Wael S. Mekky Dec, 2025

ABG INTERPRETATION

Exercises

Dr. Wael S. Mekky

Case Study 1



Patient Information:

- Age: 68 years
- History: Chronic COPD, smoker for 40 years
- Presentation: Increasing dyspnea, confusion, use of accessory muscles.

Vital signs:
RR: 30/min
SpO_2: 84% on room air
BP: 150/85
HR: 110 bpm

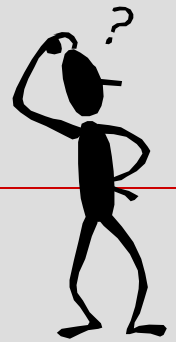
ABG Results:
pH = 7.29
$PaCO_2$ = 62 mmHg
HCO_3^- = 28 mEq/L
PaO_2 = 55 mmHg

- **What is the primary acid-base disorder?**
- **Is compensation present?**

Answer of Case Study 1

- **Primary disorder: Respiratory Acidosis**
- **Compensation: Partial metabolic compensation ($\uparrow HCO_3^-$)**

Case Study 2



Patient Information:

- Age: 24 years
- History: Type 1 DM, missed insulin doses
- Presentation: Polyuria, vomiting, abdominal pain, deep rapid breathing.

Vital signs:
RR: 34/min
BP: 90/60
HR: 125 bpm

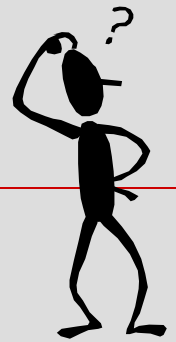
ABG Results:
pH = 7.08
$PaCO_2$ = 18 mmHg
HCO_3^- = 7 mEq/L
Glucose = 520 mg/dL

- **What is the primary acid-base disorder?**
- **Is compensation present?**

Answer of Case Study 2

- **Primary disorder: Metabolic Acidosis**
- **Compensation: Respiratory (hyperventilation ↓CO_2)**

Case Study 3



Patient Information:

- Age: 35 years
- History: Substance abuse
- Presentation: Unconscious, pinpoint pupils, respiratory depression

Vital signs:

RR: 8/min

SpO_2: 78%

BP: 100/60

ABG Results:

pH = 7.15

$PaCO_2$ = 72 mmHg

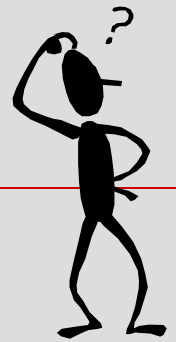
HCO_3^- = 25 mEq/L

- **What is the primary acid-base disorder?**
- **Is compensation present?**

Answer of Case Study 3

- **Primary disorder: Respiratory Acidosis**
- **Compensation: Not yet developed**

Case Study 4



Patient Information:

- Age: 55 years
- History: Gastric outlet obstruction
- Presentation: Repeated vomiting, weakness, muscle cramps

Vital signs:

BP: 95/60

HR: 105

ABG Results:

pH = 7.50

$PaCO_2$ = 48 mmHg

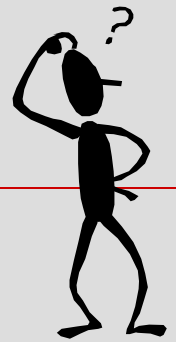
HCO_3^- = 34 mEq/L

- **What is the primary acid-base disorder?**
- **Is compensation present?**

Answer of Case Study 4

- **Primary disorder: Metabolic Alkalosis**
- **Compensation: Present but limited (hypoventilation cannot fully correct pH)**

Case Study 5



Patient Information:

- Age: 60 years
- Diagnosis: Septic shock (pneumonia source)
- Presentation: Fever, hypotension, confusion On vasopressors in ICU

ABG Results:

pH = 7.25

$PaCO_2$ = 52 mmHg

HCO_3^- = 20 mEq/L

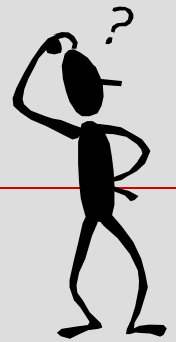
Lactate elevated

- **What is the primary acid-base disorder?**
- **Is compensation present?**

Answer of Case Study 5

- **Metabolic Acidosis & Respiratory Acidosis (hypoventilation/failure)**
- **Type: Mixed acid-base disorder**

Case Study 6



Patient Information:

- Age: 72 years
- History: Hypertension, heart failure
- Presentation: Severe dyspnea, orthopnea, pink frothy sputum

Vital signs:

RR: 34/min

SpO_2: 80% on room air

BP: 180/100

HR: 120 bpm

ABG Results:

pH = 7.48

$PaCO_2$ = 30 mmHg

HCO_3^- = 23 mEq/L

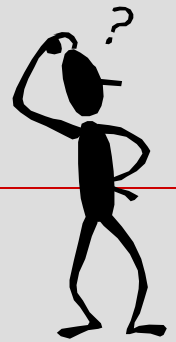
PaO_2 = 50 mmHg

- **What is the primary acid-base disorder?**
- **Is compensation present?**

Answer of Case Study 6

- **Primary disorder: Respiratory Alkalosis**
- **Compensation: None (HCO_3^- normal)**

Case Study 7



Patient Information:

- Age: 21 years
- History: Type 1 Diabetes Mellitus
- Presentation: Vomiting, abdominal pain, deep rapid breathing

Vital signs:

RR: 36/min

BP: 90/60

HR: 130 bpm

ABG Results:

pH = 7.06

$PaCO_2$ = 18 mmHg

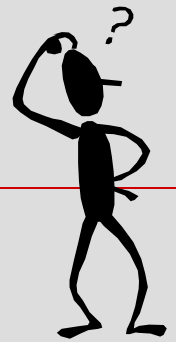
HCO_3^- = 5 mEq/L

- **What is the primary acid-base disorder?**
- **Is compensation present?**

Answer of Case Study 7

- **Primary disorder: Metabolic Acidosis**
- **Compensation: Partial Respiratory ($\downarrow CO_2$)**

Case Study 8



Patient Information:

- Age: 38 years
- History: Drug abuse
- Presentation: Decreased consciousness, shallow breathing

Vital signs:
RR: 7/min
SpO_2: 75%
BP: 100/65
HR: 90 bpm

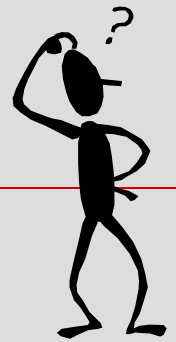
ABG Results:
pH = 7.17
$PaCO_2$ = 70 mmHg
HCO_3^- = 25 mEq/L

- **What is the primary acid-base disorder?**
- **Is compensation present?**

Answer of Case Study 8

- **Primary disorder: Respiratory Acidosis**
- **Compensation: No (HCO_3^- normal)**

Case Study 9



Patient Information:

- Age: 50 years
- History: Peptic ulcer disease
- Presentation: Persistent vomiting, weakness

Vital signs:

BP: 95/60

HR: 105

ABG Results:

pH = 7.50

$PaCO_2$ = 48 mmHg

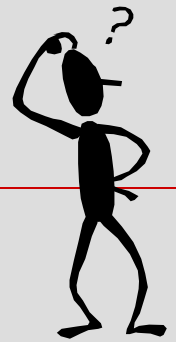
HCO_3^- = 34 mEq/L

- **What is the primary acid-base disorder?**
- **Is compensation present?**

Answer of Case Study 9

- **Primary disorder: Metabolic Alkalosis**
- **Compensation: Partial Respiratory ($\uparrow CO_2$)**

Case Study 10



Patient Information:

- Age: 65 years
- Diagnosis: Septic shock (pneumonia)
- Presentation: Fever, hypotension, confusion
- On ICU support

ABG Results:

pH = 7.22

$PaCO_2$ = 52 mmHg

HCO_3^- = 18 mEq/L

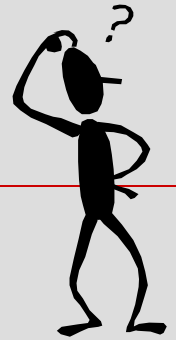
Lactate: High

- **What is the primary acid-base disorder?**
- **Is compensation present?**

Answer of Case Study 10

- **Primary disorder: Metabolic Acidosis & Respiratory Acidosis**
- **Compensation: None**

Case Study 11



Patient Information:

- Age: 70 years
- History: Chronic Kidney Disease (CKD)
- Presentation: Fatigue, confusion

ABG Results:

pH = 7.31

$PaCO_2$ = 32 mmHg

HCO_3^- = 16 mEq/L

- **What is the primary acid-base disorder?**
- **Is compensation present?**

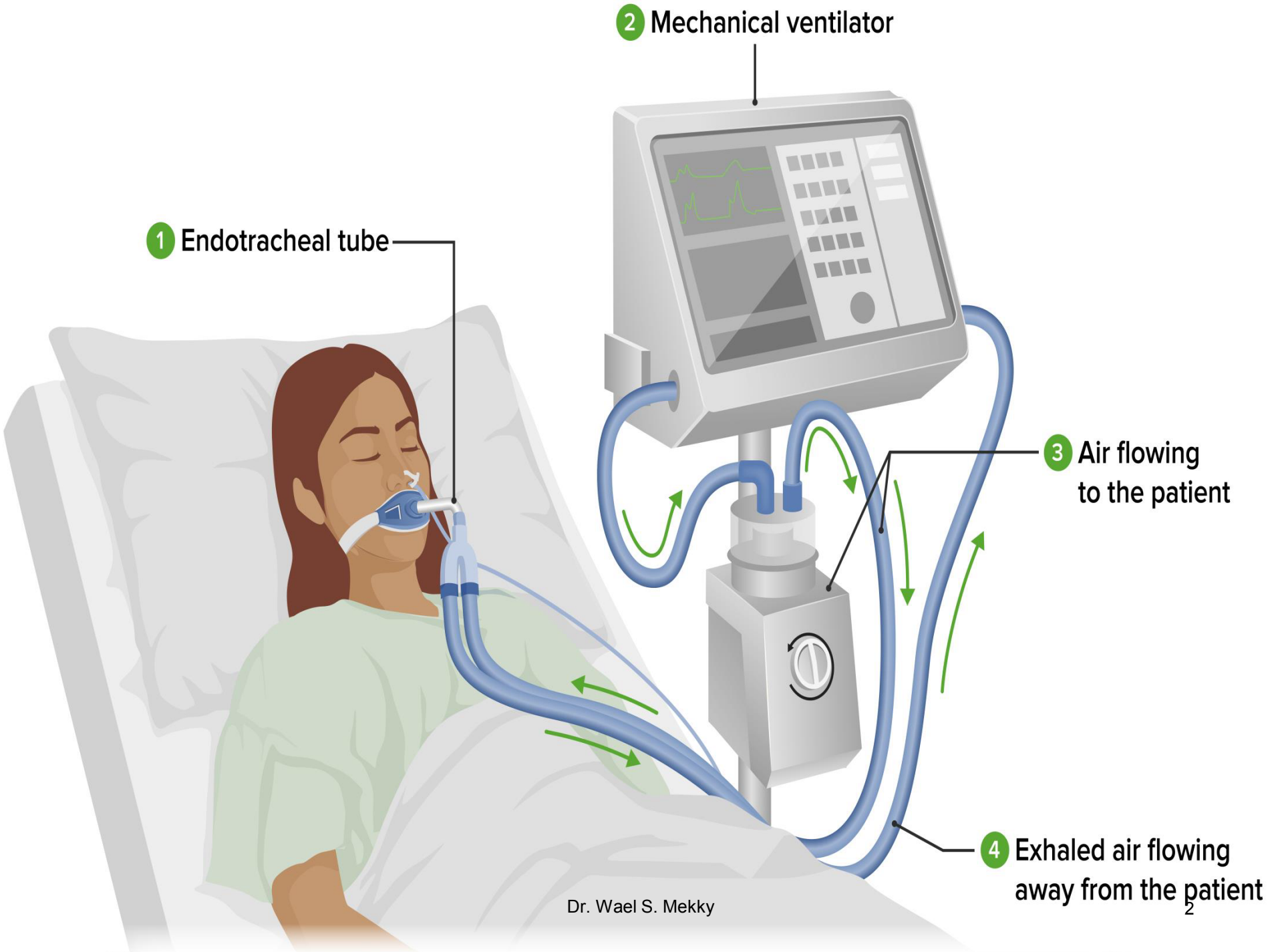
Answer of Case Study 11

- **Primary disorder: Metabolic Acidosis**
- **Compensation: Partial Respiratory ($\downarrow CO_2$)**

Critical Care Nursing

Mechanical Ventilation Part (1)

Dr. Wael Subhi Mekky


2 Mechanical ventilator
1 Endotracheal tube
3 Air flowing to the patient
4 Exhaled air flowing away from the patient

What is Mechanical Ventilation:

- Mechanical Ventilation is ventilation of the lungs by artificial means usually by a ventilator.
- Mechanical Ventilation is method to mechanically assist or replace spontaneous breathing when patients cannot do so on their own.
- A ventilator is a life support machine that generates a controlled flow of mixed air and oxygen into a patient's airway.

Purposes of Mechanical Ventilation:

- The basic effects of mechanical ventilation are improved oxygenation and carbon dioxide elimination in order to maintain or improve ventilation, & tissue oxygenation.
- To decrease the work of breathing & improve patient's comfort.

Indications For Mechanical Ventilation:

1- Acute respiratory failure due to:

- Mechanical failure, includes neuromuscular diseases as Myasthenia Gravis (1), Guillain-Barré Syndrome (2), and Poliomyelitis (failure of the normal respiratory neuromuscular system).
- Musculoskeletal abnormalities, such as chest wall trauma (flail chest)(3).
- Infectious diseases of the lung such as pneumonia, tuberculosis.

What is myasthenia gravis?

1

* **Myasthenia gravis** is a chronic autoimmune, neuromuscular disease that causes weakness in the skeletal muscles (the muscles that connect to your bones and contract to allow body movement in the arms and legs, and allow for breathing).

What is Guillain-Barré syndrome?

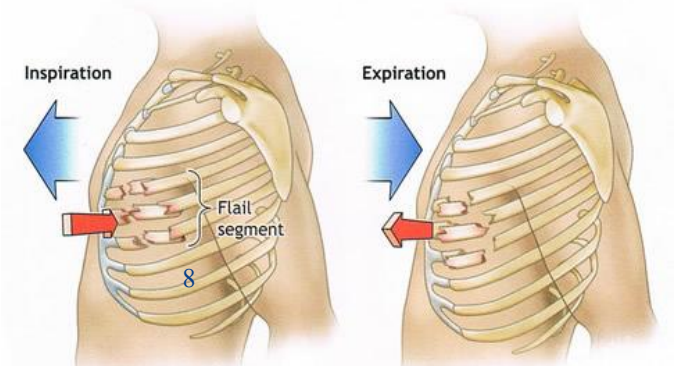
2

* **Guillain-Barré syndrome (GBS)** is a very rare and serious condition that affects the nerves. It mainly affects the feet, hands and limbs, causing problems such as numbness, weakness and pain.

What is Flail chest ?

3

* **Flail chest** is **a traumatic disorder that happens when three or more ribs located next to each other are fractured in two or more places**. That means the chest cannot expand properly and cannot properly draw air into the lungs.



Indications For Mechanical Ventilation:

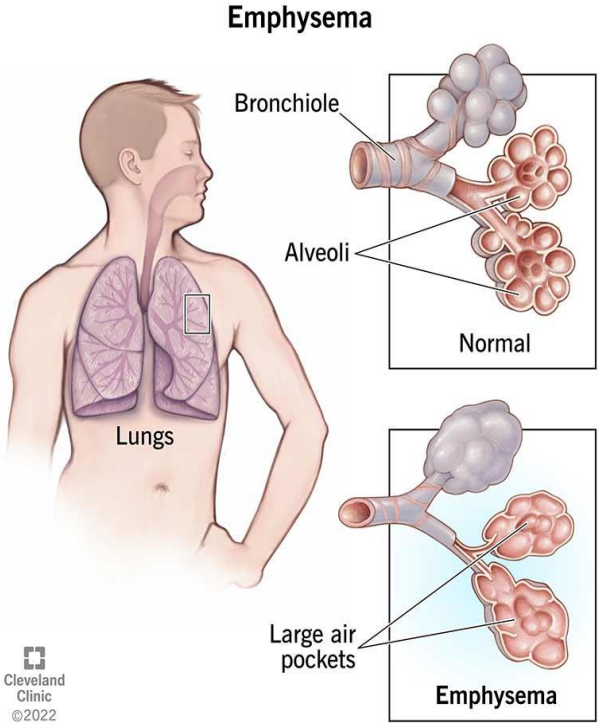
2- Abnormalities of pulmonary gas exchange as in:

- Obstructive lung disease in the form of asthma, chronic bronchitis or emphysema (4).
- Conditions such as pulmonary edema, atelectasis (5), pulmonary fibrosis.
- Patients who has received general anesthesia as well as post cardiac arrest .

4

What is Emphysema?

- **Emphysema** is a lung condition that causes shortness of breath. In people with emphysema, the air sacs in the lungs (alveoli) are damaged. Over time, the inner walls of the air sacs weaken and rupture.



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5

What is Atelectasis?

- **Atelectasis:** The collapse or closure of the lung resulting in reduced or absent gas exchange. It may affect part or all of one lung (Collapse of alveoli or lung tissue).

Indications For Mechanical Ventilation:

3- Cardiac Insufficiencies:

- Not every patient who is intubated has a primary pulmonary pathology.
- **Some cardiac diseases eliminate work of breathing as:** Cardiogenic Shock or with CHF, that the demands of the respiratory system may precipitate cardiovascular collapse.
- Therefore, supporting the patient with mechanical ventilation can reduce the demands on the heart, allowing it to recover.

Criteria for institution of ventilatory support:

Parameters	Ventilation indicated	Normal range
A- Respiratory rate (breaths/min).	**> 35**	**10-20**
B- Arterial blood Gases		
• **PH**	**< 7.25**	**7.35-7.45**
• **PaO_2 (mmHg)**	**< 60**	**75-100**
• **$PaCO_2$ (mmHg)**	**> 50**	**35-45**

Types of Mechanical ventilators:

- **Negative-pressure ventilators**
- **Positive-pressure ventilators.**

Negative-Pressure Ventilators

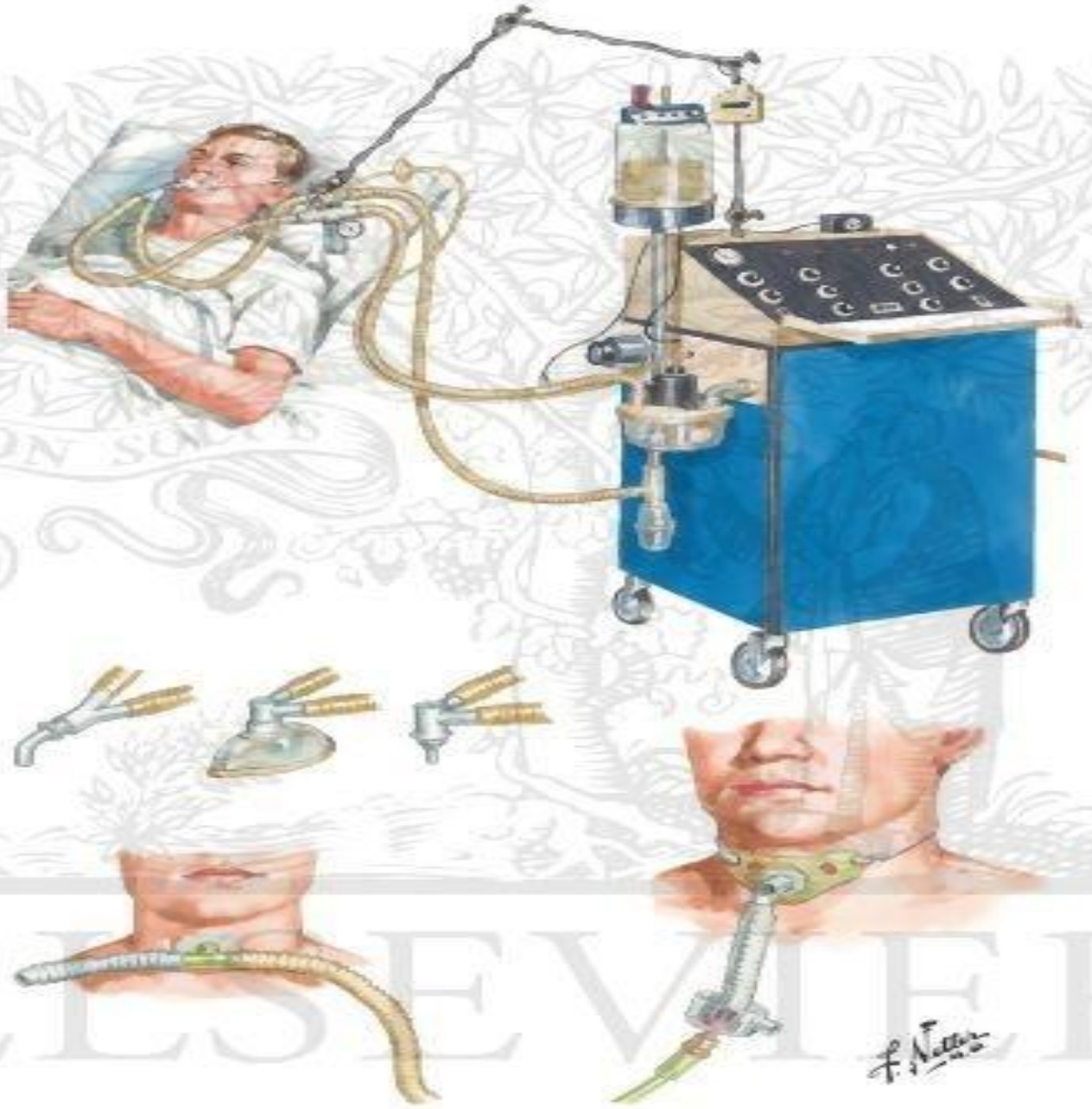
- Early negative-pressure ventilators were known as "iron lungs."
- creates a negative pressure environment around the patient's chest, thus sucking air into the lungs.
- The patient's body was encased in an iron cylinder and negative pressure was generated .
- The iron lung are still occasionally used today.

Negative-Pressure Ventilators

- Indicated for patient who have adequate lung elasticity, so it's selected for patient with neuromuscular problems, which does not require intubation of the airway.
- The use of negative-pressure ventilators is restricted in clinical practice, however, because they limit positioning and movement and they lack adaptability to large or small body chests.
- Our focus will be on the positive-pressure ventilators.

Positive-pressure ventilators

- Positive-pressure ventilators increasing the pressure in the patient's airway through an endotracheal or tracheostomy tube and thus forcing additional air into the lungs.
- Positive-pressure ventilators deliver gas to the patient under positive-pressure, during the inspiratory phase.



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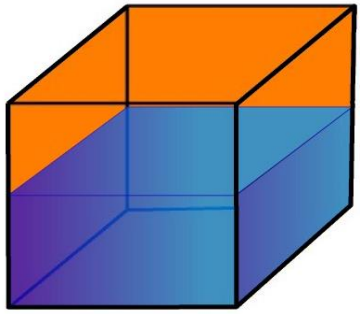
Types of Positive-Pressure Ventilators



1- Volume Ventilators.

2- Pressure Ventilators

3- High-Frequency Ventilators

1- Volume Ventilators

- The volume ventilator is commonly used in critical care settings.
- The basic principle of this ventilator is that a designated volume of air is delivered with each breath.
- In volume ventilator the machine generates flow to achieve a set volume known as

TIDAL VOLUME

Tidal Volume

V_T

Definition –

'the volume of air that is inspired or expired in a single breath during regular breathing'

1- Volume Ventilators

- The amount of volume required depends on:
  - Patient's lung compliance
  - Patient–ventilator resistance factors.
- With this mode of ventilation, a respiratory rate, inspiratory time, and tidal volume are selected for the mechanical breaths.

1- Volume Ventilators

❑Advantages of Volume Ventilation:

- Particularly beneficial when lung compliance may be an issue due to abdominal or chest surgery.
- Or when in transit and Vt cannot be monitored so closely.
- Guaranteed Minute Ventilation (Mv).

Minute Ventilation

Definition –

$\dot{M}_V$

'the total volume of gas in litres expelled from the lungs per minute'

1- Volume Ventilators

❑Disadvantages of Volume Ventilation:

- Increased monitoring of airway pressures.

- Airway pressures will increase if lung compliance decreases.

- Risk of barotrauma.

- **Barotrauma:** means injury to your body because of changes in barometric (air) or water pressure).

2- Pressure Ventilators

- The use of pressure ventilators is increasing in critical care units.
- A typical pressure mode delivers a selected gas pressure to the patient early in inspiration, and sustains the pressure throughout the inspiratory phase.
- By meeting the patient's inspiratory flow demand throughout inspiration, patient effort is reduced and comfort increased.

2- Pressure Ventilators

- Although pressure is consistent with these modes, volume is not.
- Volume will change with changes in resistance or compliance,
- Therefore, exhaled tidal volume is the variable to monitor closely.
- With pressure modes, the pressure level to be delivered is selected, and with some mode options (i.e., pressure controlled [PC]), rate and inspiratory time are preset as well.

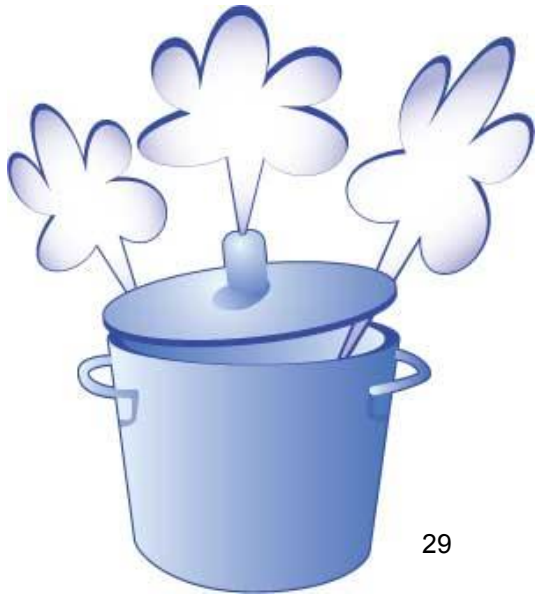
2- Pressure Ventilators

❑ Advantages of Pressure Ventilation:

- Greater control of airway pressure.
- Less risk of barotrauma.

❑ Disadvantages of Pressure Ventilation:

- No guaranteed minute ventilation.
- Increased monitoring of VT required.
- Rapid changes in the compliance can cause hypoventilation/hypoxia.



3- High-Frequency Ventilators

- High-frequency ventilators use small tidal volumes (1 to 3 mL/kg) at frequencies greater than 100 breaths/minute.
- The high-frequency ventilator accomplishes oxygenation by the diffusion of oxygen and carbon dioxide from high to low gradients of concentration.

3- High-Frequency Ventilators

- This diffusion movement is increased if the kinetic energy of the gas molecules is increased.
- A high-frequency ventilator would be used to achieve lower peak ventilator pressures, thereby lowering the risk of barotrauma.

Ventilator Mode

- The way the machine ventilates the patient.
- How much the patient will participate in his own ventilatory pattern.
- Each mode is different in determining how much work of breathing the patient has to do.

Modes of Mechanical Ventilation

A- Volume Modes:

1- Controlled Mandatory Ventilation (CMV)

2- Assist-control (A/C)

3- Intermittent Mandatory Ventilation (IMV)

4- Synchronized intermittent mandatory ventilation (SIMV)

Modes of Mechanical Ventilation

B- Pressure Modes:

1- Pressure-controlled ventilation (PCV)

2- Pressure-support ventilation (PSV)

3- Continuous positive airway pressure (CPAP)

4- Positive end expiratory pressure (PEEP)

5- Non-invasive bi-level positive airway pressure ventilation (BiPAP)

A- Volume Modes
1- Controlled Mandatory Ventilation (CMV)

- Preset rate and tidal volume
- No patient interaction with the ventilator.
- Advantage:

✓ Rest muscles from respiration

✓ Assured minute ventilation or peak pressure

- Disadvantage:

✓ The patient can not initiate a breath

✓ Awake patient tends to buck the ventilator thus requires sedation and neuro-muscular blockade.

A- Volume Modes
2- Assist Control Mode A/C

- The ventilator provides the patient with a pre-set tidal volume at a pre-set rate .
- The patient may initiate a breath on his own, but the ventilator assists by delivering a specified tidal volume to the patient. Client can initiate breaths that are delivered at the preset tidal volume.
- Client can breathe at a higher rate than the preset number of breaths/minute.

A- Volume Modes
2- Assist Control Mode A/C

- The total respiratory rate is determined by the number of spontaneous inspiration initiated by the patient plus the number of breaths set on the ventilator.
- In A/C mode, a mandatory (or "control") rate is selected.
- If the patient wishes to breathe faster, he or she can trigger the ventilator and receive a full-volume breath.

A- Volume Modes
2- Assist Control Mode A/C

Advantages:

- Often used as initial mode of ventilation when the patient is too weak to perform the work of breathing (e.g., when emerging from anesthesia).

Disadvantages:

- Hyperventilation,

A- Volume Modes
3- Intermittent Mandatory Ventilation (IMV)

- IMV assures the patient of a preset number of mechanical breathing of the preset volume to take breaths at his own rate and volume between the mandatory intervals.
- Client breaths spontaneous at any tidal volume between the mechanical breaths.
- Primary disadvantage is chance for breath stacking, therefore care should be taken to set high pressure limit properly to reduce risk of barotrauma.

A- Volume Modes

4- Synchronized Intermittent Mandatory Ventilation (SIMV)

- The ventilator provides the patient with a pre-set number of breaths/minute at a specified tidal volume and FiO_2.
- In between the ventilator-delivered breaths, the patient is able to breathe spontaneously at his own tidal volume and rate with no assistance from the ventilator.
- However, unlike the A/C mode, any breaths taken above the set rate are spontaneous breaths taken through the ventilator circuit.

A- Volume Modes

4- Synchronized Intermittent Mandatory Ventilation (SIMV)

- Adding pressure support during spontaneous breaths can minimize the risk of increased work of breathing.
- Ventilators breaths are synchronized with the patient spontaneous breathe. (no fighting)
- Used to **wean** the patient from the mechanical ventilator.
- Weaning is accomplished by gradually lowering the set rate and allowing the patient to assume more work.

A- Volume Modes

4- Synchronized Intermittent Mandatory Ventilation (SIMV)

- **Advantages:**
  - Better patient-ventilator interaction.
  - Less hemodynamic effects.
- **Disadvantages:**
  - Higher work of breathing than CMV

B- Pressure Modes

1- Pressure-controlled ventilation (PCV)

2- Pressure-support ventilation (PSV)

3- Continuous positive airway pressure (CPAP)

4- Positive end expiratory pressure (PEEP)

5- Noninvasive bilevel positive airway pressure ventilation (BiPAP)

B- Pressure Modes
1- Pressure-Controlled Ventilation Mode (PCV)

- **The PCV mode is used:**
  - If compliance is decreased and the risk of barotrauma is high.
  - It is used when the patient has persistent oxygenation problems despite a high FiO_2 and high levels of **PEEP**.
- The inspiratory pressure level, respiratory rate, and inspiratory–expiratory (I:E) ratio must be selected.

B- Pressure Modes
1- Pressure-Controlled Ventilation Mode (PCV)

- In pressure controlled ventilation the breathing gas flows under constant pressure into the lungs during the selected inspiratory time.
- The flow is highest at the beginning of inspiration(i.e when the volume is lowest in the lungs).
- As the pressure is constant the flow is initially high and then decreases with increasing filling of the lungs.
- Like volume controlled ventilation PCV is time controlled.

B- Pressure Modes
1- Pressure-Controlled Ventilation Mode (PCV)

- Advantages:

- Reduction of peak pressure and therefore decreasing the risk of barotrauma and tracheal injury.
- Effective ventilation.
- Improve gas exchange.

B- Pressure Modes
2- Pressure Support Ventilation (PSV)

- The patient breathes spontaneously while the ventilator applies a pre-determined amount of positive pressure to the airways upon inspiration.
- Pressure support ventilation augments patient's spontaneous breaths with positive pressure boost during inspiration i.e. assisting each spontaneous inspiration.
- Helps to overcome airway resistance and reducing the work of breathing.

B- Pressure Modes
2- Pressure Support Ventilation (PSV)

- Indicated for patients with small spontaneous tidal volume and difficult to wean patients.
- Patient must initiate all pressure support breaths.
- Pressure support ventilation may be combined with other modes such as SIMV or used alone for a spontaneously breathing patient.
- It is it is used primarily as a step (before CPAP) in weaning process from mechanical ventilation.

B- Pressure Modes
2- Pressure Support Ventilation (PSV)

- Advantage:

❖ Patient comfort because of Less WOB than spontaneous breathing

❖ Can used with SIMV to support spontaneous breaths

- Disadvantage:

❖ Variable tidal volume if pulmonary resistance changes rapidly

❖ If sole mode of ventilation, apnea alarm is only back-up

❖ Gas leak from circuit may interfere with cycling.

B- Pressure Modes
3-Continuous Positive Airway Pressure (CPAP)

- Used primarily in weaning process.
- Use of constant positive pressure throughout respiratory cycle for patient who is breathing spontaneously, that keeps alveoli open during inspiration and prevents alveolar collapse during expiration. Resulting in an increasing the gas exchange, and oxygenation.

B- Pressure Modes
3-Continuous Positive Airway Pressure (CPAP)

- No breaths are given by ventilator.
- Rate determined by patient.
- CPAP simply is a spontaneous breath mode, with the baseline pressure elevated above zero.
- Allows the nurse to observe the ability of the patient to breathing spontaneously while still on the ventilator.
- CPAP can be used for intubated and non-intubated patients (nasal or mask CPAP).

B- Pressure Modes
4- Positive End Expiratory Pressure (PEEP)

- Positive End Expiratory Pressure mode refers to the use of an elevated pressure during the end of expiratory phase during mandatory ventilator breath.
- It is not a stand-alone mode, but is used in conjunction with other modes as SIMV.
- When PEEP is applied to spontaneous breathing pt., the mode called CPAP.
- High level of PEEP (6-8 cm H_2O) can be used for patients with ARDS, COPD & pulmonary edema.

B- Pressure Modes
4- Positive End Expiratory Pressure (PEEP)

- Functionally this mode do as CPAP by reduce the work of breathing (at low levels) and help preserve functional residual capacity "FRC" (volume of gas remaining in the lung after a normal expiration) in case of collapsed alveoli as in atelectasis.
- PEEP is effective in an increasing intra-thoracic pressure and thus reducing venous return, cardiac output, decreasing fluid movement from the pulmonary capillaries to the alveoli, and improving oxygenation.

Uses of CPAP & PEEP

1- Prevent Atelactasis or collapse of alveoli.

2-Treat Atelactasis or collapse of alveoli.

3- Improve gas exchange & oxygenation.

4- Treat hypoxemia refractory to oxygen therapy (prevent oxygen toxicity).

5- Treat pulmonary edema (pressure help expulsion of fluids from alveoli.

B- Pressure Modes
5- Noninvasive Bilateral Positive Airway Pressure Ventilation (BiPAP)

- BiPAP is a noninvasive form of mechanical ventilation provided by means of a nasal mask or nasal prongs, or a full-face mask.
- The system allows the clinician to select two levels of positive-pressure support:
- An inspiratory pressure support level (referred to as IPAP)
- An expiratory pressure called EPAP (PEEP/CPAP level).
- BiPAP has been shown to be an effective management tool for chronic obstructive pulmonary disease and acute and chronic respiratory failure.

Common Ventilator Settings parameters/ controls

- Fraction of inspired oxygen (FIO_2)
- Tidal Volume (VT)
- Peak Flow/ Flow Rate
- Respiratory Rate/ Breath Rate / Frequency (F)
- Minute Volume (VE)
- I:E Ratio (Inspiration to Expiration Ratio)
- Sigh

Fraction of inspired oxygen (FIO_2)

- The percent of oxygen concentration that the patient is receiving from the ventilator.
  (Between 21% & 100%)

  (room air has 21% oxygen content).

- Initially a patient is placed on a high level of FIO_2 (60% or higher).

- Subsequent changes in FIO_2 are based on ABGs and the SaO_2.

Fraction of inspired oxygen (FIO_2)

- **In adult patients** the initial FiO_2 may be set at 100% until arterial blood gases can document adequate oxygenation.
- An FiO_2 of 100% for an extended period of time can be dangerous **(oxygen toxicity)** but it can protect against hypoxemia.
- **For infants**, and especially **in premature infants**, high levels of FiO_2 (>60%) should be avoided.
- Usually the FIO_2 is adjusted to maintain an SaO_2 of greater than 90%.
- **Oxygen toxicity** is a concern when **an FIO_2 of greater than 60% is required** for more than 25 hours.

Signs and symptoms of oxygen toxicity :-

1- Flushed face

2- Dry cough

3- Dyspnea

4- Chest pain

5- Tightness of chest

6- Sore throat

Tidal Volume (VT)

- The volume of air delivered to a patient during a ventilator breath. Also it is the amount of air inspired and expired with each breath.
- Usual volume selected is between 5 to 15 ml/ kg body weight).
- The large tidal volumes may aggravate the damage on the lungs. For this reason, lower tidal volume targets (6 to 8 mL/kg) are now recommended.

Peak Flow/ Flow Rate

- The speed of delivering air per unit of time, and is expressed in liters per minute.
- The higher the flow rate, the faster peak airway pressure is reached and the shorter the inspiration;
- The lower the flow rate, the longer the inspiration.

Respiratory Rate/ Breath Rate / Frequency (F)

- The number of breaths the ventilator will deliver/minute (10-16 b/m).
- Total respiratory rate equals patient rate plus ventilator rate.
- The nurse should double-checks the functioning of the ventilator by observing the patient's respiratory rate.

Minute Volume (VE)

- The volume of expired air in one minute .
- Respiratory rate times tidal volume equals minute ventilation VE = (VT x F)
- In special cases, hypoventilation or hyperventilation is desired.

For Example:

Minute Volume (VE)

In a patient with head injury, Respiratory alkalosis may be required to promote cerebral vasoconstriction, with a resultant decrease in ICP.

- In this case, the tidal volume and respiratory rate are increased (means increase of Minute volume) or (hyperventilation) to achieve the desired alkalotic pH by manipulating the $PaCO_2$.

I:E Ratio (Inspiration to Expiration Ratio):-

- The ratio of inspiratory time to expiratory time during a breath, usually set at 1:2, can be manipulated to facilitate gas exchange.

Sigh التنهيدة

- A deep breath.
- A breath that has a greater volume than the tidal volume.
- It provides hyperinflation and prevents atelectasis.
- Sigh volume :----------Usual volume is 1.5 –2 times tidal volume.
- Sigh rate/ frequency :---------Usual rate is 4 to 8 times an hour.

Peak Airway (Inspiratory) Pressures

P_{ip}

P_{ap}

P_{peak}

Definition –

'is the highest level of pressure applied to the lungs during inhalation expressed in cmh_2o'

Peak Inspiratory Pressure (PIP)

- It's a value used to set pressure alarm limits.
- In adults if the peak airway pressure is persistently above 45 cmH_2O, the risk of barotrauma is increased and efforts should be made to try to reduce the peak airway pressure.
- In infants and children it is unclear what level of peak pressure may cause damage. In general, keeping peak pressures below 30 is desirable.

Pressure Limit

- On High pressure limit is the maximum pressure the ventilator can generate to deliver the preset VT.
- Once the high pressure limit is reached, inspiration is terminated.
- Usually set 10 - 20 cm H_2O above the PIP.

Sensitivity (Trigger Sensitivity)

- The sensitivity function controls the amount of patient effort needed to initiate an inspiration.
- Increasing the sensitivity (requiring less negative force) decreases the amount of work the patient must do to initiate a ventilator breath.
- Decreasing the sensitivity increases the amount of negative pressure that the patient needs to initiate inspiration and increases the work of breathing.
- The most common setting for pressure sensitivity are -1 to -2 cm H_2O.

Ensuring Humidification and Thermoregulation

- All air delivered by the ventilator passes through the water in the humidifier, where it is warmed and saturated.
- Humidifier temperatures should be kept close to body temperature 35 °C- 37°C.
- In some rare instances (severe hypothermia), the air temperatures can be increased.
- The humidifier should be checked for adequate water levels.

Ensuring Humidification and Thermoregulation

- An empty humidifier result in drying the airway, dried secretions, mucus plugging and less ability to suction out secretions.
- Humidifier should not be overfilled as this may increase circuit resistance and interfere with spontaneous breathing.
- if moisture accumulates near the endotracheal tube, the patient can aspirate the water. The nurse and respiratory therapist jointly are responsible for preventing this.

Parameters Need To Be Adjusted To Improve Oxygenation And Ventilation

- **Oxygenation:** is commonly used to refer to oxygen saturation. So Oxygenation is monitored with pulse oximetry and by measurement of arterial blood gases.

- Oxygenation can be improved by:

✓ Increasing the FiO_2

✓ Increasing the level of the PEEP

✓ Increasing the I:E ratio

Parameters Need To Be Adjusted To Improve Oxygenation And Ventilation

- **Ventilation:** is the rate at which gas enters or leaves the lung, so it refers to the ability to 'blow off' CO_2).

- Ventilation can be improved by:

✓ Increasing the respiratory rate

✓ Increasing the tidal volume

✓ Increasing the peak pressure

TIME'S
UP!

THANK
YOU
END
SESSIONS

Critical Care Nursing

Mechanical Ventilation Part (2)

Dr. Wael Subhi Mekky

VENTILATOR ALARMS

Ventilator Alarms

- **Mechanical ventilators** comprise audible and visual alarm systems, which act as immediate warning signals to altered ventilation.
- Alarm systems can be categorized according to volume and pressure (high and low).
- High-pressure alarms warn of rising pressures.
- Low-pressure alarms warn of disconnection of the patient from the ventilator or circuit leaks.
- Ventilator alarms must never be ignored and a nurse or respiratory therapist must respond to every ventilator alarm.

Responding To Ventilator Alarms

- If an alarm sounds, respond immediately because the problem could be serious.
- Assess the patient first, while you silence the alarm.
- If you can not quickly identify the problem, take the patient off the ventilator and ventilate him with a resuscitation bag connected to oxygen source until the physician arrives.
- Ventilator malfunction is a potentially serious problem. Nurse's or respiratory therapists should perform ventilator checks every 2 to 4 hours.

Causes of Ventilator Alarms

High pressure alarm

- Increased secretions
- Kinked ventilator tubing or
- Kinked Endotracheal Tube (ETT)
- Patient biting the ETT
- Water in the ventilator tubing.

Causes of Ventilator Alarms

Low pressure alarm

- Disconnected tubing
- A cuff leak
- A hole in the tubing (ETT or ventilator tubing)
- A leak in the humidifier

Causes of Ventilator Alarms

Oxygen alarm

- The oxygen supply is insufficient or is not properly connected.

Causes of Ventilator Alarms

High respiratory rate alarm

- Episodes of tachypnea,
- Anxiety,
- Pain,
- Hypoxia,
- Fever.

Causes of Ventilator Alarms

Apnea alarm

- During weaning, indicates that the patient has a slow Respiratory rate and a period of apnea.

Causes of Ventilator Alarms

Temperature alarm

- Overheating due to too low or no gas flow.
- Improper water levels.

Complications of Mechanical Ventilation

❑ **Complications of Mechanical Ventilation are classified into:**

I- Airway Complications

II- Mechanical complications

III- Physiological Complications

IV- Artificial Airway Complications; divided into:

A- Complications related to Endotracheal Tube

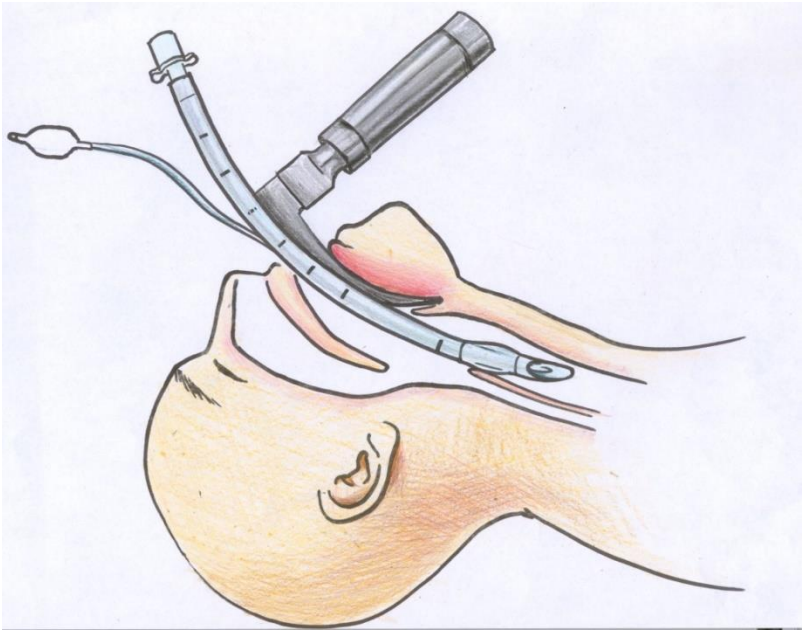
B- Complications related to Tracheostomy tube

I- Airway Complications

1- Aspiration

2- Decreased clearance of secretions

3- Nosocomial or ventilator-acquired pneumonia.



II- Mechanical complications

1- Hypoventilation with atelectasis with respiratory acidosis or hypoxemia.

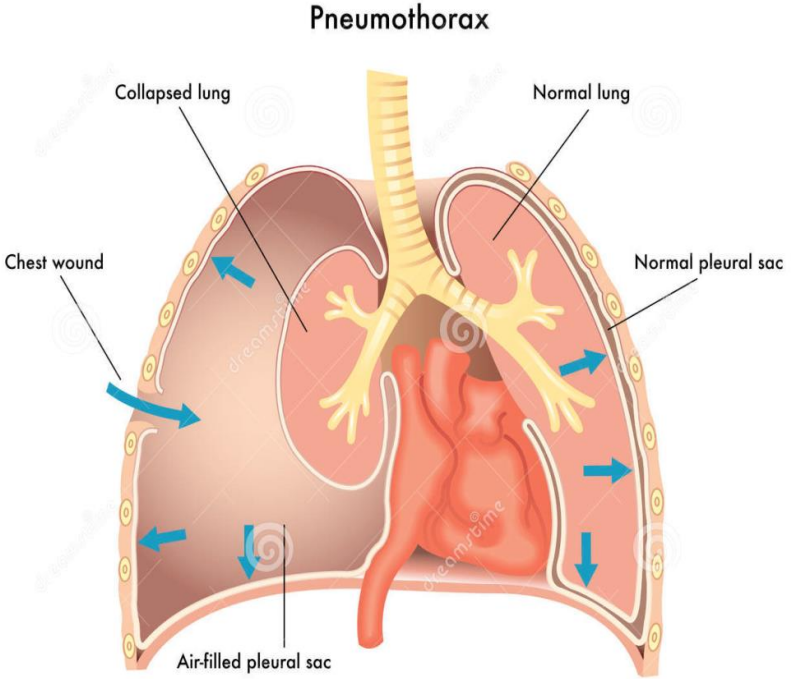
2- Hyperventilation with hypocapnia and respiratory alkalosis

3- Barotrauma:

a- Closed pneumothorax,

b- Tension pneumothorax,

c- Emphysema.



II- Mechanical complications

4- Alarm "turned off"

5- Failure of alarms or ventilator

6- Inadequate nebulization or humidification

7- Overheated inspired air, resulting in hyperthermia.

III- Physiological Complications

1- Fluid overload with humidified air and sodium chloride (NaCl) retention.

2- Depressed cardiac function and hypotension.

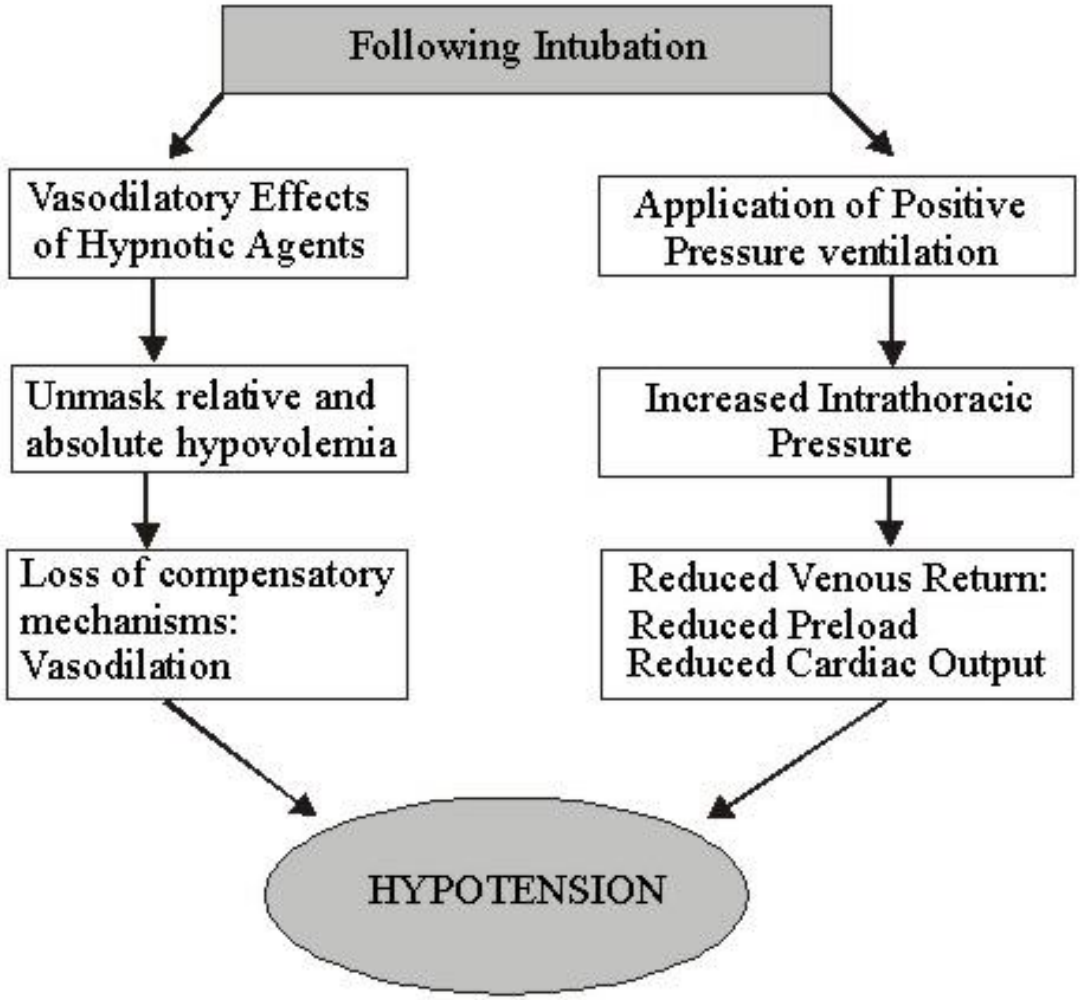
3- Stress ulcers

4- Paralytic ileus

5- Gastric distension

6- Difficult breathing pattern.

III- Physiological Complications
Hemodynamic Consequences of Intubation



Hemodynamic consequences of Intubation

NELIGAN

IV- Artificial Airway Complications

A- Complications related to Endotracheal Tube:-

1- Tube kinked or plugged

2- Cuff failure

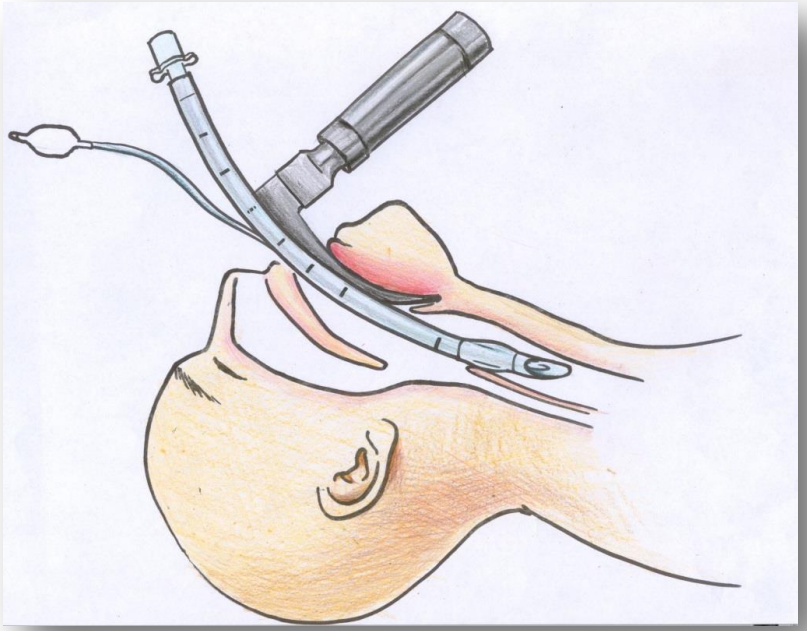
3- Tracheal stenosis or Tracheomalacia

4- lung atelectasis

5- Laryngeal edema

6- Sinusitis

7- Otitis media



IV- Artificial Airway Complications

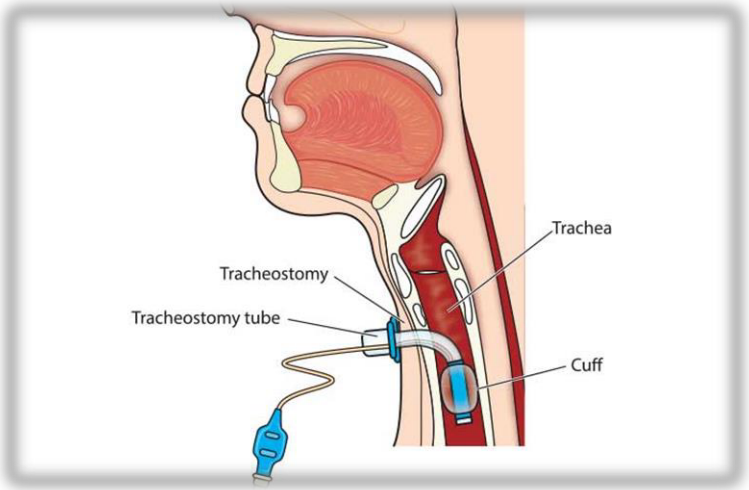
A- Complications related to Tracheostomy Tube:-

1- Acute hemorrhage at the site of tracheostomy.

2- Air embolism

3- Aspiration

4- Tracheal stenosis



5- Accidental decannulation with loss of airway.

6- Failure of the tracheostomy cuff

7- Laryngeal nerve damage.

IV- Artificial Airway Complications

A- Complications related to Tracheostomy Tube:-

8- Obstruction of tracheostomy tube

9- Pneumothorax

10- Emphysema

11- Swallowing dysfunction

12- Tracheoesophageal fistula

13- Infection.

Nursing care of patients on mechanical ventilation

Nursing Assessment:

1- Assess the patient condition

2- Assess the artificial airway (tracheostomy or endotracheal tube)

3- Assess the ventilator

Nursing care of patients on mechanical ventilation

Nursing Interventions:

1- Maintain airway patency & oxygenation

2- Promote comfort

3- Maintain fluid & electrolytes balance

4- Maintain nutritional state

5- Maintain urinary & bowel elimination

6- Maintain eye , mouth and cleanliness.

7- Maintain mobility/ musculoskeletal function.

Nursing care of patients on mechanical ventilation

Nursing Interventions:

8- Maintain safety.

9- Provide psychological support to patient

10- Facilitate communication

11- Provide psychological support & information to family

12- Responding to ventilator alarms

13- Prevent nosocomial infection

14- Documentation

Methods of Weaning

1- T-piece trial, weaning

2- Continuous Positive Airway Pressure (CPAP) weaning,

3- Synchronized Intermittent Mandatory Ventilation (SIMV) weaning,

4- Pressure Support Ventilation (PSV) weaning.

1- T-Piece trial

- It consists of removing the patient from the ventilator and having him / her breathe spontaneously on a T-tube connected to oxygen source.
- During T-piece weaning, periods of ventilator support are alternated تتناوب with spontaneous breathing.
- The goal is to progressively increase the time spent off the ventilator.

2-Synchronized Intermittent Mandatory Ventilation (SIMV) Weaning

- SIMV is the most common method of weaning.
- It consists of gradually decreasing the number of breaths delivered by the ventilator to allow the patient to increase number of spontaneous breaths.

3-Continuous Positive Airway Pressure (CPAP) Weaning

- When placed on CPAP, the patient does all the work of breathing without the aid of a back up rate or tidal volume.
- No mandatory (ventilator-initiated) breaths are delivered in this mode i.e. all ventilation is spontaneously initiated by the patient.
- Weaning by gradual decrease in pressure value.

4- Pressure Support Ventilation (PSV) Weaning

- The patient must initiate all pressure support breaths.
- During weaning using the PSV mode the level of pressure support is gradually decreased based on the patient maintaining an adequate tidal volume (8 to 12 mL/kg) and a respiratory rate of less than 25 breaths/minute.
- PSV weaning is indicated for :-

❖ Difficult to wean patients

❖ Small spontaneous tidal volume.

Extubation (Removal of ETT)

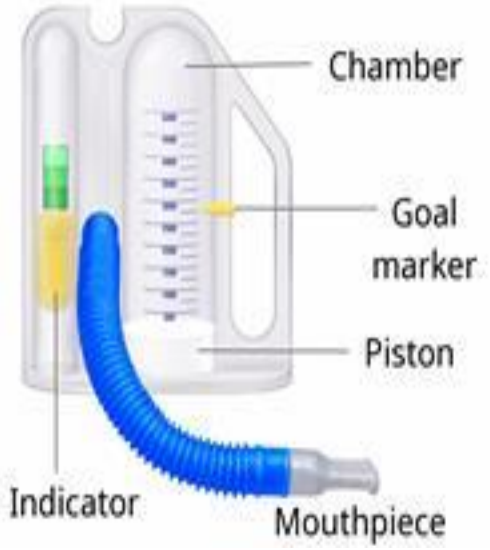
● Nursing Responsibilities During Extubation:

- Explain procedure to the pt.
- Set up new O2 delivery system (Face Mask, Nasal Canula)
- Have equipment ready to re-intubate if necessary.
- Hyperoxygenate pt./ Suction ETT and orally.
- Give patient anti-inflammatory agent as Decort or Ventolin inhalation.
- Deflate ETT cuff.
- Instruct client to take deep breath or to cough.

Extubation (Removal of ETT)

● Nursing Responsibilities During Extubation:

- Remove tube; suction any secretions in mouth and throat.
- Administer O2 via NC or FM.
- Monitor for signs of respiratory distress & upper airway obstruction.
- Monitor VS, respiratory pattern, q1hr.



- Instruct Pt. to Sit in Semi fowlers position; Take deep breaths q ½ hr; use Spirometer q 2hrs; and Limit speaking.

Criteria For Successful Weaning

- Patient is awake, alert, oriented, and cooperative
- Patient can cough & take deep breath affectively
- There is no bronchospasm
- Hemodynamically stable, adequately resuscitated.
- Arterial blood gases (ABGs) normalized or at patient's baseline:
  - $PaCO_2$ acceptable
  - PH of 7.35 – 7.45
  - PaO_2 > 60 mm Hg ,
  - SaO_2 >92%

Criteria For Successful Weaning

- Positive end-expiratory pressure (PEEP) $\leq$ **5 cm H_2O**
- **F < 25 / minute**
- **Vt 5 ml / kg**
- **VE 5- 10 L/m (f x Vt)**
- **VC > 10- 15 ml / kg**
- **PEP** (positive expiratory pressure) **> - 20 cm H2O**

(indicates patient's ability to take a deep breath & cough),

Criteria For Successful Weaning

- Chest x-ray reviewed for correctable factors,
- Major electrolytes within normal range,
- Hematocrit >25%,
- Core temperature >36°C and <39°C,
- Adequate management of pain/anxiety/agitation.
- Adequate analgesia/ sedation (record scores on flow sheet),

Role of Nurse Before Weaning:-

1- Ensure that indications for the implementation of Mechanical ventilation have improved,

2- Ensure that all factors that may interfere with successful weaning are corrected:-

- **Acid-base abnormalities**
- **Electrolyte abnormalities**
- **Fever**
- **Hyperglycemia**
- **Fluid imbalance**
- **Infection**
- **Anemia**
- **Sleep deprivation**

Role of Nurse Before Weaning:-

3- Assess readiness for weaning

4- Ensure that the weaning criteria / parameters are met.

5- Explain the process of weaning to the patient and offer reassurance to the patient.

6- Initiate weaning in the morning when the patient is rested.

7- Elevate the head of the bed & Place the patient upright.

Role of Nurse Before Weaning:-

8- Ensure a patent airway and suction if necessary before a weaning trial.

9- Provide for rest period on ventilator for 15 – 20 minutes after suctioning.

10- Ensure patient's comfort.

11- Evaluate and document the patient's response to weaning.

Role of Nurse During Weaning:-

1- Wean only during the day.

2- Remain with the patient during initiation of weaning.

3- Instruct the patient to relax and breathe normally.

4- Monitor the respiratory rate, vital signs, ABGs, and use of accessory muscles frequently.

If signs of fatigue or respiratory distress develop.

- Discontinue weaning trials.

5- Monitor the signs of weaning intolerance criteria:-

Signs of Weaning Intolerance Criteria

- Diaphoresis
- Dyspnea & Labored respiratory pattern
- Increased anxiety ,Restlessness, Decrease in level of consciousness
- Dysrhythmia, Increase or decrease in heart rate of > 20 beats /min. or heart rate > 110 b/m, Sustained heart rate >20% higher or lower than baseline.

Signs of Weaning Intolerance Criteria

- Increase or decrease in blood pressure of > 20 mm Hg
- Systolic blood pressure >180 mm Hg or <90 mm Hg
- Increase in respiratory rate of > 10 above baseline or > 30
- Sustained respiratory rate greater than 35 breaths/minute
- Tidal volume $\leq$ 5 mL/kg, Sustained minute ventilation <200 mL/kg/minute
- SaO2 < 90%, PaO2 < 60 mmHg, decrease in PH of < 7.35.
- Increase in $PaCO_2$.

Role of nurse after weaning

1- Ensure that Extubation criteria are met

2- Decannulate or extubate

3- Documentation.

Criteria for institution of ventilatory support:

Parameters	Ventilation indicated	Normal range
A- Pulmonary function studies:		
• Respiratory rate (breaths/min).	> 35	10-20
• Tidal volume (ml/kg body wt)	< 5	5-7
• Vital capacity (ml/kg body wt)	< 15	65-75
• Maximum Inspiratory Force (cm HO_2)	<-20	75-100

Complications of Mechanical Ventilation

1- Nosocomial or ventilator-acquired pneumonia

2- Barotrauma - parenchymal damage due to high airway pressure/volume

3- Hemodynamic compromise - due to high intra-thoracic pressure and reduced venous return (hypotension)

4- Inadequate Ventilation

5- Decrease lung elasticity (Ventilator Dependence)

6- Depressed cardiac function and hypotension

Complications of Mechanical Ventilation

7- Tension pneumothorax

8- Acid - Base Disturbances and O_2 toxicity

9- Aspiration

10- Psychological Complications

11- Inadequate Ventilation and Leakage

12- Stress ulcers

13- Paralytic ileus

14- Gastric distension

15- Accidental decannulation with loss of airway.

Three Types Of Positive-Pressure Ventilators:

- Ventilators are classified according to how the inspiratory phase ends. The factor which terminates the inspiratory cycle reflects the machine type.
- They Are Three Types:

1- Pressure cycled ventilator

2- Volume cycled ventilator

3- Time cycled ventilator

1- Pressure-Cycled Ventilator

- Rarely used. Sometimes used post-anesthesia, for short periods, and for respiratory therapy.

- Pushes air into lungs until a predetermined airway pressure is reached.
- Although pressure is consistent with these modes, volume is not. Volume will change with changes in resistance or compliance, therefore, exhaled tidal volume is the variable to monitor closely.

2- Volume-Cycled Ventilator

- Today volume cycled ventilators is the most commonly used among positive pressure ventilator.
- The volume of air delivered with each inspiration is preset regardless of pressure needed to deliver volume.
- Once the preset volume is delivered to the patient, the ventilator cycles off and exhalation occurs passively.

2- Volume-Cycled Ventilator

- The peak inspiratory pressure (PIP) must be monitored in volume modes because it varies from breath to breath. Pressure limits set so as not to over expand the lungs, that the amount of pressure required to deliver the set volume depends on:

- Patient's lung compliance

- Patient–ventilator resistance factors.

- Constant Tidal volume delivered regardless of compliance of lungs or chest wall, and airway resistance (advantage)

3- Time-Cycled Ventilator

- Tidal volume and pressure is variable according to patient and/or ventilators.
- Ventilator pushes air into lungs until preset time has elapsed. The volume of air patient receives is regulated by the length of inspiration and flow rate of the air.
- Time cycled machines are not used in adult critical care settings. They are used in pediatric intensive care areas.

Good Luck